

CHAPTER 21

Moderation and Mediation in Interindividual Longitudinal Analysis

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INTRODUCTION

In 1986, Reuben Baron and David Kenny published their classic article on the distinction between mediator and moderator variables in social psychological research (Baron & Kenny, 1986). This particular paper has been incredibly influential on the way data have been analyzed in the social sciences and, according to the Web of Science, has been cited more than 24,000 times to date. In fact, this is the most highly cited paper written by any social psychologist (Nosek et al., 2010), and its citation count is still climbing by more than 1000 each year. This

particular article has even spawned its own spinoff literature, with a number of recent articles including Baron and Kenny in their titles, specifically written in response to and elaborating upon the initial Baron and Kenny work, including *Reconsidering Baron and Kenny* (Zhao, Lynch, & Chen, 2010), *Reflections on the Baron and Kenny Model of Statistical Mediation* (Pardo & Román, 2013), and two additional works with *Beyond Baron and Kenny* in their titles (Hayes, 2014; Krause et al., 2010).

Perhaps the popularity and influence of this particular paper is due to that fact that it strives to distinguish between *mediation* and *moderation*, two “often-confused

functions of third variables” (Baron & Kenny, 1986, p. 1173), both conceptually and in terms of the analyses necessary to test them. Both mediation and moderation involve systems of three variables. Each has an initial or focal predictor variable (which we will refer to as X) and an outcome or criterion variable (which we will refer to as Y). In addition, there is a third variable—the mediator (M) or moderator (Z)—involved in the system. Baron and Kenny (1986) state that a moderator “is a qualitative...or quantitative...variable that affects the direction and/or strength of the relation between an independent or predictor variable and a dependent or criterion variable” (p. 1174) and that it “partitions a focal independent variable into subgroups that establish its domains of maximal effectiveness in regard to a given dependent variable” (p. 1173). In other words, a moderator affects the association between two variables such that the impact of the predictor on the criterion varies depending on the level or value of the moderator (Holmbeck, 1997). Baron and Kenny (1986) stated that a mediator, on the other hand, “represents the generative mechanism through which the focal independent variable is able to influence the dependent variable of interest” (p. 1173). The key features defining such a mediator model are that (1) the predictor influences the mediator, and (2) the mediator, in turn, influences the outcome variable (Holmbeck, 1997; MacKinnon & Dwyer, 1993).

In the nearly 30 years since the publication of the Baron and Kenny article, the analytic literatures on both mediation and moderation have continued to grow. For example, recommendations for the post hoc probing of moderated effects have been extensively developed (Aiken & West, 1991), as have optimal methods for statistical testing and finding confidence intervals for mediated effects (MacKinnon, Fritz, Williams, & Lockwood, 2007; MacKinnon, Lockwood, Hoffman, West, & Sheets, 2002; MacKinnon, Lockwood, & Williams, 2004; Preacher & Hayes, 2004, 2008). In addition, the collection of longitudinal data has become more prevalent as new analytic approaches well-suited to such data have been developed and researchers have come to realize the benefit of investigating how individuals change across time (Willett, 1997). In this chapter, we will discuss how both mediation and moderation can be explored in longitudinal data in which the focal predictors, outcomes, and moderators and mediators are repeatedly measured over time.

We will begin by discussing how the concepts of mediation and moderation feature in developmental psychopathology research and outlining the typical analysis methods used to identify moderation and mediation in

cross sectional data. Because these methods and their application in particular literatures has been covered extensively elsewhere (e.g., Dearing & Hamilton, 2006; Frazier, Tix, & Baron, 2004; Gogineni, Alsup, & Gillespie, 1995; Rose, Holmbeck, Coakley, & Franks, 2004) our discussion will be relatively brief. Then we will explain the possible benefits of collecting longitudinal observations and discuss two analytic frameworks—multilevel modeling and structural equation modeling—that are often used to model longitudinal data. Then we will bring these together and discuss in detail various models designed to test moderation and mediation hypotheses in longitudinal data, including a lag-as-moderator approach, *interactions* between time and time-varying covariates in multilevel models, a mediation *change over time* model, autoregressive and latent growth models (as well as a combination of the two), a latent change score approach, and an exponential decay model. This is followed by a discussion of how modern methods of determining causal inference may be applied and extended in the longitudinal context. Finally, we conclude with a section on promising directions for future research on mediation and moderation in longitudinal data.

THE ROLE OF MODERATION AND MEDIATION ANALYSES IN DEVELOPMENTAL PSYCHOPATHOLOGY RESEARCH

Several of the foundational ideas in developmental psychopathology involve the types of associations that can be appropriately tested with models that include moderated and mediated effects. A primary goal—if not *the* primary goal—of developmental science is to understand how processes unfold over time, and it is precisely this idea of *process* that guides methodology designed to detect mediation. In fact, one of the earliest presentations of methodology to detect longitudinal mediation effects described the procedure as process analysis (Judd & Kenny, 1981). The Judd and Kenny article is particularly important as it describes the importance of longitudinal mediation relations, while the Baron and Kenny article does not. The often used geographic analogy of *developmental pathways* implies that earlier variables affect later variables in series and readily maps onto mediational logic where initial variables predict mediators that, in turn, predict outcomes. The sort of sequential effects in a mediation analysis could also be used to model the spreading influence across domains or systems in developmental cascades (Masten & Cicchetti, 2010) and chain

reactions (Rutter, 1999). In addition, mediation analyses can be useful in exploring how interventions designed to change or prevent maladaptive behavior achieve their effects. Intervention versus control is the initial variable, constructs that are targeted by the intervention are the mediators, and the behavior is the outcome in such an analysis.

The idea that contextual effects influence individual development is a natural fit for a moderation framework, which can model how characteristics of the individual interact with aspects of the physical and social environment to produce adaptive and maladaptive behavior. The emphasis on risk and resilience that features prominently in the developmental psychopathology literature also suggests an analytic framework involving moderator variables. Vulnerabilities and protective factors from multiple domains serve as moderators and interact with stressors to produce a range of behavioral outcomes. Any particular outcome is probably not a simple function of a single isolated factor but rather the result of a combination of multiple biological, social/cultural, and psychological factors, and a framework that can accommodate such interactions is necessary to model such influences appropriately. The concept of multifinality—the idea that a given risk factor could lead to a variety of different outcomes depending on other aspects of the system (Cicchetti & Rogosch, 1996)—is necessarily going to involve some moderator variables that determine the manner in which the risk factor is ultimately expressed behaviorally.

Indeed, modeling a process as complex as human development may ultimately require models that combine aspects of both mediation and moderation. Viewing the individual as a complex system with variables at multiple levels exerting mutual and potentially reciprocal effects might lead to models that examine the effects of change at one level of the system on mediators at other levels that ultimately affect behavioral outcomes. One might also test whether the effects of focal variables at one level might depend upon the state of moderator variables at other levels. Adding to this the longitudinal aspect of processes unfolding over time results in an incredibly intricate framework of influences that will require careful consideration, clever study design, and sophisticated analytical tools to explore and ultimately understand. The focus of this chapter is to provide developmental psychopathology researchers with information about analytic techniques involving mediation and moderation in longitudinal data that might prove useful in understanding how developmental processes result in adaptive and maladaptive behavior.

ECLS-K DATA EXAMPLES

The various moderation and mediation analyses described in this chapter will be illustrated with examples drawn from the Early Childhood Longitudinal Study–Kindergarten Cohort (ECLS-K) data. This data set includes a wealth of information on a nationally representative sample of children who entered kindergarten in the 1998–1999 school year and were followed through the eighth grade. ECLS-K participants were selected via a multistage probability sampling design (U.S. Department of Education, National Center for Educational Statistics, 2009). One hundred geographical areas consisting of counties or groups of counties were initially selected, and a total of 1,280 public and private schools were sampled from within these areas. A target of approximately 24 kindergarteners enrolled at each school were then selected for the study. In total, the original sample at kindergarten fall included 21,260 children.

Study participants were assessed at kindergarten fall and spring, first grade fall and spring, third grade spring, fifth grade spring, and eighth grade spring. Because the first grade fall assessment involved only a small subsample of participants and many of the measures of interest here were not assessed at eighth grade spring, we focus on the remaining five assessments. For analyses in which equally spaced time points are desirable, we will focus on the first through fifth grade assessments, which occurred every 2 years. At each assessment, data were gathered from children, parents, and teachers. Our illustrative analyses utilize demographic information, child academic assessments, and teacher questionnaire data.

To further unify the illustrative analyses, we will focus on mediation and moderation effects involving attention-deficit/hyperactivity disorder (ADHD). A number of research efforts have examined ADHD in the ECLS-K data set (e.g., Barnard-Brak & Brak, 2011; Morgan, Staff, Hillemeier, Farkas, & Maczuga, 2013; Schneider & Eisenberg, 2006; Stone, Brown, & Hinshaw, 2010). Several of these have focused primarily on diagnoses of ADHD within the sample. We focus instead on reported ADHD symptom levels, which provide different (and, some would argue, better) information than diagnosis indicators (Currie & Stabile, 2006, 2009; Fletcher & Wolfe, 2008).

A modified version of the Social Rating Scale (adapted by ECLS-K with permission; Gresham & Elliott, 1990) was administered to teachers at each time point to assess children's social and behavioral functioning in the classroom. Four of the five subscales of the Social Rating Scale are used in our illustrative analyses: Approaches to Learning

(LEARN, six items), Externalizing Behavior (EXTERN, five items), Interpersonal Skills (INTERP, five items), and Internalizing Behavior (INTERN, four items). These subscales are highly reliable, with split half reliabilities ranging from .76 (EXTERN at third grade spring) to .92 (INTERN at fifth grade spring; Tourangeau, Nord, Lê, Sorongon, & Najarian, 2009). A detailed examination of the associations between these subscales and ADHD diagnosis measures in the ECLS-K data set showed that approaches to learning and externalizing behavior were the strongest predictors of diagnosis (Elder, 2010). The Approaches to Learning subscale includes items that rate the child's attentiveness, task persistence, eagerness to learn, learning independence, flexibility, and organization. The Externalizing Behavior subscale includes items that indicate the frequency with which a child argues, fights, gets angry, acts impulsively, and disturbs ongoing activities. Thus, we will use these particular variables as proxies for the inattentive (LEARN) and impulsive/hyperactive (EXTERN) symptoms of ADHD.

The ECLS-K data includes a number of academic performance measures. We utilize the IRT theta estimates for age-graded reading (READ) and mathematics (MATH) performance. The reliability of these scores ranged from .91 (for math at kindergarten fall) to .96 (for reading at kindergarten spring; Tourangeau et al., 2009).

Our illustrative analyses in the ECLS-K data will examine a number of moderation and mediation effects involving ADHD that have been previously identified in the literature. For example, we will examine how the association between ADHD symptoms and academic performance changes over time (or, alternately, as children grow older; Frazier, Youngstrom, Glutting, & Watkins, 2007; Kawabata, Tseng, & Gau, 2012). We will also examine potential moderating effects of gender on the association between ADHD and negative academic and social effects (Gaub & Carlson, 1997; Gershon & Gershon, 2002; Lahey et al., 2007; Lee, Lahey, Owens, & Hinshaw, 2008; Owens, Hinshaw, Lee, & Lahey, 2009).

Our primary mediation example is drawn from a recent article examining possible mediators of the association between ADHD and depression (Humphreys et al., 2013). Specifically, we will examine whether interpersonal skills and academic performance in reading and math mediate the association between our two ADHD symptom measures and depression. Such a mediated effect has been suggested as an explanation for the comorbidity between ADHD and depression that fits within a developmental psychopathology perspective (Drabick & Kendall, 2010). ADHD may lead to difficulties in social and academic functioning, and over time, failure to achieve success in

these areas may lead to depression. Interpersonal skills and depression will be measured by the appropriate subscales of the Social Rating Scale. The Interpersonal Skills subscale (INTERP) rates a child's skill in forming and maintaining friendships, getting along with people who are different, comforting or helping others, expressing feelings and ideas in positive ways, and showing sensitivity to the feelings of others. The Internalizing Behavior subscale (INTERN) includes items rating the apparent presence of anxiety, loneliness, low self-esteem, and sadness. Admittedly, this is a confounded measure of depression, but it is the best proxy for this construct in the public use data set.

Because we wish to focus the reader's attention on similarities and differences between different methods of estimating longitudinal moderation and mediation, our analyses of the ECLS-K data are somewhat oversimplified. In particular, they do not address the complexly multilevel nature of the data, and they do not include sampling weights. Our analyses of the longitudinal ECLS-K data incorporate the nesting of multiple observations over time within individuals, but they do not explicitly model higher levels of nesting, for example individuals nested within classrooms, classrooms nested within schools, and schools nested within the counties that make up the primary sampling units of the study. Moreover, we did not make use of the multiple weights included in the ECLS-K data set designed to adjust for oversampling of Asians and Pacific Islanders and survey nonresponse (Tourangeau et al., 2009) to produce estimates that characterize the U.S. population of kindergarteners from the 1998–1999 cohort. As a consequence, similarity among students within classes and schools may produce some downward bias in the standard errors reported here, and our results are not necessarily representative of the whole of the U.S. population of students.

BASIC MODERATION

A situation in which the effect of one predictor varies depending on the value of another defines moderation. Moderation hypotheses are common in the social sciences and can be tested using a number of different frameworks, including one-way analysis of variance (ANOVA), regression, and structural equation modeling. ANOVA-based methods for testing moderation were developed prior to those in other areas; however, these methods are appropriate only for categorical predictors. Application of ANOVA-based methods to continuous predictors requires artificial dichotomization of the continuum into high and low groups, which is not ideal because it unnecessarily

reduces the power to detect the moderated effect and because it can produce spurious main and moderated effects under certain conditions (Maxwell & Delaney, 1993). Hence we will focus here on moderation within the regression framework.

The most typical approach to testing moderation effects with regression simply represents the synergistic effect between two predictors as a cross-product term and includes the cross-product in a regression equation simultaneously with its lower order component vectors. If we wish to test whether the effect of a predictor variable X on an outcome variable Y is moderated by Z , we would fit an equation:

$$Y = b_0 + b_1X + b_2Z + b_3XZ + e \quad (21.1)$$

and examine the test of the b_3 coefficient to determine whether the effect of X on Y is moderated as predicted. The analysis is carried out in the same way regardless of whether X and Z are continuous measures or measured dichotomies. If one or more of the predictors is a categorical variable with more than two levels, a series of coded vectors will be required to represent it, and the test of the moderated effect will involve more than a single vector. In such a case, each vector coding the first predictor will need to be multiplied by each vector coding the second predictor, and the whole set of cross-products should be tested using a hierarchical regression approach to determine the R^2 change associated with the addition of the cross-products to the model (Cohen, Cohen, West, & Aiken, 2003).

Because cross-product vectors created to represent moderation effects are often highly collinear with their component vectors, it is typically recommended that the researcher center continuous predictor variables at their means prior to creating the cross-product. This will reduce the correlation between the vectors by eliminating what is known as nonessential collinearity (Aiken & West, 1991) among the predictor vectors, while leaving the associations between the predictors and the outcome unaltered. This can improve estimation as well as provide meaningful interpretations for lower order coefficients in the regression analysis. For example, if we left both X and Z in their raw metric in equation 21.1, the intercept of equation 21.1 would represent expected Y values for a subject with zeros on both the X and Z measures. Zero might not represent a particularly meaningful value on the X and Z scales, and it might actually be off-scale (e.g., if X and Z were 1–7 Likert-type measures). Similarly, the b_1 coefficient in equation 21.1 would represent the expected change in Y

associated with a one unit increase in X when $Z = 0$ and the b_2 coefficient would represent the expected change in Y associated with a one unit increase in Z when $X = 0$. If we were to center each of the two predictors around their respective means, however, and create the cross-product from these centered variables, the intercept will represent the expected outcome for an “average” subject with mean values on both X and Z , the b_1 coefficient would represent change associated with X when Z is at its mean, and b_2 would represent change associated with Z when X is at its mean.

In the ECLS-K data, we tested whether the association between teachers’ ratings of ADHD symptoms (LEARN and EXTERN, the predictors) and interpersonal skill (INTERP), reading (READ) and math performance (MATH), and internalizing behavior (INTERN) outcomes at kindergarten fall might be moderated by child gender (GENDER). The ADHD symptom predictors were centered at their means, and GENDER was coded such that males = 0 and females = 1. Regression results are reported in Table 21.1. As the coefficient b_1 indicates the association between the ADHD symptom predictor and the outcome at GENDER = 0 (male), the second column of Table 21.1 was labeled as the ADHD symptom effect for males (e.g., b_1 LEARN_{Male}). The coefficient b_2 in the third column (e.g., b_2 Gender) represents the difference between females and males in the outcome variable at the mean of the ADHD symptoms.

The fourth column of Table 21.1 presents the tests of the b_3 interaction coefficients in each of the various analyses. Only 2 of the 8 interactions were significant. Specifically, LEARN and EXTERN interacted with GENDER to predict READ and INTERP, respectively. These significant coefficients associated with the cross-product terms indicate that, in fact, the association between attentive symptoms and reading performance and the association between impulsive/hyperactive symptoms and interpersonal skill ratings is different for boys and girls. Both effects are slightly stronger for girls than for boys.

Higher order interactions involve synergistic effects among three or more variables. For example, we might predict that an XZ interaction is significant at only one of the two levels of a dichotomous variable W . Or if W is a continuous variable, we might hypothesize that the XZ interaction becomes more pronounced at higher levels of W . We would fit

$$Y = b_0 + b_1X + b_2Z + b_3W + b_4XZ + b_5XW + b_6ZW + b_7XZW + e \quad (21.2)$$

TABLE 21.1 Coefficients and Standard Errors for Moderation Models

Outcome	b ₁ LEARN _{Male}	b ₂ GENDER	b ₃ Interaction	LEARN _{Female}
INTERP	0.6475*** (.0074)	0.0395*** (0.0072)	0.0063 (0.0108)	0.6538*** (0.0079)
READ	0.2966*** (0.0077)	−0.0002 (0.0075)	0.0314** (0.0112)	0.3280*** (0.0082)
MATH	0.3257*** (0.0068)	−0.0816*** (0.0066)	−0.0051 (0.0099)	0.3206*** (0.0072)
INTERN	−0.2935*** (0.0079)	0.0384*** (0.0078)	0.0073 (0.0116)	−0.2862*** (0.0084)
	b ₁ EXTERN _{Male}	b ₂ GENDER	b ₃ Interaction	EXTERN _{Female}
INTERP	−0.5217*** (0.0083)	0.0705*** (0.0083)	−0.0541*** (0.0132)	−0.5757*** (0.0103)
READ	−0.1095*** (0.0082)	0.0551*** (0.0081)	−0.0038 (0.0129)	−0.1132*** (0.0100)
MATH	−0.1266*** (0.0074)	−0.0289*** (0.0074)	−0.0013 (0.0117)	−0.1279*** (0.0091)
INTERN	0.2324*** (0.0080)	0.0203* (0.0080)	−0.0116 (0.0127)	0.2208*** (0.0099)

and examine the test of the b_7 coefficient associated with the three-way interaction to determine whether the interactive effect of XZ , in fact, varies depending on the value of W . Tests of whether the two gender interactions identified above might differ for families with varying incomes in the ECLS-K data found no significant three-way interactions when income was represented continuously (using the natural log of the WKINCOME variable) or dichotomously (using a median split on the WKINCOME variable), suggesting that the interactive effects remained stable across the range of income.

Significant interactions are typically followed by post hoc tests of simple effects. These examine the effect of one variable at particular values of the other. When both variables involved in the interaction are categorical, the post hoc analyses involve splitting the sample into groups defined by the levels of one variable and examining the effect of the other variable within each of these groups. Tests are typically conducted using model sums of squares produced in the group-specific analyses and residual sums of squares from the full model in which the interaction was initially estimated (Pedhazur, 1997). If one variable is continuous and the other categorical, the data splitting approach could be used to examine the effect of the continuous predictor within the various groups defined by the levels of the categorical variable, or the effect of the categorical variable could be examined at particular values of the continuous predictor. The Pothoff extension of the Johnson-Neyman technique can be used to find the ranges of the continuous predictor where the categorical variable has a significant effect, or specific values on the continuum can be selected at which to test the significance

of the categorical distinction using a recentering strategy. Both approaches are discussed in detail in Aiken and West (1991).

When both variables involved in the interaction are continuous, the appropriate post hoc analyses involve testing the simple slope of one variable at particular values of the other. Ideally the simple slopes would be calculated and tested at meaningful points along the continuum of the other continuous variable. Often simple effects are simply examined at the mean and values ± 1 standard deviation from the mean of a variable, which might be considered average, high, and low levels of the construct. Some care should be taken with this approach, however, as it is possible that one of the extreme values might fall outside the range of actual values if the variable is highly skewed.

The various terms in equation 21.1 can be regrouped as

$$Y = (b_0 + b_2Z) + (b_1 + b_3Z)X + e \quad (21.3)$$

to show that that $b_1 + b_3Z$ provides an estimate of the simple slope of Y on X at a particular value of Z . The coefficient b_1 estimates the simple slope of Y on X when $Z = 0$, which will be at the mean of Z if centered predictors were used in the regression analysis. Values of Z (or centered Z , if centered predictor variables were used) can simply be substituted into equation 21.3 to calculate simple slopes at high and low Z values. The standard error of these simple slopes can be calculated using the typical equation for a linear combination of coefficient values:

$$SE_{b_1+b_3} = \sqrt{s_{b_1}^2 + 2Zs_{b_1,b_3} + Z^2s_{b_3}^2} \quad (21.4)$$

where $s_{b_1}^2$ is the sampling variance of b_1 , $s_{b_3}^2$ is the sampling variance of b_3 , and s_{b_1, b_3} is the sampling covariance between b_1 and b_3 (Aiken & West, 1991). The matrix of sampling covariances of the coefficient estimates can be obtained from most statistical programs that estimate ordinary least squares (OLS) regression. The square roots of the diagonal elements of this matrix are the standard errors of the coefficients reported in typical regression output.

We examined simple effects associated with the previously identified significant gender interactions in the ECLS-K data using a recoding strategy. Examination of the lower order coefficient for b_1 in Table 21.1 shows that the effect of approaches to learning on reading performance for boys (i.e., when GENDER = 0) is 0.2966, which is significant at $p < .0001$. This indicates that for every 1-point increase in the approaches to learning subscale score, we should expect a 0.2966 point increase in reading performance. We can then calculate the effect for girls by combining the b_1 and b_3 coefficients: $b_1 + b_3 = 0.2966 + 0.0314 = 0.3280$. So for girls, for every 1-point increase in the approaches to learning subscale score, we should expect a 0.328 point increase in reading performance. The significance of this value could be tested by computing the standard error of the linear combination $b_1 + b_3$ as described in equation 21.2. Alternately, GENDER could be recoded such that females = 0 and males = 1, the cross-product recomputed and the analysis rerun to produce an estimate of the LEARN effect for girls as the b_1 coefficient, shown in column 5 of Table 21.1. The standard error of this simple effect for girls is .0082 and the effect is significant at $p < .0001$. Similarly, we see that the simple effect of externalizing on interpersonal skill is -0.5217 for boys and -0.5757 for girls, both significant at $p < .0001$.

BASIC MEDIATION MODEL

The basic framework of a single mediator model at a single time point (i.e., a cross sectional mediation model) is shown in the two panels of Figure 21.1. The top panel shows the path diagram for a single predictor variable X and its effect on an outcome variable Y . The coefficient c on the path linking X to Y represents the total effect of the predictor on the outcome variable. The lower panel of Figure 21.1 shows the path diagram for this association when including the mediator variable. The coefficient a represents the effect of the predictor X on the mediator M , and the coefficient b represents the effect of the mediator M on the outcome Y ,

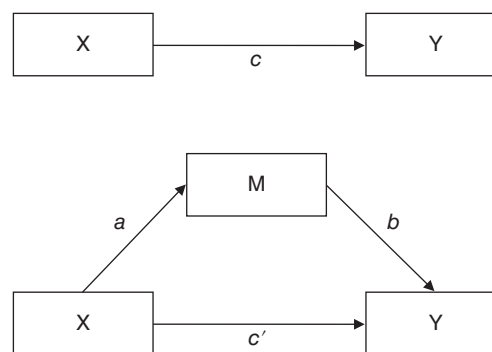


Figure 21.1 Path diagrams for the mediation model.

over and above the effect of the predictor. The coefficient c' on the path linking X to Y represents the direct effect of the predictor on the outcome variable, over and above the effect of the mediator.

The estimates of the path coefficients in these diagrams can be straightforwardly obtained through a series of three OLS regression analyses or equivalently through fitting two structural equation models to the manifest variables. The regression model corresponding to the top panel of Figure 21.1 would be

$$Y = b_0 + cX + e \quad (21.5)$$

The coefficient associated with the X predictor in this equation estimates the c path coefficient representing the total effect of X on Y . Two regression analyses are necessary to produce estimates of the coefficients in the lower panel of Figure 21.1. The first of these predicts the mediator variable from the initial predictor:

$$M = b_0 + aX + e \quad (21.6)$$

The coefficient associated with the X predictor in this equation estimates the a path coefficient linking the predictor to the mediator. The second regression equation predicts the outcome variable from the predictor and the mediator simultaneously:

$$Y = b_0 + c'X + bM + e \quad (21.7)$$

The coefficient associated with the X predictor in this equation estimates the c' path coefficient representing the direct effect of X on Y . This is the effect of the predictor on the outcome that *cannot* be attributed to the mediator. The coefficient associated with the M mediator in this equation

estimates of the b path coefficient linking M to Y , over and above the effect of X .

Baron and Kenny's (1986) steps approach to testing mediation specifies a number of conditions that must be met to establish mediation:

1. The independent variable must affect the mediator.
2. The predictor must affect the outcome.
3. The mediator must affect the outcome.
4. The effect of the predictor on the outcome must be less when the mediator is present in the analysis than when it is absent.

Each of these steps involves coefficients from the three regression equations specified above. Step 1 requires examination of the significance of the a path coefficient from equation 21.6. Step 2 requires examination of the significance of the c path coefficient for the total effect from equation 21.5. Step 3 requires examination of the significance of the b path in equation 21.7. Then Step 4 involves comparing the c and c' coefficients from equations 21.5 and 21.7. If $|c'| < |c|$, the final step has been met, and according to this approach, mediation has been identified. Once mediation has been established, the significance of the c' coefficient in equation 21.7 can be examined to determine whether mediation is complete (i.e., no significant direct effect once the mediator has been taken into account) or partial (a significant direct effect persists).

Two point estimates of the mediated or indirect effect can be produced from the set of coefficient estimates in equations 21.5–21.7. The first such point estimate is that implied by Step 4 of the Barron and Kenny approach: $c - c'$. Using this difference as a point estimate of the mediated effect makes intuitive sense: when the effect of X on Y not due to the mediator (i.e., the direct effect or the c' estimate) is subtracted from the total effect of X on Y (i.e., the c estimate), the difference logically must be the effect of X on Y that is due to the mediator. A second point estimate of the mediated effect is the product of the a and b path coefficients. Using this product as a point estimate of the mediated effect makes intuitive sense for this function as well: if the a path conveys the influence of X on M and the b path similarly conveys the influence of M on Y , their product naturally conveys the influence of X (at the beginning of the system) to Y (at the end of the system). With simple single-level data and continuous M and Y measures, these two point estimates are algebraically equivalent (MacKinnon, Warsi, & Dwyer, 1995).

Several researchers have pointed out flaws in the steps approach to mediation. The first of these is that Step 2,

which specifies that the total effect of X on Y must be significant, is not strictly necessary (MacKinnon et al., 2002; Zhao et al., 2010) for a mediated or indirect effect to be present. There are situations where the direct and indirect effects might have different signs (inconsistent mediation or suppression, see MacKinnon, Krull, & Lockwood, 2000), which might combine to produce a nonsignificant total effect that would not meet the criteria for Step 2. Such a situation may also prove problematic for Step 4, which specifies that the direct effect must be smaller in magnitude than the total effect. Consider, for example, a situation in which $a = 2$, $b = -1$, and $c' = 2$. The total effect c would be equal to the sum of the indirect effect ab and direct effect c' . This would be $c = ab + c' = 2 * -1 + 2 = 0$, and the condition of Step 2 would not be met. Moreover, in this case $|c'| > |c|$, as $2 > 0$, and the condition of Step 4 would also not be met, and a researcher applying the steps approach would erroneously conclude that no mediation was present.

Another apparent difficulty of strictly applying the steps approach is that at Step 4, any difference in the appropriate direction could be deemed evidence of mediation, but given ever-present sampling variability, a stricter criterion requiring a significant difference seems more appropriate. An approximate standard error can be derived via first-order Taylor series expansion or the multivariate delta method (MacKinnon, 2008):

$$s_{ab} = \sqrt{b^2 s_a^2 + a^2 s_b^2} \quad (21.8)$$

where s_a^2 and s_b^2 are the sampling variances of the regression estimates of a and b . These can be obtained by squaring the standard errors of a and b produced in all standard regression output or by examining the diagonal elements of the co/variance matrix of the estimates which can generally be requested as additional output in statistical software for regression estimation. This particular form of the standard error of the mediated effect is often referred to as the Sobel standard error (Sobel, 1982, 1986). Baron and Kenny (1986) suggested a slightly more complex approximate standard error, which can be derived via second order Taylor series expansion and represents the exact variance under the condition of independence:

$$s_{ab} = \sqrt{b^2 s_a^2 + a^2 s_b^2 + s_a^2 s_b^2} \quad (21.9)$$

A statistical test of the mediated effect can be conducted by dividing either the ab or the equivalent $c - c'$ estimate of

the mediated effect by the selected approximate standard error. This should be distributed as a z in large samples, and thus values more extreme than ± 1.96 would be considered significantly different from zero at $p < .05$. However, this large sample z test assumes that the product ab is normally distributed, and the ab estimate of mediation generally is not (MacKinnon et al., 2007). This test has been found to be overly conservative (i.e., to have low power) in simulation studies (MacKinnon et al., 2002). This may or may not be problematic, depending on the particular circumstances in which the test is employed. If a researcher uses this test in an existing data set and finds a significant mediated effect, the fact that the test is conservative and therefore somewhat underpowered does not diminish the significance of the finding in any way, although a confidence interval formed around the significant point estimate of the mediated effect will still be too wide and indicate more uncertainty in the value than is truly warranted. In contrast, a nonsignificant result might indicate a null effect or might simply be the result of an underpowered test. In addition, using this test to determine the necessary number of subjects for a future study in an a priori power analysis would result in an unnecessarily large sample size recommendation.

Because of this distributional difficulty, several alternative approaches have been suggested. One such alternative involves looking not at the distribution of ab , but at the product $z_a z_b$, where z_a and z_b are z scores formed as the ratios of the coefficient estimates to their standard errors. The $z_a z_b$ product is known to have a complex nonnormal distribution, but values obtained in such an analysis can be compared to tabled values (Meeker, Cornwell, & Aroian, 1981) and asymmetric confidence limits can be obtained from software written especially for this purpose (MacKinnon et al., 2007; Tofighi & MacKinnon, 2011). An alternate approach involves a resampling technique known as bootstrapping. Bootstrapping involves repeated resampling with replacement (Efron & Tibishirani, 1993). For example, if a researcher is bootstrapping cases from a sample of 100, many (typically over 1000) new samples of 100 cases are drawn from the original sample. These samples may differ from the original in that a given case may appear more than once in each bootstrap sample. Then the statistic of interest may be calculated for each bootstrap sample. These are ordered and values of the statistic associated with the desired percentiles (often the 2.5th and 97.5th for a 95% confidence interval) define the lower and upper limits of a confidence interval. This procedure may be refined using a procedure known as bias correction to adjust for the fact that the statistic obtained in the original analysis may not fall at the 50th

percentile of the distribution of bootstrap values (Efron & Tibishirani, 1993). In the case of a mediated effect, the cases themselves are resampled and ab estimates produced from regressions involving X , Y , and M in each of the new bootstrap samples are obtained (MacKinnon et al., 2004; Preacher & Hayes, 2004).

The simple single-mediator model in Figure 21.1 can be expanded to multivariate situations that include multiple predictors and multiple mediators. With multiple predictor and mediator variables, it is common to run an initial set of analyses examining mediated effects of each predictor–mediator pairing individually, then to run a combined model to examine the *unique* effects of each variable over and above the others. For example, if we were to expand Figure 21.1 to include three mediators, equation 21.5 examining the effect of the single predictor on the single outcome would remain unchanged, producing a single estimate of the total effect c . The X to M associations examined with equation 21.6 would require three separate regression equations, each predicting a single mediator from the predictor X , producing three unique a coefficients, a_1 , a_2 , and a_3 . Equation 21.7 would need to include all three mediators simultaneously along with the predictor to produce three unique b coefficients b_1 , b_2 , and b_3 and a single estimate of the direct effect c' . The $c - c'$ point estimate of mediation in this case represents the total mediated effect and the $a_1 b_1$, $a_2 b_2$, and $a_3 b_3$ point estimates represent specific mediated effects unique to each of the three mediators.

A set of cross sectional mediation analyses was conducted in the ECLS-K data. These analyses tested whether the effect of ADHD symptoms on depression was mediated by academic performance and interpersonal skill at the initial kindergarten fall assessment. Specifically, teacher ratings of approaches to learning and externalizing behavior were treated as initial predictor variables, theta scores on math and reading and teacher ratings of interpersonal skill served as mediators, and teacher ratings of internalizing behavior were the outcome. Regression equations similar to those in equations 21.5 to 21.7 were first fitted to each predictor–mediator pair separately, then all predictors and mediators were entered simultaneously into a multiple mediator model. The results of these analyses are summarized in Table 21.2.

The third and fourth columns of Table 21.2 present a and b coefficient estimates, with their standard errors below in parentheses. The results of these analyses show that both approaches to learning and externalizing behaviors significantly predict each of the three mediators, and each of the three mediators significantly affects the internalizing

TABLE 21.2 Coefficients, Standard Errors, Tests of Mediation, and Bootstrap Confidence Intervals for Single- and Multiple-Mediator Models

X	M	Single-mediator models				Multiple-mediator models			
		<i>a</i>	<i>b</i>	Med(Z test)	99% bootstrap CI: LL, UL	<i>a</i>	<i>b</i>	Med (Z test)	99% bootstrap CI: LL, UL
LEARN	INTERP	0.6564*** (0.0053)	-0.1515*** (0.0083)	-0.0994*** (0.0055)	-0.1145, -0.0842	0.5213*** (0.0058)	-0.1282*** (0.0088)	-0.0668*** (0.0046)	-0.0808, -0.0550
LEARN	READ	0.3115*** (0.0055)	-0.0234** (0.0081)	-0.0073** (0.0025)	-0.0147, -0.0014	0.3427*** (0.0064)	0.0042 (0.0115)	0.0014 (0.0039)	-0.0085, 0.0112
LEARN	MATH	0.3113*** (0.0049)	-0.0422*** (0.0091)	-0.0131*** (0.0028)	-0.0212, -0.0072	0.3405*** (0.0056)	-0.0626*** (0.0130)	-0.0213*** (0.0044)	-0.0334, -0.0094
EXTERN	INTERP	-0.5545*** (0.0064)	-0.2348*** (0.0073)	0.1302*** (0.0043)	0.1183, 0.1411	-0.2755*** (0.0060)	-0.1282*** (0.0088)	0.0353*** (0.0025)	0.0280, 0.0433
EXTERN	READ	-0.1197*** (0.0062)	-0.1315*** (0.0076)	0.0157*** (0.0012)	0.0137, 0.0199	0.0638*** (0.0067)	0.0042 (0.0115)	0.0003 (0.0007)	-0.0016, 0.0023
EXTERN	MATH	-0.1227*** (0.0056)	-0.1699*** (0.0084)	0.0208*** (0.0014)	0.0165, 0.0236	0.0596*** (0.0059)	-0.0626*** (0.0130)	-0.0037*** (0.0009)	-0.0063, -0.0017

outcome. The *ab* point estimates and the Sobel standard errors of each mediated effect appear in the fifth column of the table. When examined in separate analyses, each of these mediated effects is significant. In this particular case, the bootstrap 99% confidence intervals in the sixth column of the table are very close to those that would be computed using the normal theory and Sobel standard error provided in the previous column. The fact that zero is not included in any of these confidence intervals indicates significance at $p < .01$ and corresponds to the z test results.

The remaining columns of the table present the results of a multiple predictor–multiple mediator analysis in which all predictors and mediators were analyzed simultaneously. A path diagram for this particular model is shown in Figure 21.2. The two initial predictor variables, approaches to learning and externalizing behavior, are highly negatively correlated ($r = -.50$, $p < .0001$). The two academic mediators are strongly positively correlated ($.77$, $p < .0001$), and these two measures are moderately positively correlated with interpersonal skill (.27 for math

and .25 for reading, both $ps < .0001$). The *a* path estimates representing the unique effects of each of these predictors (over and above the others) on each of the three mediators appear in column 7 of Table 21.2, and all six of these remain significant, though the effects of externalizing behavior seem somewhat attenuated (when compared with the separate analyses reported in column 3 in the table), and the effects of externalizing on reading and math have, in fact, changed sign. The *b* path estimates representing unique effects of a particular mediator are provided in column 8. When all variables are in the model simultaneously, the path representing the unique effect of reading on internalizing is not significant (over and above the effects of math, interpersonal skill, approaches to learning, and externalizing behavior). The *b* paths from interpersonal skill and math to internalizing have also reversed sign in the analyses that simultaneously include the externalizing predictor. The z tests of point estimates of the mediated effects in column 9 and the bootstrap confidence intervals in column 10 indicate that the effects of approaches to learning and externalizing behavior on internalizing behavior are significantly—and uniquely—mediated by math and interpersonal skill but not reading.

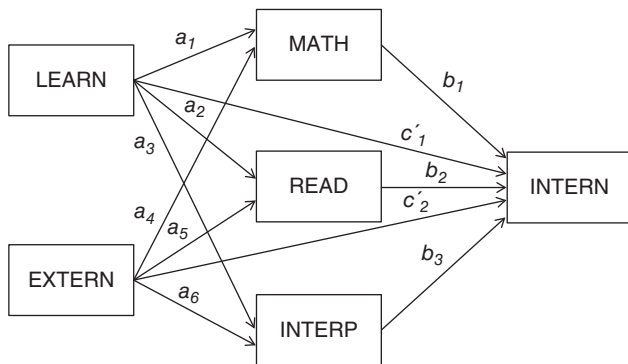


Figure 21.2 Path diagram for the two-predictor, three-mediator in the ECLS-K data.

COMBINING BASIC MEDIATION AND MODERATION

Several researchers have described approaches that combine moderation and mediation into a single framework to examine more complex associations among sets of variables (e.g., Edwards & Lambert, 2007; Fairchild & MacKinnon, 2009; Hayes, 2013; MacKinnon, 2008; Muller, Judd, & Yzerbyt, 2005; Preacher, Rucker, & Hayes, 2007). Such approaches typically consider situations involving both

moderated mediation and mediated moderation. Moderated mediation, or moderation of an indirect effect (Fairchild & MacKinnon, 2009), is moderation of the ab estimate of a mediated effect by a fourth variable, Z . Z could exert its moderating influence on the a path from X to M , on the b path from M to Y , or both (though it should be noted that circumstances exist when moderation of both paths does not necessarily imply moderation of the ab product). Mediated moderation, or the mediation of a moderator effect (Fairchild & MacKinnon, 2009), involves an exploration of the mechanism through which an interaction between two predictors X and Z (i.e., the XZ cross-product) exerts its effect on Y .

The framework described in Fairchild and MacKinnon (2009) attempts to unify the various proposed approaches by providing a single comprehensive presentation that subsumes the specific differing approaches as special cases. Figure 21.3 presents the path diagram for this combined framework. The regression equations necessary to produce the various path coefficients are straightforward generalizations of each of the three regression equations necessary for a simple mediation analysis. For example, equation 21.5 in which the outcome is regressed on the focal predictor becomes

$$Y = b_0 + c_1X + c_2Z + c_3XZ + e \quad (21.10)$$

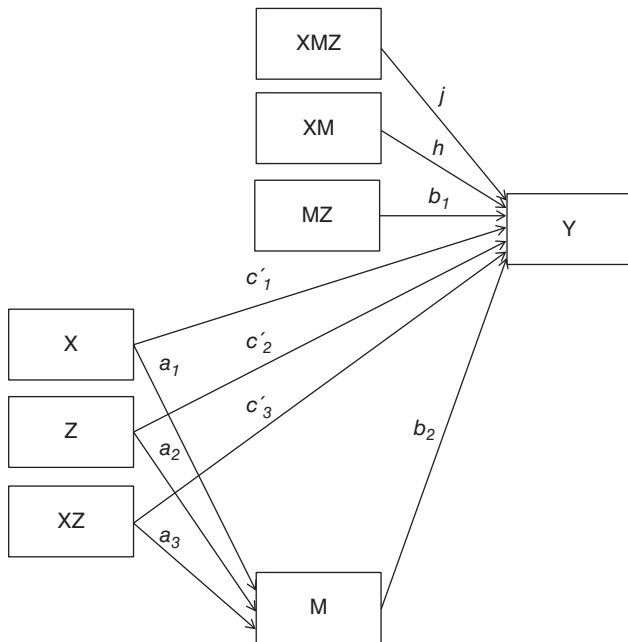


Figure 21.3 Path diagram of the Fairchild and MacKinnon (2009) approach to moderated mediation and mediated moderation.

The c coefficients in this equation thus provide total effect analogs for each of two predictors, X and Z , as well as their interaction XZ . The coefficient c_1 is the effect of X on Y when $Z = 0$, the coefficient c_2 is the effect of Z on Y when $X = 0$, and the coefficient c_3 is the interaction of X and Z on the outcome. Typically, all predictor variables in such an analysis would be centered around their means prior to estimation, which would make the c_1 and c_2 coefficients the average effects of X and Z on the outcome, respectively.

The generalization of equation 21.6 is

$$M = b_0 + a_1X + a_2Z + a_3XZ + e \quad (21.11)$$

This equation provides a path coefficient estimates for X , Z , and their interaction. The coefficient a_1 is the average effect of X on the mediator M , the coefficient a_2 is the average effect of Z on the mediator, and the coefficient a_3 is the interaction of X and Z on the mediator variable.

The generalization of equation 21.7 is

$$Y = b_0 + c'_1X + c'_2Z + c'_3XZ + b_1M + b_2MZ + hXM + jXMZ + e \quad (21.12)$$

The c' coefficients in this equation provides direct effect analogs for X , Z , and their interaction XZ . The coefficients c'_1 and c'_2 are the average direct effects of X and Z on the outcome, and the coefficient c'_3 is the direct effect of the XZ interaction on the outcome. The b_1 coefficient is the average effect of M on Y , and the remaining paths b_2 , h , and j estimate possible moderation of this effect by Z , X , and the XZ interaction, respectively.

Moderation of an indirect effect, in this framework, would involve the a_3 or b_2 coefficients. Fairchild and MacKinnon (2009) suggested that a joint test of these two coefficients may be acceptable to identify moderated mediation (i.e., mediation is deemed significant if both the a and b paths are significant). When the moderator is dichotomous, this effect can be more rigorously tested by calculating the ab point estimate of the mediated effect within each group and testing the significance of the difference between them using a pooled standard error. Details of this procedure as well as that for a continuous moderator are discussed in Fairchild and MacKinnon (2009). Mediation of a moderator effect could be estimated as the a_3b_1 product.

Table 21.3 presents the results of six analyses examining whether (1) the mediated effects of ADHD symptoms on internalizing behavior through academic performance and

TABLE 21.3 Coefficients, Standard Errors, and Tests of Moderated Mediation and Mediated Moderation

<i>X</i>	<i>M</i>	a_3	b_2	b_1	Med_{Male}	Med_{Female}	Moderated mediation test	Mediated moderation test
LEARN	INTERP	0.0018 (0.0102)	−0.0239 (0.0084)	−0.1551*** (0.0081)	−0.0948*** (0.0078)	−0.1051*** (0.0078)	0.0103 (0.0110)	−0.0003 (0.0016)
LEARN	READ	0.0324** (0.0109)	−0.0365* (0.0165)	−0.0281*** (0.0082)	−0.0033 (0.0034)	−0.0122** (0.0038)	0.0089† (0.0051)	−0.0009* (0.0004)
LEARN	MATH	−0.0042 (0.0097)	−0.0260 (0.0176)	−0.0407*** (0.0088)	−0.0086* (.0040)	−0.0171*** (0.0044)	0.0086 (0.0060)	0.0002 (0.0004)
EXTERN	INTERP	−0.0545*** (0.0128)	−0.0326* (0.0142)	−0.2399*** (0.0071)	0.1164*** (0.0058)	0.1450*** (0.0064)	−0.0286*** (0.0086)	0.0131*** (.00031)
EXTERN	READ	−0.0031 (0.0127)	−0.0394* (0.0155)	−0.1401*** (0.0078)	0.0127*** (0.0015)	0.0171*** (0.0020)	−0.0044† (0.0025)	0.0004 (0.0018)
EXTERN	MATH	−0.0009 (0.0116)	−0.0384* (0.0164)	−0.1719*** (0.0082)	0.0191*** (0.0019)	0.0246*** (0.0025)	−0.0055† (0.0030)	0.0002 (0.0020)

interpersonal skill might be moderated by gender, and (2) the moderation effect of gender on the association between ADHD symptoms and internalizing might be mediated by academic performance and/or interpersonal skill. The a_3 and b_2 coefficients that suggest moderated mediation appear in columns 3 and 4, and the statistical test appears in column 8. The joint test would suggest moderation of the mediated effect by gender for the effect of LEARN on INTERN through READ and the effect of EXTERN on INTERN through INTERP. The test of the difference between the ab estimates of mediation for boys and girls indicates that the latter effect is significant at $p < .001$, but the former is only marginally significant. The a_3 and b_1 coefficients that comprise the a_3b_1 estimate of mediated moderation appear in columns 3 and 5 of Table 21.3, and the test of mediated moderation is provided in column 9. These results indicate that the interactive effect of LEARN and GENDER on INTERN is mediated by reading performance, and the interactive effect of EXTERN and GENDER on INTERP is mediated by interpersonal skill.

LONGITUDINAL DATA

The discussion of methods to detect mediation and moderation above are specific to cross sectional data. The introduction of longitudinal data, with repeated measurements of the same variables over time, provides several benefits that may help to elucidate the associations of interest to a researcher. Many of these are discussed in MacKinnon (2008), with attention to how these are relevant to mediated effects in particular. First, longitudinal data provide information about the temporal precedence of the variables involved in the association. This is especially salient when the researcher wishes to establish causal

pathways in a mediation model. Causes must precede their effects, and mediation models necessarily involve either implicit or explicit causal claims. In cross sectional data, if the initial predictor is randomly assigned (e.g., randomization to treatment and control groups), then logic dictates that X precedes M and Y in the model. For nonrandomly assigned variables and for the M to Y relation, however, it is often difficult to establish the ordering of the variables, and thus it is common to acknowledge the possibility of reverse or reciprocal causation in such situations. With repeated measurements of the various constructs in longitudinal data, however, the researcher can explore lagged associations and changes in variables over time. If X at Time 1 predicts M at Time 2, which in turn predicts Y at Time 3, it is more likely that the earlier measures cause the later measures than vice versa (though it is still possible that some third variable causes both the earlier and later measures). Testing moderation does not necessarily involve ordering variables as in a mediation model, but here still it can be useful to explore lagged associations and change over time. If an interaction between two variables at Time 1 predicts an outcome at a later time point or change in the outcome over time, this makes a stronger argument for possible causality than examination of all variables cross sectionally.

A second benefit of the repeated measurement of constructs in longitudinal data is that it allows the researcher to disentangle processes occurring within and between individuals. In cross sectional data, the only differences that can be detected are those between different individuals at the single time point. With two or more observations, however, it is possible to calculate intraindividual change, as well as differences between individuals. It may well be that the predictors of individual change are quite different from the

predictors of differences between individuals, hence disentangling the within-person and between-person effects will provide a more fully accurate model of how real-world processes operate to produce their effects on behavior.

A third benefit of longitudinal data is that having repeated measurements of variables allows the researcher to rule out some possible alternative explanations for the effect of one variable on another. In cross sectional data, it is possible to argue that virtually any association between two variables is, in fact, due to an unmeasured and omitted third variable that affects them both. With longitudinal data, however, a researcher can covary previous measures of an outcome construct when predicting the current measure of that construct. This sort of autoregressive effect essentially removes the influence of any stable person characteristics that might influence the current outcome measure, as each individual serves as his or her own control. While one could still argue that some characteristic of the individual influences both the Time 1 and Time 2 measurement of a construct, it would be difficult to assert that this stable characteristic influences change in the construct over time. However, it is still possible that some time-varying confound provides a reasonable alternative account of a significant effect, so though use of a longitudinal design guards against some kinds of third variable explanations, it is not a universal panacea.

Another benefit of longitudinal data was noted by Gollob and Reichardt (1991), who point out that the effect that one variable has on another may not be instantaneous. It may take some time for certain types of variables to exert their effects, which cross sectional data may not be able to capture. Examining lagged associations in longitudinal data or changes over time may be more illuminating in such cases.

The concepts of stability, stationarity, and equilibrium are helpful in describing longitudinal data, longitudinal relationships, and the models used to examine them. Stability refers to the extent to which some feature of the data stays the same across time. That feature may be a variable mean, a rank ordering of variable values, a trend over time, or a periodic process. Stationarity refers to the extent to which relations between variables stay the same across time. Equilibrium refers to the extent to which the pattern of variances and covariances among variables stays the same over time. Equilibrium is related to both stability and stationarity, but a stable and stationary model is not necessarily at equilibrium (Cole & Maxwell, 2003). A cross sectional model can accurately assess relations among variables only after equilibrium has been reached (Maxwell & Cole, 2007). In longitudinal data, stability can be assessed

by measuring the dependency of an observation at a given time point on its value at the previous time point; stationarity can be evaluated by assessing invariance of relations among variables across time points; and some information about equilibrium can be obtained by examining the variance–covariance matrix among the variables across time points (MacKinnon, 2008).

The analysis of longitudinal data requires an analytic framework that can accommodate the unique features of such a data structure, most importantly the dependence that arises from the within-person stability of repeated measurements of an outcome measure assessed at multiple time points. Such dependence violates assumptions regarding independent errors that are necessary for many traditional statistical model techniques (e.g., GLM), resulting in downwardly biased standard errors, inflated test statistics, and inflation of the Type I error rate when such methods are improperly applied to longitudinal data. Two alternative analytic frameworks—multilevel modeling and structural equation modeling—are able to appropriately model the within-person similarity and thus are frequently used to analyze data from developmental studies. Below we briefly discuss these two frameworks, how cross sectional moderation and mediation are incorporated into each, and what types of models are commonly used to analyze longitudinal data with each approach.

MULTILEVEL MODELING

When data are nested or clustered, application of standard regression analysis techniques can be problematic. For example, when individual subjects are nested within groups, it is quite possible that outcome measures for individuals within the same group may be more similar than a random grouping of individuals would be. This similarity could arise from a number of different sources, including shared group experiences (e.g., students within the same elementary school classroom all share the same teacher), interaction among group members (e.g., students within the classroom talk to each other, which might result in more similar attitudes), or nonrandomly distributed background variables (e.g., students within a school may all live in the surrounding neighborhood, which may be relatively homogeneous in terms of socioeconomic status). Such data would have a positive intraclass correlation (ICC). Such within-group similarity (or between-group mean differences) would not be modeled within a single-level regression analysis, resulting in downwardly biased standard errors and overly large

test statistics, ultimately increasing the probability of making a Type I error (alpha inflation) and concluding that a significant effect exists more often than warranted (Barcikowski, 1981). Multilevel models—also known as mixed models (Laird & Ware, 1982), random coefficient models (Longford, 1993), or hierarchical linear models (Raudenbush & Bryk, 2002)—are explicitly designed to accommodate such nested data structures.

A multilevel model can be conceptualized as a series of models at the various levels of the data. For students nested within schools, the lower level (L1) model for students would be specified as

$$\text{L1: } Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + r_{ij} \quad (21.13)$$

where Y_{ij} , X_{ij} , and r_{ij} are the outcome score, predictor score, and residual, respectively, for individual i within school j . β_{0j} and β_{1j} are intercept and slope values for school j . Thus, one can think of this part of the model as a series of regressions within each of the j groups.

Each of the coefficients of the L1 model is treated as a criterion measure in higher or upper level (L2) models. In the students within schools example, these would be specified at the school level:

$$\begin{aligned} \text{L2: } \beta_{0j} &= \gamma_{00} + u_{0j} \\ \beta_{1j} &= \gamma_{10} + u_{1j} \end{aligned} \quad (21.14)$$

The β_{0j} intercept for school j is a function of the overall intercept for the sample (γ_{00}) plus a group-specific deviation u_{0j} . Similarly the β_{1j} slope of Y on X is a function of the overall slope for the sample (γ_{10}) plus a group specific deviation u_{1j} . Including error terms at both L1 (r_{ij}) and L2 (the u 's) appropriately models the within-group similarity, thus providing accurate tests of significance of predictor effects.

The multiple equations at L1 and L2 may be algebraically combined into a single equation:

$$Y_{ij} = \gamma_{00} + \gamma_{10}X_{ij} + u_{0j} + u_{1j}X_{ij} + e_{ij} \quad (21.15)$$

The single-equation form of the model is typically estimated using an iterative procedure that alternates between generalized least squares estimates of fixed effects (γ_{00} and γ_{10}) and maximum likelihood estimates of the variance components. The L2 u error terms are assumed to be distributed $\sim N(\mathbf{0}, \boldsymbol{\tau})$, and thus the variance components include τ_{00} (the variance of u_{0j}), τ_{11} (the variance of u_{1j}), τ_{01} (the covariance between u_{0j} and u_{1j}), as well

as σ^2 (the variance of r_{ij}). Estimating the combined equation rather than performing the two-step procedure implied by the multiple equation form (known as slopes and intercepts-as-outcomes; see Burstein & Miller, 1980) allows for more appropriate weighting of large versus small groups in the analysis and permits estimation even when some groups are rank deficient (i.e., when some groups contain too few individuals to run the within-group regression analysis). This particular analysis, also known as a random coefficient regression model (Raudenbush & Bryk, 2002), essentially describes the distribution of intercepts and slopes across groups, estimating an average intercept γ_{00} , the variability of intercepts τ_{00} , an average slope γ_{10} , and the variability of slopes τ_{11} . These averages are not the simple arithmetic averages that one could obtain from the two-step procedure but rather an Empirical Bayesian weighting that shrinks less reliable estimates (i.e., those from smaller groups) toward common group estimates to obtain a more representative measure of central tendency.

Another common application of multilevel modeling involves repeated measures data where the same constructs are assessed on multiple occasions for a set of individuals. In this scenario, L1 is the repeated measures (within-person) level, and L2 is the individual (between-person) level. The multiple outcome measurements provided by a given individual are more likely to be similar than a random grouping of measurements, and thus we would expect a positive ICC value. When the repeated measurements are considered simple replications and no systematic trend across time is expected, the multilevel framework discussed above can be readily adapted. Such analyses are common, for example, in daily diary studies (see Bolger & Laurenceau, 2013) where days are nested within individuals. Often no trends are necessarily expected across the relatively brief spans of such studies; the multiple days are included simply to obtain many measures of predictor and outcome variables and to allow the researcher to distinguish between within- and between-person effects.

The data for a multilevel analysis of repeated measures is typically organized as a long file with one observation for each of the assessments. The multiple-equation form of the model would be

$$\begin{aligned} \text{L1: } Y_{ti} &= \pi_{0i} + \pi_{1i}X_{ti} + e_{ti} \\ \text{L2: } \pi_{0i} &= \beta_{00} + r_{0i} \\ \pi_{1i} &= \beta_{10} + r_{1i} \end{aligned} \quad (21.16)$$

One might think of the first equation as a regression across timepoints t for each individual i , providing individual-specific intercept and slope coefficients that serve as outcomes in the individual-level L2 equations.

The single-equation form of the model is

$$Y_{ti} = \beta_{00} + \beta_{10}X_{ti} + r_{0i} + r_{1i}X_{ti} + e_{ti} \quad (21.17)$$

This model provides estimates of the overall expected value of the outcome measure when the X predictor is equal to zero (β_{00}), the variance in this value across individuals ($\tau_{00} = \text{var}(r_{0i})$) the overall association between X and Y (β_{10}), the variance in this association across individuals ($\tau_{11} = \text{var}(r_{1i})$), as well as the residual variance among time points within an individual ($\sigma^2 = \text{var}(e_{ij})$).

Moderation in Multilevel Models

The aforementioned multilevel equations could be expanded to accommodate a higher level predictor of both intercepts and slopes. This would be a group-level predictor in the scenario with individuals nested within groups and an individual-level predictor in the repeated measures scenario. Going forward with the repeated measures case, the lower level equation would remain unchanged:

$$\text{L1: } Y_{ti} = \pi_{0i} + \pi_{1i}X_{ti} + e_{ti} \quad (21.18)$$

and each of the L2 equations would be expanded to incorporate the individual-level predictor:

$$\begin{aligned} \text{L2: } \pi_{0i} &= \beta_{00} + \beta_{01}W_i + r_{0i} \\ \pi_{1i} &= \beta_{10} + \beta_{11}W_i + r_{1i} \end{aligned} \quad (21.19)$$

The single-equation form of the model becomes

$$Y_{ti} = \beta_{00} + \beta_{01}W_i + \beta_{10}X_{ti} + \beta_{11}W_iX_{ti} + r_{0i} + r_{1i}X_{ti} + e_{ti} \quad (21.20)$$

This model estimates four fixed effects. The β_{00} estimate is the expected value of the outcome when both the L1 and L2 predictors are equal to 0. The β_{01} estimate is the effect of the individual-level predictor W on the outcome when the repeated measures predictor X is equal to 0 (i.e., the simple effect of W). The β_{10} estimate is the overall effect of the repeated measures predictor X on the outcome when the individual-level predictor W is equal to 0 (i.e., the simple effect of X). The β_{11} estimate is the cross-level interaction between the repeated measures predictor X

and the individual level predictor W . Thus this estimate represents (1) the extent to which the association between X and Y changes for each unit increase in W , and/or (2) the extent to which the association between W and Y changes for each unit increase in X . Cross-level interactions are not entirely unique to multilevel models, but they arise most naturally within the multiple equation form within this framework. The t test of the β coefficient associated with the cross-product (i.e., β_{11}) is thus often an appropriate test of moderation when predictor and moderator variables are assessed at different levels of a hierarchical data structure.

In addition to cross-level interactions, multilevel models can accommodate single-level interactions at either the lower or the higher level of the data structure. The techniques involved are essentially identical to the cross-product methods used in single-level regression models. Moderation of the effect of a repeated measures variable by another repeated measures variable would require expanding the L1 equation to include the second variable and the L1 cross-product vector:

$$\text{L1: } Y_{ti} = \pi_{0i} + \pi_{1i}X_{ti} + \pi_{2i}Z_{ti} + \pi_{3i}X_{ti}Z_{ti} + e_{ti} \quad (21.21)$$

and additional L2 equations would need to be specified for the two new L1 coefficients:

$$\begin{aligned} \text{L2: } \pi_{2i} &= \beta_{20} + r_{2i} \\ \pi_{3i} &= \beta_{30} + r_{3i} \end{aligned} \quad (21.22)$$

The β_{20} coefficient represents the simple effect of the Z predictor when X equals 0, and the test of the β_{30} coefficient tests the overall moderated effect. Note that if an L2 predictor was added to the L3 equation for π_{3i} , the associated coefficient would represent the three-way interaction among L1 X , L1 Z , and the L2 predictor. The fact that this is a three-way interaction would be apparent in the single-equation form of the model.

Single-level moderation at L2 would involve expanding (at least) the first of the L2 equations (which predicts the L1 intercept) to include the second individual-level predictor and the cross-product vector:

$$\text{L2: } \pi_{0i} = \beta_{00} + \beta_{01}W_i + \beta_{02}Z_i + \beta_{03}W_iZ_i + r_{0i} \quad (21.23)$$

The β_{02} coefficient represents the simple effect of Z when W equals 0, and the test of the β_{03} coefficient tests the moderated effect. Note that if the individual-level Z predictor and the WZ cross-product are added to other L2 equations

(which predict L1 slope coefficients associated with particular L1 variables) the associated coefficients would represent three-way interactions among L2 W , L2 Z , and the particular L1 variable whose L1 parameter estimate serves as the outcome for the L2 equation. Calculation, testing, and plotting simple slopes for interactions in multilevel models is carried out in exactly the same way as in single-level regression models.

The issue of centering, which features prominently in discussions of interactions within the single-level regression framework, is no less important when considering multilevel models. Centering serves the same two purposes, providing meaningful interpretations of lower order coefficients and increasing the stability of model estimation. In fact, the latter probably provides greater benefit in multilevel models than in single-level regression analyses, as estimation is much more difficult in the multilevel case. When considering an interaction between two L2 predictor variables, centering is essentially carried out just as in single-level regression: subtracting the mean of each L2 predictor from each observation of that predictor and creating the cross-product from these centered variables reduces nonessential collinearity and allows interpretation of lower order effects of one variable at the mean of the other. One slight wrinkle does emerge: the analyst must decide whether to calculate the mean of the L2 variable to use in the centering as a simple unweighted function, counting each L2 score equally, or as a weighted function which adjusts for the number of L1 units within each L2 unit (i.e., individuals within groups or observations within individuals). Of course, there will be no difference between the two when the number of L1 units per L2 unit is identical across all groups/individuals.

When one or both of the variables involved in the interaction is at L1, however, centering becomes a slightly more complex issue. Consider first the situation with individuals nested within groups. Kreft, de Leeuw, and Aiken (1995) discussed two possible means to use in centering L1 predictors. Centering around the grand mean (CGM) simply involves subtracting the single grand mean value of the L1 predictor from each predictor observation. Centering within contexts (CWC) involves subtracting the mean of each group from the predictor observations within each group. Thus while GGM is a simple linear transformation of the raw variable score, CWC is not. Both provide meaningful (though different) zero points at which to interpret lower order coefficients, and both confer some computational advantage, though this is typically greater for CWC, which orthogonalizes effects at L1 and L2. Additional discussion of centering issues for multilevel models with individuals nested

within groups can be found in Hofmann and Gavin (1998) and Enders and Tofghi (2007). Repeated measurements of time-varying predictors can be centered with CGM or CWC analogs that subtract the mean value of the L1 predictor for the entire sample or a person-specific mean from each observed score. Hoffman and Stawski (2009) explain how such centered L1 variables can be entered into multilevel models that simultaneously include the means as L2 predictors to disentangle between-person and within-person effects in repeated measures data.

Mediation in Multilevel Models

Mediation in multilevel models is tested in essentially the same matter as for single-level regression models. Each of the three regression models necessary for producing the coefficients for a mediation analysis may be recast as a multilevel model, including an L2 intercept error term to appropriately model clustering of individuals within groups or repeated measures within individuals. Details are provided in Krull and MacKinnon (1999, 2001). The multiple equation forms of such models would differ depending on the level of measurement of X and M . Krull and MacKinnon (2001) adopted a model naming convention related to the sequential ordering of variables in a mediation model, where $X \rightarrow M \rightarrow Y$. In a $1 \rightarrow 1 \rightarrow 1$ model, the X , M , and Y variables would each be measured at the lower level of the data hierarchy. So the initial variable, the mediator, and the outcome variable would all be characteristics of individuals in multilevel data with individuals nested within groups or time-varying characteristics measured on multiple occasions over a period of time in multilevel data with repeated measures nested within individuals. In a $2 \rightarrow 1 \rightarrow 1$ model the initial predictor would be a L2 variable, while the mediator and outcome would be L1 constructs. In a $2 \rightarrow 2 \rightarrow 1$ model, both the initial predictor and mediator are measured at L2, while the outcome is at L1. Models including a link from a lower level to a higher level are not estimable within a multilevel modeling framework, but these could potentially be handled within a multilevel structural equation model (Preacher, Zyphur, & Zhang, 2010).

Fitting a $2 \rightarrow 2 \rightarrow 1$ mediation model would involve running a regression model on L2 data to obtain an estimate of the a path linking predictor to mediator and a multilevel model of the form

$$\begin{aligned} \text{L1: } Y_{ij} &= \beta_{0j} + e_{ij} \\ \text{L2: } \beta_{0j} &= \gamma_{00} + c'X_j + bM_j + u_{0j} \end{aligned} \quad (21.24)$$

to obtain an estimate of the b coefficient. These would be multiplied to produce the ab estimate of the mediated effect, and this can be tested using the Sobel standard error described earlier for single-level mediation models or more precisely by developing bootstrap confidence intervals (Pituch, Stapleton, & Kang, 2006).

For a $2 \rightarrow 1 \rightarrow 1$ model, the multilevel model equations for obtaining the a path are

$$\begin{aligned} \text{L1:} \quad & M_{ij} = \beta_{0j} + e_{ij} \\ \text{L2:} \quad & \beta_{0j} = \gamma_{00} + aX_j + u_{0j} \end{aligned} \quad (21.25)$$

And the equations for the b path are

$$\begin{aligned} \text{L1:} \quad & Y_{ij} = \beta_{0j} + \beta_{1j}M_{ij} + e_{ij} \\ \text{L2:} \quad & \beta_{0j} = \gamma_{00} + c'X_j + u_{0j} \\ & \beta_{1j} = b \end{aligned} \quad (21.26)$$

Note the similarity in form for the b path in the $2 \rightarrow 2 \rightarrow 1$ model and the a path in the $2 \rightarrow 1 \rightarrow 1$ model, both of which involve $2 \rightarrow 1$ links.

For a $1 \rightarrow 1 \rightarrow 1$ model, the multilevel equations for the a path are

$$\begin{aligned} \text{L1:} \quad & Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + e_{ij} \\ \text{L2:} \quad & \beta_{0j} = \gamma_{00} + u_{0j} \\ & \beta_{1j} = a \end{aligned} \quad (21.27)$$

And the multilevel equations for the b path are

$$\begin{aligned} \text{L1:} \quad & Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + \beta_{2j}M_{ij} + e_{ij} \\ \text{L2:} \quad & \beta_{0j} = \gamma_{00} + u_{0j} \\ & \beta_{2j} = b \end{aligned} \quad (21.28)$$

Again, note the similarity between the multilevel equations for these paths and those for the b path in the $2 \rightarrow 1 \rightarrow 1$ model, which also involves a $1 \rightarrow 1$ link. Equations for producing c path estimates are not provided here but can be found in Krull and MacKinnon (2001).

Two complications arise when working with multilevel rather than single-level equations for estimating the various paths in a mediation model. The first is that the two alternate point estimates of the mediated effect, ab and $c - c'$, are generally not identical in the multilevel case.

The reasons for this are discussed in Krull and MacKinnon (1999). The differences are usually quite small and become smaller with increasing sample size. The second complication is unique to the $1 \rightarrow 1 \rightarrow 1$ model. The previously specified equations allow for random intercepts to accommodate within group similarity, but slopes of all other L1 coefficients are specified as fixed. It may, in fact, be that the effects of the initial predictor on the mediator (i.e., the a path) or the effect of the mediator on the outcome (i.e., the b path) vary across L2 units. This would mean that different groups have different a and/or b paths in multilevel data with individuals nested within groups or that different individuals have different a and/or b paths in multilevel data with repeated measures nested within individuals. To appropriately model such data, it will be necessary to include random errors on the L2 equations predicting the L1 slope coefficients.

If it should be the case that both the a path and the b path in a $1 \rightarrow 1 \rightarrow 1$ model randomly vary across L2 units, it is possible that these a and b coefficients might covary. In this case, the point estimate of the mediated effects is not equal to ab but rather

$$Med = ab + \sigma_{ab} \quad (21.29)$$

which involves the covariance between the a and b estimates (Kenny, Korchmaros, & Bolger, 2003). Moreover, the approximate standard error is not equal to equation 21.8 but rather

$$s_{Med} = \sqrt{b^2\sigma_a^2 + a^2\sigma_b^2 + \sigma_a^2\sigma_b^2 + 2ab\sigma_{ab} + \sigma_{ab}^2} \quad (21.30)$$

which involves the covariance between the a and b estimates as well as the sampling variance of the covariance estimate (Kenny et al., 2003).

Finding an estimate of the covariance between the a and b estimates is challenging because these are typically produced in two completely separate analyses. Bauer, Preacher, and Gil (2006) presented a method that fits both the a and b paths in a single analysis, which allows the covariance and its corresponding sampling variance to be estimated. Their approach involves stacking the data so that each L1 unit has two observations with identical predictors and creating a generic outcome variable. The outcome variable is set equal to the M variable on the first observation for each L1 unit, and the outcome variable is set equal to the Y variable on the second observation. Then two selection variables (S_m and S_y) are created to distinguish between the first ($S_m = 1, S_y = 0$) and second

($S_m = 0$, $S_y = 1$) observations for each case. A multilevel model with a L1 equation of the form:

$$\begin{aligned} \text{L1: } Z_{ti} &= S_{mti}(1 + a_i X_{ti}) + S_{yti}(1 + b_i M_{ti} + c'_i X_{ti}) + e_{Zti} \\ &= S_{mti} + a_i S_{mti} X_{ti} + S_{yti} + b_i S_{yti} M_{ti} \\ &\quad + c'_i S_{yti} X_{ti} + e_{Zti} \end{aligned} \quad (21.31)$$

is then fit to the stacked data, and each of the five L1 coefficients is allowed to vary randomly at L2. This model essentially fits multilevel models for the a path and the b and c' paths simultaneously. To model the error variance appropriately, it will be necessary to allow the estimation of two separate values, corresponding to the observations with different values on one of the two indicator variables. Further details on how to implement the Bauer et al. (2006) method in a number of statistical software programs can be obtained from the website of the article's first author. More information on implementation of the model in the Mplus programming language is also available (MacKinnon, 2008; Preacher et al., 2010). The multilevel mediation model is summarized in MacKinnon and Valente (2014) and a Bayesian formulation for the multilevel mediation model is in Yuan and MacKinnon (2009).

MULTILEVEL MODELING APPROACHES FOR LONGITUDINAL DATA

The dominant multilevel model for the analysis of longitudinal data is the multilevel growth model (Hox, 2010; Raubenbush & Bryk, 2002; Singer & Willett, 2003). Growth in this sense refers to any sort of change over time, including increasing values, decreasing values, or trends that change direction. For longitudinal data that involve an intervention delivered at a particular point in time, an exponential decay model might provide a useful alternative that is especially well-suited for examining the change that follows the delivery of the intervention. Both approaches are discussed in detail below.

Multilevel Growth Modeling

A multilevel growth model is very similar to the two-level model for repeated measurements nested within individuals shown in equations 21.16 and 21.17. The primary difference is that the growth model includes at least one variable representing time at L1. If a single time variable is linearly

coded and no covariates are included in the model, the multiple equation specification for the model would be

$$\begin{aligned} \text{L1: } Y_{ti} &= \pi_{0i} + \pi_{1i} \text{time}_{ti} + e_{ti} \\ \text{L2: } \pi_{0i} &= \beta_{00} + r_{0i} \\ \pi_{1i} &= \beta_{10} + r_{1i} \end{aligned} \quad (21.32)$$

where time_{ti} is the time relevant variable. Each individual-specific growth parameter π is a function of the overall estimate of the growth parameter β in the sample and individual specific deviation r from that value. The single-equation form of the model is

$$Y_{ti} = \beta_{00} + \beta_{10} \text{time}_{ti} + r_{0i} + r_{1i} \text{time}_{ti} + e_{ti} \quad (21.33)$$

β_{00} estimates is the overall intercept, which is the expected value of the outcome variable when time is at its zero point. β_{10} estimates the overall linear slope estimate. The τ matrix contains the variances and covariances of the L2 errors r , and thus τ_{00} is the variance of individual-specific intercepts, τ_{11} is the variance of individual-specific slopes, and τ_{01} is the covariance between them. This particular parameter estimate is more often interpreted in growth models than in other multilevel models that do not involve time, because it can indicate whether individuals are becoming more similar (compensatory growth) or different (fan-spread growth) in their outcome variable scores.

More complex forms of growth can be modeled by adding additional variables coding time information to the L1 model. For example, higher order polynomial trends can be fitted by adding powered functions of the linear time variable. Piecewise models can also be fitted by adding additional coded vectors to produce models that estimate either (1) the slopes of the different time segments, or (2) one slope and the difference between slopes (Raudenbush & Bryk, 2002). Both individual and time-varying covariates can be added to the model. Individual covariates are added to the L2 specification as predictors of intercepts and/or slopes. For example, a gender variable could be added to the L2 equations in 21.32 to test whether intercepts and slopes of the outcome variable differed for males and females. A time-varying covariate could be added to the L1 specification either to adjust for its effect when estimating growth or to examine its effect over and above growth. A decision would need to be made about whether the effect of this variable would vary across individuals (in which case the L2 equation associated with this variable

would include an r error term) or would be assumed fixed (no L2 error term).

Most information relevant to multilevel models in general applies equally to multilevel growth models, with just a couple of notable exceptions. First, the time variable in a growth model is typically “centered” on a meaningful value—often the initial time point of the study—rather than on the mean *per se*. Centering time at its mean would simply put the zero point in the middle of the study period, which may not be the most substantively meaningful point at which to evaluate effects. Second, typical independence and homoscedasticity assumptions about L1 errors may not be realistic in longitudinal data. L1 error variances may increase or decrease over time, and values measured at later time points may depend on previous values measured at earlier time points. Most multilevel modeling software includes options for specifying heteroscedastic and autoregressive error structures that might be better suited for growth models.

Multilevel Exponential Decay Model

In considering models for longitudinal data, a primary issue is the shape of the trajectory across time. As discussed by Fritz (2014), often when researchers encounter data that exhibit any nonlinearity, a quadratic growth model is chosen without regard for the implications to the underlying variables or theory. For longitudinal psychological data, especially following an intervention or treatment, Fritz suggests that an exponential decay model has several advantages over a quadratic model. First, the exponential decay function is more realistic for many variables in psychology because it approaches an asymptote. One can imagine many scenarios where an individual is given a treatment, causing a change in the individual’s underlying construct, which slowly returns to the individual’s pretreatment level over time. At the same time, the individuals in the control group remain unchanged, a scenario illustrated in Figure 21.4. If the treatment had a lasting effect on the individual, the individual’s score would still likely decrease across time, but not back to his or her pretreatment level (i.e., the construct would asymptote at a different value than the pretreatment level). While a quadratic growth model can approximate the exponential decay present in the figure, a quadratic function at some point must reach a minimum and then begin to grow towards infinity, an unlikely scenario given most variables in psychology. For this reason, many researchers have already used an exponential decay function (e.g., Blozis, 2004, 2007; Blozis, Conger, & Harring, 2007; Browne, 1993; Cudeck, 1996).

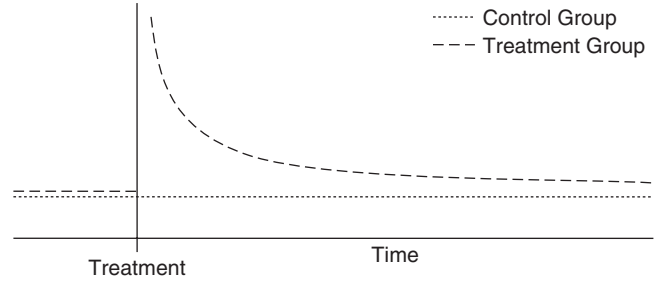


Figure 21.4 Illustration of a treatment group decaying exponentially across time while the control group remains constant.

Another reason for choosing an exponential decay model is that exponential decay is a parsimonious function requiring the estimation of three parameters:

$$Y_i = \beta_2 + (\beta_0 - \beta_2) \exp[\beta_1 t] + r_i \quad (21.34)$$

where t is time, β_0 is the initial state value at $t = 0$, β_2 is the asymptote value as time goes to infinity, and β_1 is the rate of change value, which is constrained to be less than zero to ensure that the asymptote exists (Pinheiro & Bates, 2000). While a quadratic model also contains three parameters, the interpretations of the coefficients from the exponential decay model are likely to be more useful to a researcher than those from a quadratic model.

Analyzing longitudinal data using an exponential decay model requires the use of a nonlinear multilevel model. Using notation from Raudenbush and Bryk (2002; see also Davidian & Giltinan, 1995), the exponential decay model is

$$\begin{aligned} \text{L1: } Y_{ti} &= \beta_{2i} + (\beta_{0i} - \beta_{2i}) \exp[\beta_{1i} t] + r_{ti} \\ \text{L2: } \beta_{0i} &= \gamma_{00} + u_{0i} \\ \beta_{1i} &= \gamma_{10} + u_{1i} \\ \beta_{2i} &= \gamma_{20} + u_{2i} \end{aligned} \quad (21.35)$$

where $time_i$, β_{0i} , β_{1i} , and β_{2i} have the same interpretation as in equation 21.34, r_{ti} are the Level 1 residuals, and the u_{ji} 's are random effects distributed $\sim N(0, \tau)$. As in linear multilevel models, time-varying predictors can be added at L1, while time-invariant predictors are added at L2. The inclusion of L2 predictors may be especially interesting for exponential decay because they test whether variance in the initial value, rate of change, and asymptote values can be predicted by another variable. Two of the most widely used software packages for estimating nonlinear multilevel

models in the social sciences are NLME (R Development Core Team, 2013) and SAS PROC NL MIXED (SAS Institute, 2014), although many other statistical packages for linear multilevel models also allow the estimation of nonlinear multilevel models.

STRUCTURAL EQUATION MODELING

Structural equation modeling is a multivariate analytic technique for exploring patterns of association among variables. A typical structural equation model (SEM) has two components: (1) a measurement model that specifies how observed indicator variables are related to unobserved latent constructs; and (2) a structural model that specifies how the various observed and latent variables are related to one another. Structural equation modeling employs a matrix representation to specify a number of simultaneous equations that denote the associations among observed and latent variables, and the model can be fitted using estimation methods that maximize the fit between the observed covariance matrix of the variables and the covariance matrix implied by the model. In depth treatment of this topic can be found in a number of texts, including Bollen (1989) and Kline (2011).

Diagrams are often used to depict the variables and relations specified in an SEM. By convention, observed variables are typically represented as squares or rectangles, and latent variables are typically represented as circles or ovals. Single-headed arrows represent directional causal relations between variables and double-headed arrows represent nondirectional correlational associations. The estimates produced in an SEM typically include factor loadings relating observed indicators to latent factors, path coefficients relating variables and factors to one another, variances and covariances of errors/disturbance terms. It is also possible to estimate means of latent variables, although these parameters are not typically part of standard SEM estimation. Mean structure can be indicated in SEM diagrams by arrows from a triangle representing a constant value of 1, though often such effects are not noted in the diagrams. In addition to parameter estimates, SEM programs provide a number of absolute and relative fit indices that allow researchers to examine how well the proposed model fits the data. These include a chi-square test of overall model fit, the root mean square error of approximation (RMSEA; Steiger, 1990), and the comparative fit index (CFI; Bentler, 1990). Particular aspects of the model can be tested by examining the size of the estimated coefficients relative to their standard errors or

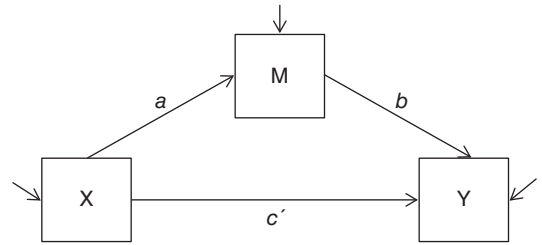


Figure 21.5 SEM mediation model for manifest X , M , and Y variables.

by comparing the fit of a series of models in which specific parameters, originally freely estimated, are systematically constrained.

Mediation in SEM

Specification and estimation of mediation in SEM is straightforward. Figure 21.5 depicts a simple SEM for a single mediator model. All three variables in this model are observed, and thus this particular SEM is a simple path analysis and does not include a measurement model. Fitting the model produces estimates of the a and b path coefficients and their standard errors. These can be used to test mediation in much the same way as with regression analysis. Some SEM programs offer specific options for testing mediated (or indirect) effects, using either delta standard error method tests or bootstrapping methods.

If variables are measured with multiple indicators, SEM can be used to examine mediation among the underlying latent constructs. A measurement model can be specified to define the initial, mediator, and outcome factors from their various observed indicator variables (as shown in Figure 21.6), and the structural model specifies the appropriate paths between the latent constructs to test the mediated effect. The latent constructs are theoretically error-free, and the path coefficients will not suffer from the attenuation that occurs in estimated associations between observed measurements. Thus the mediated effects among latent constructs should be stronger and more easily detected than mediation among observed variables.

Moderation in SEM

There are a number of different approaches for examining moderation within an SEM. If the moderating variable is observed and categorical, the typical method involves fitting a multiple group model (Jöreskog, 1971). Such a model essentially fits a separate SEM within each level of the categorical moderator. For example, to examine

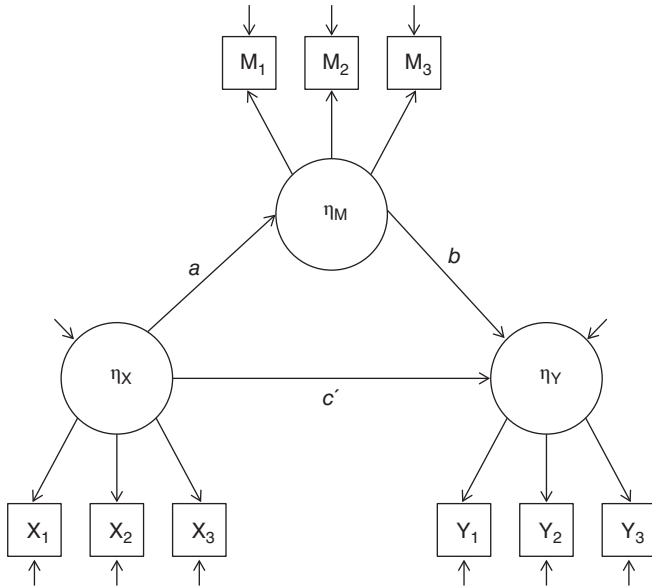


Figure 21.6 Mediation model with latent X , M , and Y variables.

whether the association between impulsive/hyperactive symptoms and interpersonal skill is moderated by gender, an SEM that allows different coefficients for the $\text{EXTERN} \rightarrow \text{INTERP}$ path would be fitted. The chi-square value for this model would be compared with that from a nested model that constrained the coefficients in the two groups to be equal. A significant chi-square difference would indicate that model fit improves when the two coefficients differ, supporting the hypothesis that the association is moderated by gender.

If the two variables involved in the moderated effect are observed, but neither is categorical, a multiplicative cross-product variable can be created and entered as an additional observed variable in the SEM. This cross-product variable then (1) is allowed to covary with the two component variables from which it was produced, and (2) predicts the outcome variable as shown in Figure 21.7. A significant path coefficient from the cross-product variable to the outcome measure would indicate moderation.

A moderated effect involving latent variables is somewhat more difficult to model. The most common approach to representing a latent variable interaction involves specifying an additional latent variable—essentially the cross-product of the two latent variables of interest—with cross-products of the original variable indicators serving as indicators of the interaction construct. This latent cross-product variable represents the interactive effect on the outcome measure when fitted with the two original latent variables simultaneously in the model. Specification

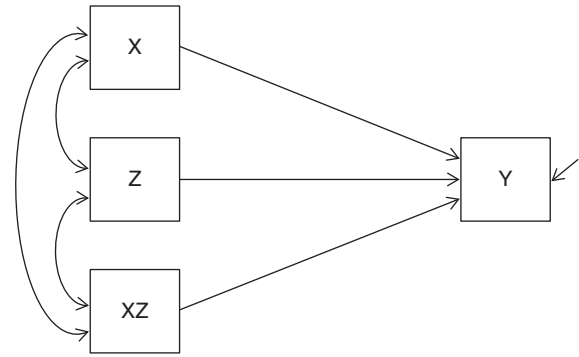


Figure 21.7 An SEM with an interaction between manifest variables.

and fitting of this model can be difficult, as it may involve complex nonlinear constraints, include many highly correlated variables, and violate normality assumptions. A number of different methods have been proposed to address these issues, but no clear consensus as to the best method for analyzing latent variable interactions has emerged. The various issues and proposed solutions are summarized in Kline (2011, Ch. 12).

STRUCTURAL EQUATION MODELS FOR LONGITUDINAL DATA

When longitudinal data are to be used in an SEM, the data set is typically organized in wide format (as opposed to the long format necessary for multilevel modeling), with one line of data per individual subject and multiple uniquely named variables representing the repeated measures of each construct. A number of different types of SEMs can be used to model longitudinal data, including autoregressive panel models, latent growth models, and latent change models. It is also possible to fit an exponential decay model within the SEM framework.

Autoregressive Model

In an autoregressive panel model (Duncan, 1969; Humphreys, 1960; Jöreskog, 1979; Wiley & Wiley, 1970), a measure of a given construct at a particular time point is presumed to be a function of value of that construct at the previous time point. The top series in Figure 21.8 illustrates an autoregressive model for an X variable, and the lower series illustrates a similar model for a Y variable. The arrows from time t to $t + 1$ within each series (i.e., X_1 to X_2 and X_2 to X_3 ; Y_1 to Y_2 and Y_2 to Y_3) are the autoregressive coefficients representing stability

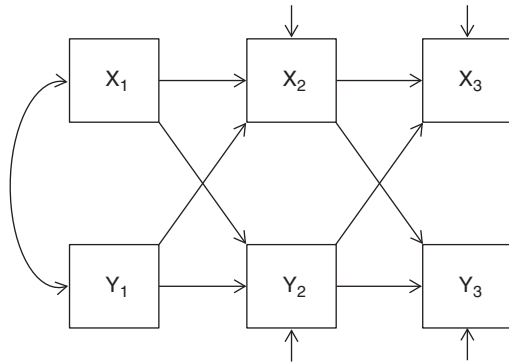


Figure 21.8 Autoregressive model with cross-lagged paths.

over time. Such models can be estimated with a series of OLS regression models or in a single simultaneous SEM. Autoregressive models for particular variables can be combined to form more complex models. For example, Figure 21.8 includes two types of cross-lagged paths: (1) paths from earlier measures of X to later measures of Y , and (2) paths from earlier measures of Y to later measures of X . Testing such a cross-lagged model—usually by fitting a series of SEMs that constrain particular sets of paths to be equal to zero or equal to each other (see Martens & Haase, 2006)—can allow a researcher to determine whether one set of effects leads the other or if there is reciprocal causality between the two constructs.

Latent Growth Curve Model

Another common SEM for longitudinal data is the latent growth curve (LGC) model (Duncan, Duncan, & Strycker, 2006). In such a model, multiple measurements of a given construct over time serve as indicator variables for latent growth factors. While typical SEM measurement models estimate the factor loadings that relate the latent variable to the observed indicators, the loadings in a latent growth model are fixed to define the shape of the growth trajectories. For example, Figure 21.9 presents a linear growth trajectory modeled based on three measurement time points. The factor loadings on the intercept factor are set to 1. The factor loadings on the linear slope factor are set to represent the amount of time elapsed since the first measurement occasion while setting the loading for the time point represented by the intercept factor to 0. Therefore, in the ECLS-K data, if we model linear growth from the kindergarten fall to fifth grade spring with kindergarten fall as the intercept, the first loading would be set equal to zero (essentially erasing the arrow) for the kindergarten fall assessment and the second loading would be set equal to .5, indicating that, on average, half of a year has elapsed

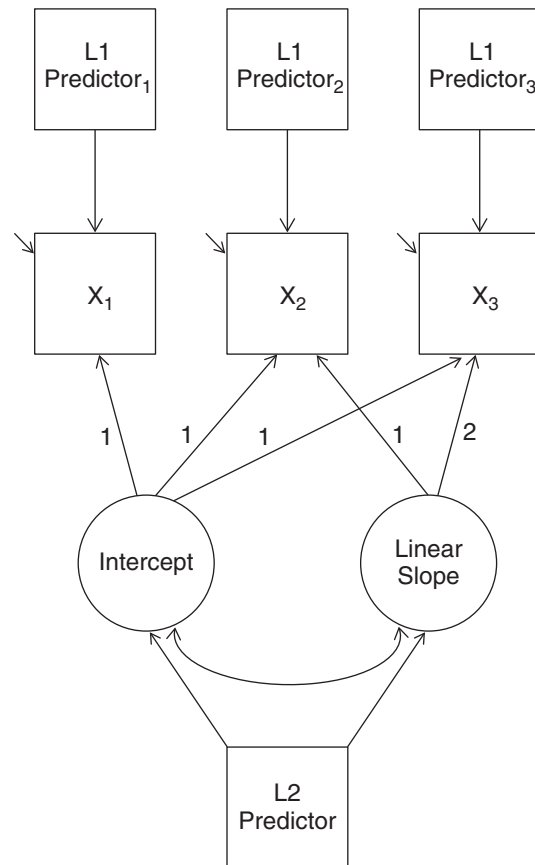


Figure 21.9 SEM latent growth curve model with L1 and L2 predictions.

at kindergarten spring. The other loadings would be set equal to 1.5, 3.5, and 5.5 to represent elapsed time at the first, third, and fifth grade spring assessments. If we model a quadratic growth trajectory, another growth factor, a quadratic growth factor, is added to the linear growth model in Figure 21.9, with the factor loadings equal to the squared linear values: 0, .25, 2.25, 12.25, and 30.25.

Estimating an SEM for a latent growth model produces means and variances for each of the latent growth factors. The means describe the normative growth trajectory. In a linear growth model, means of the intercept and linear growth factors would represent the expected value of the construct at the initial time point and the linear growth rate, respectively. The estimated variances of the growth factors represent differences between individuals in intercepts and linear rates of change. The covariance between intercepts and linear slopes is also estimated, along with an estimate of the error variance associated with the measurement of the outcome variable at each time point.

Both time-varying and individual (invariant over time) covariates can be added to a latent growth model.

Individual covariates are added as predictors of the latent growth factors, as shown in the lower portion of Figure 21.9. A time-varying covariate measured at a given time point can be added as a predictor of the outcome construct at that time point, as shown in the upper portion of Figure 21.9. Alternately, a researcher could let a time-varying covariate measured at an earlier time point predict the outcome construct measured at later time point, introducing an aspect of temporal precedence and a prospective lagged effect.

SEM and multilevel models can produce equivalent estimates of growth model parameters (Curran, 2003). However, the two modeling frameworks differ somewhat in their defaults for model specification and estimation, and thus results from the two approaches may initially appear to differ. Specifically, multilevel models estimate a single L1 variance component (assuming homogenous error variances across time points), while SEMs usually estimate time point-specific error variances. Also, multilevel programs will usually constrain negative variance estimates to zero, while SEM programs will allow the negative estimate. However, if program defaults are appropriately changed to reflect common settings, results produced with multilevel models and SEMs will be essentially identical.

Latent Change Model

Another type of latent variable model for investigating longitudinal data is the latent change score (LCS) model, also referred to as the latent difference score model (Ferrer & McArdle, 2003; McArdle, 2001; McArdle & Hamagami, 2001; McArdle & Nesselroade, 2003). An example of a four-wave LCS model is shown in Figure 21.10, where the η variables represent the error-free true score for each measurement wave, the Δ variables represent the latent change scores, and the ξ_1 variable is a second-order slope variable that represents the change in the latent change scores over time or, in other words, the change in the rate of change in score over time. Because each η variable is represented by only one observed score, the paths between all of the observed Y variables and the η variables are constrained to be equal to 1. A measurement model can be added to the LCS model to incorporate multiple indicators; Figure 21.11 shows an example of a four-wave multiple-indicator LCS model.

McArdle and Hamagami (2001) described the Δ variables as being a rate of change for a given time lag such that for a time lag of 1,

$$\Delta_t = \eta_t - \eta_{t-1} \quad (21.36)$$

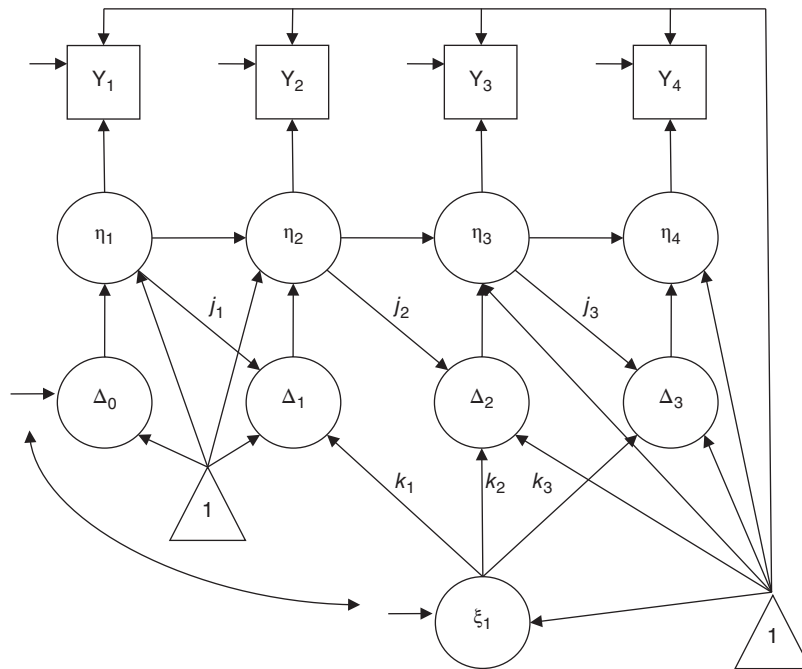


Figure 21.10 A four-wave latent change score model.

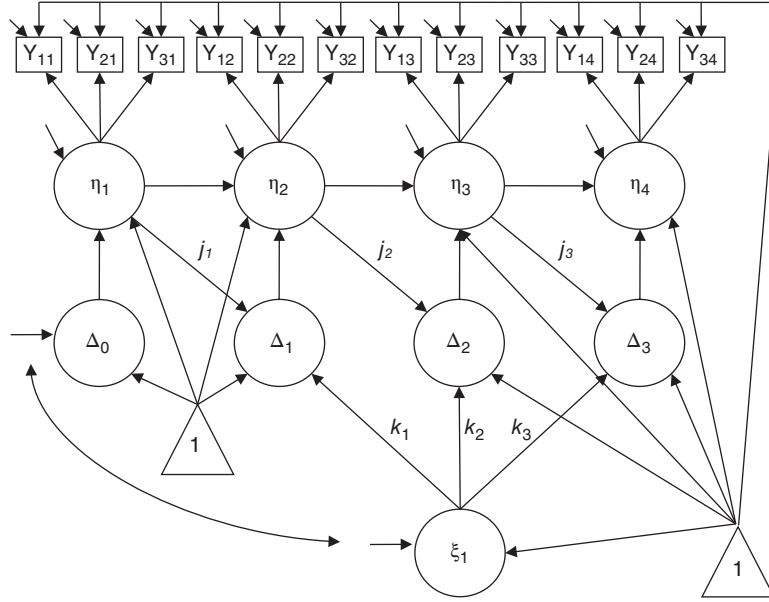


Figure 21.11 Multiple-indicator latent change score model.

except for Δ_0 . An individual latent change score is then equal to

$$\Delta_t = k_t \xi_1 + j_t \eta_{t-1} \quad (21.37)$$

where j_t is the loading between the latent change score at time t to the latent construct score at time $t - 1$, and k_t is the loading of the latent change score on the ξ_1 variable. Assuming equally spaced measurement, the values of k_t are usually constrained to be equal across waves, as are the values of j_t , but unequal spacing of measurements can be taken into account by adjusting these constraints to reflect the actual spacing. In addition, Ferrer and McArdle (2003) described a cross-lagged version of the LCS model where the change score at time t predicts the observed score at time $t + 1$.

Equation 21.37 shows that the latent change score Δ_t is proportional to the score at the previous time point, $j_t \eta_{t-1}$, and the latent slope, $k_t \xi_1$. That is, an individual's change score at time t is a function of the overall change across time and the individual's score at the previous time point, which is why McArdle and Hamagami (2001) called this model a dual change score model. Constraining either the j paths, k paths or both to be equal to 0 changes the interpretation of the model. For instance, if the k paths are set to 0, the model becomes a proportional change score model because an individual's change score at time t only depends on their score at the previous time point, which will not be the same for each individual or time point. Alternatively

if the j paths are set to 0, the model becomes a constant change score model because an individual's score at time t only depends on the latent slope, which is the same for all individuals across all time points. If the k and j paths are both set to 0, then the latent change score is not proportional to the previous score or related to the latent slope. When this occurs, there is no change in the construct scores across time, so the model becomes a no-change score model.

Much like a linear LGC model, the mean of Δ_0 can be considered as a latent intercept because it is the average score across individuals at baseline and the mean of ξ_1 can be considered a latent slope because it is the average change in change scores across time. Overall model fit can be determined by typical SEM fit indices such as the chi-square goodness of fit test, the RMSEA, and the CFI. Stationarity in the LCS model can be investigated by comparing models that constrain the autoregressive paths, j paths, and k paths of the model to be equal across waves with models where the constraint is lifted, using a series of nested model chi-square change tests.

Exponential Decay Structural Equation Model

A latent variable version of the exponential decay nonlinear mixed effects model can be specified using the general method for nonlinear latent variable growth curve models described by Browne (1993; see also Blozis, 2004, 2007;

Blozis et al., 2007). The nonlinear LGC model for exponential decay is similar to the LGC model and is of the form

$$\mathbf{X} = \boldsymbol{\tau}_x + \boldsymbol{\Lambda}_x \boldsymbol{\xi} + \boldsymbol{\delta} \quad (21.38)$$

where the vector $\boldsymbol{\xi}$ contains the latent growth parameters, the paths in the loading matrix $\boldsymbol{\Lambda}_x$ are constrained to specific values, and the $\boldsymbol{\tau}_x$ vector contains the means of the observed variables. Also present is the vector $\boldsymbol{\kappa}$, which contains the means of the latent variables, in this case the growth parameters. The difference between a linear LGC model and one for exponential decay is in the interpretation of the means of the latent variables and in the values of the constrained path loadings in the $\boldsymbol{\Lambda}_x$ matrix.

To construct an exponential decay LGC model using Browne's (1993) method, illustrated in Figure 21.12, the loadings in the $\boldsymbol{\Lambda}_x$ matrix must be set equal to the first partial derivatives of the exponential decay function

$$\lambda_{0t} = \frac{d}{d\beta_0} f(t, \boldsymbol{\theta}) = \text{Exp}[t\beta_0] \quad (21.39)$$

$$\lambda_{1t} = \frac{d}{d\beta_1} f(t, \boldsymbol{\theta}) = (\beta_0 - \beta_2)t \text{Exp}[t\beta_1] \quad (21.40)$$

$$\lambda_{2t} = \frac{d}{d\beta_2} f(t, \boldsymbol{\theta}) = 1 - \text{Exp}[t\beta_2] \quad (21.41)$$

where

$$f(t, \boldsymbol{\theta}) = \beta_2 - (\beta_0 - \beta_2) \text{Exp}[t\beta_1]$$

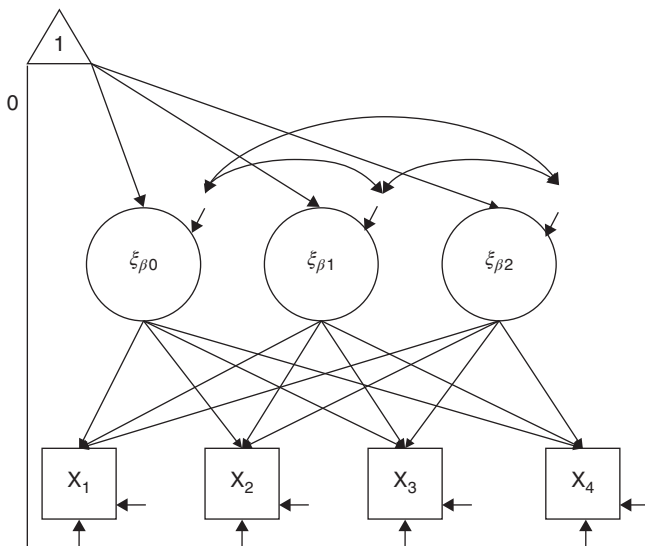


Figure 21.12 A four-wave exponential decay latent growth curve model.

This defines the vector of latent variables to be $\boldsymbol{\xi} = [\xi_{\beta_0} \ \xi_{\beta_1} \ \xi_{\beta_2}]'$ and the vector of means for the latent variables to be $\boldsymbol{\kappa} = [\beta_0 \ \beta_1 \ \beta_2]'$ where β_0 , β_1 , and β_2 have the same interpretations as in equation 21.34.

The latent exponential decay model created using the Browne (1993) method has two problems that arise during estimation. The first is that the mean of one of the latent variables must be constrained to be equal to zero. The second problem is that, as shown in equations 21.39 to 21.41, to constrain the loadings in the $\boldsymbol{\Lambda}_x$ matrix correctly, the values of the means of the latent variables must already be known prior to estimating the model. Both of these problems may be circumvented, however, by setting β_1 equal to 0 and then simultaneously estimating the path loadings and the means of the latent variables using nonlinear constraints, for example using the MODEL CONSTRAINT command in Mplus.

MODELS FOR MODERATION AND MEDIATION IN LONGITUDINAL DATA

The next section of the chapter describes several models that incorporate mediation and moderation into the analysis of longitudinal data. Most employ the multi-level and SEM frameworks for model specification and estimation. Some of these approaches (e.g., the lag as moderator model, time-varying covariate interactions within multilevel models, and the mediation over time model) make use of the repeated measurements to explore how certain effects and associations change over time. Other approaches, especially those examining mediation using the SEM framework, focus more on the inherently time-ordered aspect of longitudinal data to establish the temporal precedence of effects necessary to make a causal argument about how predictor variables exert their effects on the outcomes of interest.

Lag as Moderator Model

Selig, Preacher, and Little (2012) developed a moderator model unique to longitudinal data. Specifically, they propose the lag as moderator (LAM) approach to examine time-dependent associations. It is possible that the strength of the association between two variables assessed at different occasions will depend on the length of time elapsed between the occasions. Their presentation assumes just a single measurement of two variables for each subject, but variability of the time lag between the two measurements across subjects.

The general approach involves using the length of the lag between measurement occasions as a predictor in a regression model. If the focal variables are X (predictor) and Y (criterion), the model would include X , the lag, and the interaction between X and the lag as predictors:

$$\hat{Y} = b_0 + b_1X + b_2Lag + b_3 \quad (21.42)$$

In this equation, the b_1 coefficient represents the association between X and Y when $Lag = 0$ (i.e., simultaneous estimation), the b_2 coefficient represents the expected change in Y for a 1 unit increase in the length of the lag, and b_3 represents the linear change in the association between X and Y that accompanies a 1 unit increase in the length of the lag. Strategic centering of the lag variable can change the meaning of the b_1 coefficient to test the association between X and Y if a particular lag is believed to be particularly important.

Selig et al. pointed out that in many cases, change in the X – Y association as a function of the lag will not be linear. This nonlinearity might be captured with a straightforward polynomial specification (here, a quadratic form):

$$\hat{Y} = b_0 + b_1X + b_2Lag + b_3Lag^2 + b_4XLag + b_5XLag^2 \quad (21.43)$$

or with a more complex nonlinear parameterization:

$$\hat{Y} = b_0 + \left[\alpha_{Y \cdot X} - (\alpha_{Y \cdot X} - \alpha_0) \left(\frac{Lag}{\alpha_{Lag}} - 1 \right)^2 \right] X \quad (21.44)$$

Though the latter expression requires nonlinear regression techniques for the estimation of its parameters, it provides estimates with more meaningful interpretations than those of the quadratic model (Cudeck & du Toit, 2002). α_0 is the expected association between X and Y when $Lag = 0$, α_{Lag} is the value of the lag at which the maximal X – Y association is obtained, and $\alpha_{X \cdot Y}$ is that maximum association. Alternately, an exponential model might be specified:

$$\hat{Y} = b_0 + (ae^{bLag})X \quad (21.45)$$

to represent a functional relationship that approaches an asymptote rather than symmetric form (i.e., any increase to a maximum must necessarily be followed by a decrease) implied by a quadratic specification.

Selig et al. provide an empirical example illustrating the LAM approach using variables from the Early Head Start Research and Evaluation study (Department of Health

and Human Services, Administration for Children and Families, 1996–2001; available from the Inter-university Consortium for Political and Social Research website: www.icpsr.org/). As is often the case, the original intention was to collect the observational data at a single fixed lag; however, practical difficulties resulted in variation in time elapsed between measurement occasions across subjects. The LAM analysis examined effect of lag length on the association between the quality of stimulation provided to a child in his or her home at approximately 14 months and a measure of cognitive, language, and social development at approximately 24 months of age. Though the nominal lag between the 14-month and the 24-month observations was 10 months (and, in fact, the average lag was 10.08 months), there was considerable variability in the actual length of the lag, which ranged from 2.95 to 16.98 months ($SD = 1.74$ months).

Initial examination of the data consisted of a series of scatterplots windowing different lag lengths and estimation of a regression slope within each window. A plot of these slope values against the average lag for each window showed a clear decrease in the strength of association with longer lags. To capture this change, Selig et al. fit a linear and an exponential LAM model, and both showed significant lag moderation. The two models seemed to fit the data equally well for the range of lags actually present in the data, but the authors pointed out that for even longer lags the linear model predicts a negative association between the two variables while the exponential model predictions approach a lower asymptote of zero. They cautioned, however, that their example data did not involve random assignment of the lag length, and thus interpretations must be circumspect.

We used the LAM model to examine the effect of the lag between the measurement of ADHD symptoms at kindergarten fall and measurement of interpersonal skills, academic performance, and internalizing behavior at kindergarten spring. The lag between the two assessment occasions ranged from 3.8 to 9.3 months, with an average of 6.1 months. Linear and quadratic effects of lag length were tested. Only two interactions between predictor values and lag length emerged, both linear. The negative association between kindergarten fall EXTERN and kindergarten spring INTERP was stronger for shorter than for longer lags ($b_3 = 0.3340$, $SE = 0.1177$, $p < .01$), and the positive association between kindergarten fall EXTERN and kindergarten spring INTERN was stronger for longer than for shorter lags ($b_3 = 0.2268$, $SE = 0.1063$, $p < .05$).

Selig et al.'s LAM approach was designed for a very specific data structure, with only two observations of

measurement per subject, each variable assessed at only one of the measurement occasions, and variation in the lag between measurement occasions across subjects either produced intentionally (e.g., through random assignment of lag lengths) or as the result of vagaries in the implementation of a fixed-lag design. The single-level regression estimation that they recommend, whether linear or nonlinear, requires a single outcome observation for each subject. It should be possible, however, to adapt their framework for estimation as a multilevel model, which will widen the range of data structures to which the LAM approach might be applied. Moreover, such applications would convert what Selig et al. acknowledge as an interindividual approach to what is often an intraindividual question into an appropriately matched intraindividual analysis.

When the predictor variable is time-invariant but the criterion is assessed at multiple later time points, each subject shows intraindividual variation in the predictor-criterion lag. The single equation form of a multilevel model would closely parallel those specified by Selig et al., simply adding L2 error terms to model the potential unmeasured similarity among the repeated criterion observations obtained from a given individual. At L1 (the repeated measures level), the equation would be

$$Y_{ti} = \beta_{0i} + \beta_{1i}\text{Lag}_{ti} + e_{ti} \quad (21.46)$$

and at L2 (the individual level), we would have

$$\begin{aligned} \beta_{0i} &= \gamma_{00} + \gamma_{01}X_i + u_{0i} \\ \beta_{1i} &= \gamma_{10} + \gamma_{11}X_i + u_{1i} \end{aligned} \quad (21.47)$$

The γ_{11} coefficient captures the lag moderation effect in this model.

Results of fitting such models in the ECLS-K data are provided in Table 21.4. These models predicted kindergarten spring through fifth grade spring assessments of interpersonal skills, academic performance, and internalizing behavior from kindergarten fall measures of ADHD symptoms. Initial models included interactions between ADHD symptoms and both linear and quadratic lag variables. If the interaction between the ADHD symptoms variable and the quadratic lag was not significant, the model was refitted with only the interaction involving the linear lag. Most of the final models included significant interactions with the quadratic lag function; however, the model predicting READ from EXTERN included only an interaction with linear lag. All of the models indicated that

TABLE 21.4 Coefficients and Standard Errors for Multilevel LAM Models

Predictor	Outcome	Simple effect at Lag = 0	Predictor x Lag	Predictor x Lag ²
LEARN	INTERP	0.6204*** (0.0050)	-0.1982*** (0.0060)	0.0261*** (0.0011)
LEARN	READ	0.3262*** (0.0052)	-0.0451*** (0.0030)	0.0029*** (0.0005)
LEARN	MATH	0.3231*** (0.0046)	-0.0484*** (0.0025)	0.0059*** (0.0004)
LEARN	INTERN	-0.2690*** (0.0052)	0.0665*** (0.0054)	-0.0103*** (0.0010)
EXTERN	INTERP	-0.5390*** (0.0060)	0.0866*** (0.0066)	-0.0105*** (0.0012)
EXTERN	READ	-0.1469*** (0.0061)	0.0146*** (0.0013)	NS
EXTERN	MATH	-0.1242*** (0.0054)	0.0146*** (0.0027)	-0.0021*** (0.0005)
EXTERN	INTERN	0.2084*** (0.0057)	-0.0508*** (0.0059)	0.0078*** (0.0011)

the strongest effects of the kindergarten fall ADHD symptoms variables were the contemporaneous ones on the outcome variables at kindergarten fall with Lag = 0. The magnitude of the association between kindergarten fall EXTERN and READ declined linearly through the fifth grade spring assessment. For the other predictor/outcome combinations, the magnitude of the association declined quickly at the kindergarten spring and first grade spring assessments and gradually leveled off. Some of the models, in fact, indicate a reversal of the original trend near the fifth grade assessment, but this seems more likely an artifact of the necessary symmetry in quadratic models than a real pattern in the data, suggesting that exponential models might be a more appropriate choice for modeling some of these particular associations.

Multilevel Growth Models with Time-Varying Covariate Interactions

The multilevel growth model presents an interesting opportunity for examining a particular kind of moderation in longitudinal data. This is moderation of the effect of a L1 predictor variable on a L1 outcome variable by time. That is, one can test whether the association between repeatedly measured predictor and outcome constructs changes systematically over the course of a longitudinal study. Many theories would predict change in the strength of association between two variables during the course of development or when transitioning from one developmental stage to another. Judicious coding of time trends and modeling interactions between predictors and the time-related variables will allow such effects to be directly tested.

For example, consider a simple linear multilevel growth model

$$\text{L1: } Y_{ti} = \beta_{0i} + \beta_{1i} \text{time}_{ti} + e_{ti} \quad (21.48)$$

$$\begin{aligned} \text{L2: } \beta_{0i} &= \gamma_{00} + u_{0i} \\ \beta_{1i} &= \gamma_{10} + u_{1i} \end{aligned}$$

Combined:

$$Y_{ti} = \gamma_{00} + \gamma_{10} \text{time}_{ti} + u_{0i} + u_{1i} \text{time}_{ti} + e_{ti}$$

This model estimates the linear trend in the outcome variable over time. If a researcher is interested in the effect of a time point-specific predictor variable (X_{ti} , a time-varying covariate), it can be added to the L1 specification

$$\text{L1: } Y_{ti} = \beta_{0i} + \beta_{1i} \text{time}_{ti} + \beta_{2i} X_{ti} + e_{ti} \quad (21.49)$$

with an associated L2 equation

$$\text{L2: } \beta_{2i} = \gamma_{02} \quad (21.50)$$

Combined:

$$Y_{ti} = \gamma_{00} + \gamma_{10} \text{time}_{ti} + \gamma_{02} X_{ti} + u_{0i} + u_{1i} \text{time}_{ti} + e_{ti} \quad (21.51)$$

With this specification, the time trend (γ_{01}) is adjusted for variation in the time-varying predictor and γ_{02} represents the effect of the predictor over and above the time trend.

Now consider adding the interaction between the L1 predictor and the linear time variable:

$$\begin{aligned} \text{L1: } Y_{ti} &= \beta_{0i} + \beta_{1i} \text{time}_{ti} + \beta_{2i} X_{ti} \\ &\quad + \beta_{3i} X_{ti} \text{time}_{ti} + e_{ti} \\ \text{L2: } \beta_{3i} &= \gamma_{30} \end{aligned} \quad (21.52)$$

Combined:

$$Y_{ti} = \gamma_{00} + \gamma_{10} \text{time}_{ti} + \gamma_{02} X_{ti} + \gamma_{30} X_{ti} \text{time}_{ti} + u_{0i} + u_{1i} \text{time}_{ti} + e_{ti}$$

As Singer and Willett (2003, p. 171) noted, the coefficient associated with the interaction between a time-varying predictor and the time variable (here, γ_{03}) can be interpreted in two ways. One interpretation is that the rate of change

over time in the outcome variable depends on the level of the time-varying predictor. The second interpretation, and the one that we will focus on here, is that the effect of the time-varying predictor on the outcome changes over time. The equation above specifies linear change in the association between the time-varying predictor and the outcome, but (as discussed in the original Baron and Kenny (1986) paper), a more (e.g., quadratic) or less (e.g., step) complex function could be used to model a hypothesized pattern of change.

Hedeker (2004) discussed such an effect in the context of a longitudinal psychiatric study originally described in Reisby et al. (1977). This study examined the association between imipramine and desipramine plasma levels (the time-varying predictors) and clinical response (the outcome variable) over time in a sample of depressed inpatients. Because the effectiveness of antidepressants is not immediate but develops over time, an exploration of such moderated effects seems warranted. The model included a linear time trend, main effects of the imipramine and desipramine plasma levels, and interactions between linear time and the plasma levels. The model including the two interactions fit better than the model without the interactions. Examination of specific coefficients showed that the imipramine level by time interaction was not significant in this sample, but the desipramine interaction was significant, indicating that, in fact, the effect of this particular antidepressant on clinical response became more pronounced over the course of the 5-week study.

Table 21.5 reports the results of analyses in the ECLS-K data that included interactions between time-varying predictors and time. In particular, we explored whether the effects of the two ADHD symptoms variables on interpersonal skill, academic performance, and internalizing behavior change between fall of kindergarten and spring of fifth grade. Multilevel growth models incorporating an interaction between an ADHD symptoms variable and functions of the time variable were fitted for each possible pairing of LEARN and EXTERN with INTERP, MATH, READ, and INTERN. In each model, the base growth curve for the outcome construct was allowed to be quadratic with random effects for all growth parameters, and any nonsignificant variances were trimmed.

As Table 21.5 shows, the interaction between LEARN and linear time on INTERP is significant. This indicates that the contemporaneous positive association between the approaches to learning score and interpersonal skill ratings become stronger over time. For example, as indicated in column 3 of Table 21.5, the expected association between these two variables at kindergarten fall (i.e., when

TABLE 21.5 Coefficients and Standard Errors for Multilevel Models Including Time-Varying Predictor by Time Interactions

Predictor	Outcome	Simple effect of predictor	Linear time interaction	Quadratic time interaction
LEARN	INTERP	0.6365*** (0.0035)	0.0070*** (0.0013)	NS
LEARN	READ	0.1602*** (0.0035)	-0.0293*** (0.0030)	0.0029*** (0.0005)
LEARN	MATH	0.1312*** (.0029)	-0.0436*** (0.0026)	0.0064*** (0.0004)
LEARN	INTERN	-0.2632*** (0.0037)	-0.0114*** (0.0014)	NS
EXTERN	INTERP	-0.5120*** (0.0052)	-0.0787*** (0.0056)	0.0093*** (.0010)
EXTERN	READ	-.0234*** (0.0039)	-0.0247*** (0.0033)	0.0034*** (.0005)
EXTERN	MATH	-0.0315*** (.0037)	-.0035** (.0012)	NS
EXTERN	INTERN	0.2242*** (0.0049)	0.0369*** (0.0054)	-0.0040*** (.0010)

Time = 0) is .6365 and can be interpreted in much the same way as an unstandardized regression coefficient: for each one unit increase in LEARN at kindergarten fall, the expected value of interpersonal skill at that time point increases by .6365 points. For each additional year that passes, the contemporaneous association is expected to increase by .0070 points (column 4). So at fifth grade spring, the expected value of interpersonal skills increases by .6750 (i.e., .6365 + .0070 * 5.5) for each one unit increase in LEARN. Similarly, Table 21.5 shows that the negative effects of LEARN on INTERN and the negative effects of EXTERN on MATH become significantly more pronounced over time. The predicted associations between the ADHD predictor variables and the outcome variables at each time point correspond fairly well—but not perfectly—to coefficients produced via OLS regressions conducted on data from each time point separately. Although this model may not precisely reproduce the regression coefficients at each time point, it does allow for a single parameter test of the change in such coefficients across time that is not available from analyses conducted on data from each measurement occasion separately.

Five of the eight analyses presented in Table 21.5 involve a significant interaction between the time-varying ADHD symptoms predictor and the quadratic time function, and for all of these, the interactions involving the linear and quadratic time functions are of opposite sign. This indicates that the initial changes in the magnitude of the ADHD symptoms variable and the outcome will eventually change direction. For example, the coefficients for the effect of LEARN on MATH in columns

3 and 4 of Table 21.5 indicate a positive association at kindergarten fall that initially decreases in magnitude over time. However, because of the positive sign of the quadratic interaction in column 5, this trend reaches a minimum around third grade spring, and the magnitude of the association increases thereafter. The maxima or minima for the other four associations with quadratic time interactions, however, reach their minima or maxima much closer to the end of the grade range included in the data. Care should be taken to ensure that these reversals are a true feature of the data and not just an artifact of the necessarily symmetric time function used in this particular example.

The model specification in equation 21.50 as well as those in Hedeker (2004) and the previous ECLS-K analyses assume that the effects of the time-varying covariates are fixed. That is, the association between the time-varying covariate and the outcome measure is the same for all subjects. Moreover, the extent to which the covariate's association with the outcome changes over time is also assumed to be the same for all subjects. However, with a sufficiently large sample, it may be possible to explore whether, in fact, these effects differ across individuals. This can be accomplished by adding error terms u_{2i} and u_{3i} to the L2 equations predicting the β_{2i} and β_{3i} coefficients, respectively. Then the variance of u_{2i} estimates (τ_{22}) indicates the extent to which individuals vary in the association between their predictor and outcome values when time is at its zero point, and the variance of the u_{3i} estimates (τ_{33}) indicates the extent to which individuals vary in how much this association changes over time.

Mediation Change over Time Model

The mediation change over time (M-COT) model builds upon the idea of creating interactions between predictors and the explicit time variable in a multilevel growth model to examine possible change in both the a and b paths of a mediation model over time. In contrast to other longitudinal mediation models that will be presented below, the focus of this model is not on establishing the temporal precedence of effects necessary for causal pathways. Rather, the M-COT model is largely descriptive and explores how the various paths in the basic model of Figure 21.1, specifically the a path from X to M and the b path from M to Y , change over the course of a longitudinal study. The $X \rightarrow M$ and $M \rightarrow Y$ relations within the model might be cross sectional, or X might be assessed prior to M , and M might be assessed prior to Y . In the former case, the M-COT model would simply describe how the contemporaneous associations between variables

change over time. In the latter case, the M-COT model would describe how the lagged relations among variables change as time progresses.

This discussion will concentrate on the M-COT model for exploring cross sectional mediation in longitudinal data. This model is appropriate for data with repeated measures of the outcome variable Y . The initial variable X and the mediator variable M could also be measured repeatedly or could be time-invariant individual variables. The following model assumes a $1 \rightarrow 1 \rightarrow 1$ model with all three variables measured repeatedly at the same time points. The M-COT model can be specified as a two-level random coefficient (i.e., multilevel) model with time points nested within individuals (Arruda & Krull, 2013). For a simple model that examines linear change over time in the a and b paths, the multiple equation form of the model to estimate a paths would be:

$$\begin{aligned} \text{L1: } M_{ti} &= \beta_{0i} + \beta_{1i}time_{ti} + \beta_{2i}X_{ti} \\ &\quad + \beta_{3i}time_{ti}X_{ti} + e_{ti} \\ \text{L2: } \beta_{0i} &= \gamma_{00} + u_{0i} \quad \beta_{2i} = \gamma_{20} \\ \beta_{1i} &= \gamma_{10} + u_{1i} \quad \beta_{3i} = \gamma_{30} \end{aligned} \quad (21.53)$$

and the single-equation form would be

$$\begin{aligned} M_{ti} &= \gamma_{00} + \gamma_{10}time_{ti} + \gamma_{20}X_{ti} + \gamma_{30}time_{ti}X_{ti} \\ &\quad + u_{0i} + u_{1i}time_{ti} + e_{ti} \end{aligned} \quad (21.54)$$

In this particular model, the γ_{20} coefficient estimates the association between X and M when time is at its zero point. The γ_{30} coefficient estimates how much this association changes for each additional unit of time.

The multilevel equation to generate β path estimates would be:

$$\begin{aligned} \text{L1: } Y_{ti} &= \beta_{0i} + \beta_{1i}time_{ti} + \beta_{2i}X_{ti} + \beta_{3i}time_{ti}X_{ti} \\ &\quad + \beta_{4i}M_{ti} + \beta_{5i}time_{ti}M_{ti} + e_{ti} \\ \text{L2: } \beta_{0i} &= \gamma_{00} + u_{0i} \quad \beta_{2i} = \gamma_{20} \quad \beta_{4i} = \gamma_{40} \\ \beta_{1i} &= \gamma_{10} + u_{1i} \quad \beta_{3i} = \gamma_{30} \quad \beta_{5i} = \gamma_{50} \end{aligned} \quad (21.55)$$

in the multiple equation form, and

$$\begin{aligned} Y_{ti} &= \gamma_{00} + \gamma_{10}time_{ti} + \gamma_{20}X_{ti} + \gamma_{30}time_{ti}X_{ti} \\ &\quad + \gamma_{40}M_{ti} + \gamma_{50}time_{ti}M_{ti} + u_{0i} + u_{1i}time_{ti} + e_{ti} \end{aligned} \quad (21.56)$$

in the single equation form. The γ_{40} coefficient in this particular model estimates the association between M and Y when time is at its zero point. The γ_{50} coefficient estimates how much this association changes for each additional unit of time.

Combining the appropriate coefficients from the two models provides an expression for how the mediated effect ab changes over time (Arruda & Krull, 2014a). The additional subscripts M and Y in the following equations indicate whether a given coefficient came from the first multilevel equation (M) regressing M on X or the second multilevel equation (Y) regressing Y on X and M simultaneously. The combined expression is simply the product of the terms relevant to the a effect in equation 21.54 and those relevant to the b effect in equation 21.6:

$$ab_{ti} = (\gamma_{20M} + \gamma_{30M}time_{ti})(\gamma_{40Y} + \gamma_{50Y}time_{ti}) \quad (21.57)$$

Manipulating and rearranging this expression gives

$$\begin{aligned} ab_{ti} &= \gamma_{20M}\gamma_{40Y} + (\gamma_{20M}\gamma_{50Y} + \gamma_{30M}\gamma_{40Y})time_{ti} \\ &\quad + \gamma_{30M}\gamma_{50Y}time_{ti}^2 \end{aligned} \quad (21.58)$$

The first term in this expression, $\gamma_{20M}\gamma_{40Y}$, estimates the mediated effect (i.e., the product of the a and b paths) when time is at its zero point. The second term, $\gamma_{20M}\gamma_{50Y} + \gamma_{30M}\gamma_{40Y}$, estimates the instantaneous linear change in the mediated effect when time is at its zero point. The third term, $\gamma_{30M}\gamma_{50Y}$, estimates the quadratic change in the mediated effect over time and indicates how the instantaneous linear change in the mediated effect itself changes over time.

Approximate standard errors for the three terms in equation 21.58 can be obtained via the multivariate delta method (Arruda & Krull, 2014a). Specifically, the standard errors are as follows:

$$SE(\gamma_{20M}\gamma_{40Y}) = \sqrt{\gamma_{20M}^2\sigma_{\gamma_{40Y}}^2 + \gamma_{40Y}^2\sigma_{\gamma_{20M}}^2} \quad (21.59)$$

$$\begin{aligned} SE(\gamma_{20M}\gamma_{50Y} + \gamma_{30M}\gamma_{40Y}) &= \sqrt{\gamma_{20M}^2\sigma_{\gamma_{50Y}}^2 \\ &\quad + \gamma_{50Y}^2\sigma_{\gamma_{20M}}^2 + \gamma_{40Y}^2\sigma_{\gamma_{30M}}^2 + \gamma_{30M}^2\sigma_{\gamma_{40Y}}^2} \\ &\quad + (2 * (\gamma_{20M}\gamma_{30M}\sigma_{\gamma_{40Y}\gamma_{50Y}}) + (\gamma_{40Y}\gamma_{50Y}\sigma_{\gamma_{20M}\gamma_{30M}})) \end{aligned} \quad (21.60)$$

$$SE(\gamma_{30M}\gamma_{50Y}) = \sqrt{\gamma_{30M}^2\sigma_{\gamma_{50Y}}^2 + \gamma_{50Y}^2\sigma_{\gamma_{30M}}^2} \quad (21.61)$$

These standard errors can be used to conduct hypothesis tests or develop confidence intervals for each of the terms in equation 21.58. Thus we can explicitly test the trend in ab over time.

One useful way to think about this model is to consider the ultimate form of change in the ab estimate of the mediated effect that will result from combining different trends in a and b . As currently specified, three types of trends are possible for the a path: the a path could be zero (both the γ_{20M} and γ_{30M} coefficients are nonsignificant), a nonzero constant over time (the γ_{20M} coefficient is significant but the γ_{30M} coefficient is nonsignificant), or a linearly increasing function of time (the γ_{30M} coefficient is significant, regardless of the significance of the γ_{20M} coefficient). The same three trends are also possible for the b path (i.e., zero—both the γ_{40Y} and the γ_{50Y} coefficients are nonsignificant, constant—the γ_{40Y} coefficient is significant but the γ_{50Y} coefficient is nonsignificant, or linearly changing—the γ_{50Y} coefficient is significant, regardless of the significance of the γ_{40Y} coefficient). Then there are nine possibilities for the multiplicative combination, as shown in Table 21.6. Of course, whenever the nature of either of the two paths is zero, the multiplicative function ab will also be zero. Thus five of the nine combinations ultimately produce no mediated effect. Only four combinations produce nonzero values for the ab function over time, and these correspond to the four terms in equation 21.58.

With this limited set of possibilities (i.e., allowing only zero, constant, and linear trends for a and b), the only way that the ab product could indicate a constant mediated effect over time is if both a and b are nonzero constants. There are two ways that ab could show a pattern of linear change over time, specifically (1) if a is a nonzero constant and b changes linearly over time, or (2) if b is a nonzero constant and a changes linearly over time. The product ab will change quadratically over time only if both a and b change linearly.

By substituting each of the unique time values into the equation, it is possible to calculate the estimated mediated

effect ab at each time point. Standard errors of these time point-specific estimates can be calculated using the multivariate delta method. Simulation studies (Arruda & Krull, 2013, 2014b) comparing the time point-specific estimates of ab from the M-COT model with those generated by fitting a series of OLS regression models to the data from each time point found that both approaches generated unbiased estimates of ab at each time point. When the data generating process for a and b was truly linear over time (i.e., when the analysis model form matched the true model), however, time point-specific estimates produced by the M-COT model had smaller standard errors than those generated using standard regression techniques to estimate mediated effects at each time point separately. Thus the M-COT approach is more highly powered than the regression approach. Across the range of simulated conditions, the standard errors of mediated effects at specific time points were 14-47% smaller in the M-COT model, resulting in 25-89% more power to detect nonzero effects than using standard regression techniques (Arruda & Krull, 2014b).

The M-COT model was used to assess changes in mediated effects over time in the ECLS-K data. Results of separate analyses in which the ADHD symptoms variables predicted interpersonal skill and academic performance mediators and an internalizing behavior outcome are presented in Table 21.7. The negative effect of LEARN on INTERN through INTERP has a negative linear trend and no significant quadratic trend, so this effect become increasingly pronounced over time. Similar patterns hold for the effects on LEARN on INTERN through READ and MATH, but these include a significant positive quadratic component, so the increasingly negative mediated effects level off at later grades for these academic performance mediators. The effects of EXTERN on INTERN through the three mediators all show similar patterns, with positive mediated effects at kindergarten fall and significant linear and quadratic trends. This indicates that the mediated effects become stronger over time, and the rate of increase speeds up at later grades.

TABLE 21.6 Combinations of Significant and Nonsignificant Coefficients That Produce Various Trends of the Mediated Effect over Time

γ_{20M}	γ_{30M}	a path trend	γ_{40Y}	γ_{50Y}	b path trend	ab trend
NS	NS	Zero	NS	NS	Zero	Zero
NS	NS	Zero	Sig	NS	Constant	Zero
NS	NS	Zero	NS or sig	Sig	Linear	Zero
Sig	NS	Constant	NS	NS	Zero	Zero
Sig	NS	Constant	Sig	NS	Constant	Constant
Sig	NS	Constant	NS or sig	Sig	Linear	Linear
NS or sig	Sig	Linear	NS	NS	Zero	Zero
NS or sig	Sig	Linear	Sig	NS	Constant	Linear
NS or sig	Sig	Linear	NS or sig	Sig	Linear	Quadratic

TABLE 21.7 Coefficients, Standard Errors, and Mediated Effect Trends for the M-COT Model

<i>X</i>	<i>M</i>	<i>a</i> at K fall	<i>a</i> linear trend	<i>b</i> at K fall	<i>b</i> linear trend	Med at K fall	Med linear trend	Med Quad Trend
LEARN	INTERP	0.6365*** (0.0035)	0.0070*** (0.0013)	-0.1333*** (0.0053)	-0.0027 (0.0020)	-0.0849*** (0.0034)	-0.0027* (0.0013)	-1.89E ⁻⁰⁵ (1.44E ⁻⁰⁵)
LEARN	READ	0.2457*** (0.0042)	-0.0213*** (0.0012)	-0.0010 (0.0036)	-0.0283*** (0.0019)	-0.0003 (0.0009)	-0.0069*** (0.0005)	0.0006*** (5.25E ⁻⁰⁵)
LEARN	MATH	0.1915*** (0.0033)	-0.0174*** (0.0010)	-0.0125** (0.0041)	-0.0247*** (0.0015)	-0.0024** (0.0008)	-0.0045*** (0.0003)	0.0004*** (3.59E ⁻⁰⁵)
EXTERN	INTERP	-0.5352*** (0.0043)	-0.0297*** (0.0016)	-0.1798*** (0.0046)	-0.0066*** (0.0018)	0.0962*** (0.0026)	0.0089*** (0.0011)	0.0002*** (5.45E ⁻⁰⁵)
EXTERN	READ	-0.0396*** (0.0047)	-0.0058*** (0.0013)	-0.0681*** (0.0036)	-0.0451*** (0.0019)	0.0027*** (0.0004)	0.0022*** (0.0003)	0.0003*** (5.97E ⁻⁰⁵)
EXTERN	MATH	-0.0315*** (0.0037)	-0.0035** (0.0012)	-0.0892*** (0.0040)	-0.0329*** (0.0015)	0.0081*** (0.0004)	0.0014*** (0.0002)	0.0001** (3.98E ⁻⁰⁵)

The M-COT model is unique among techniques designed to assess mediation in longitudinal data because it assesses mediation cross sectionally, but the cross sectional associations are not necessarily stable over time. It is assumed that there is some short-term process that has reached equilibrium at each measurement occasion, but a longer-term process that is not stationary. This model might prove useful in explaining conflicting results in developmental studies that have examined mediation at different ages or across different timespans. The model specified here is a $1 \rightarrow 1 \rightarrow 1$ mediation model, but the general framework could be easily adapted to accommodate individual-level predictor and mediator variables at L2. The model was also limited to linear trends over time for both the *a* and *b* paths. A simulation study is currently underway to examine more complex trends for *a* and *b* as well as ideal methods for determining the form of the *a* and *b* trends in real world data where the true data generating process is not known. The current model allows intercepts and time trends to vary across individuals but assumes that the *a* and *b* paths are fixed. If the data warrant, these assumptions could be removed by incorporating methods similar to those of Bauer et al. (2006). Finally, critiques about assessing mediation cross sectionally could be addressed by estimating lagged relations among variables (e.g., $X_1 \rightarrow M_2 \rightarrow Y_3$; $X_2 \rightarrow M_3 \rightarrow Y_4$; $X_3 \rightarrow M_4 \rightarrow Y_5$) rather than the cross sectional associations as shown here.

Longitudinal Mediation in Autoregressive Panel Models

When *X*, *M*, and *Y* are measured repeatedly over time, autoregressive panel data models can be extended to the three variable case to investigate longitudinal mediation. When the autoregressive mediation model involves two waves of data, fitting the autoregressive models to estimate mediation effect is somewhat limited, in that either the relation between *X* and *M* or the relation between *M* and *Y*

is contemporaneous. Gollob and Reichardt (1991) argued that it takes time for a cause to have an effect and thus, at least $k + 2$ waves of data are needed to estimate indirect effects in a model with a chain of *k* mediators. Thus, with a single mediator, data should ideally be collected at least across three waves or time points, so *X* can affect *M* and *M* can affect *Y* in time. Otherwise, some parameter constraints or assumptions are needed, based on theory and prior knowledge (Gollob & Reichardt, 1991), such as assumptions of stationarity and knowledge of test-retest reliability.

Figure 21.13 presents an example of the autoregressive mediation model with two waves of data, where the effect of *X* on *M* and the effect of *M* on *Y* are specified as lag-1 effects in an SEM. The autoregressive models described in this section are based on the observed variables, ignoring the measurement errors. With inclusion of multiple

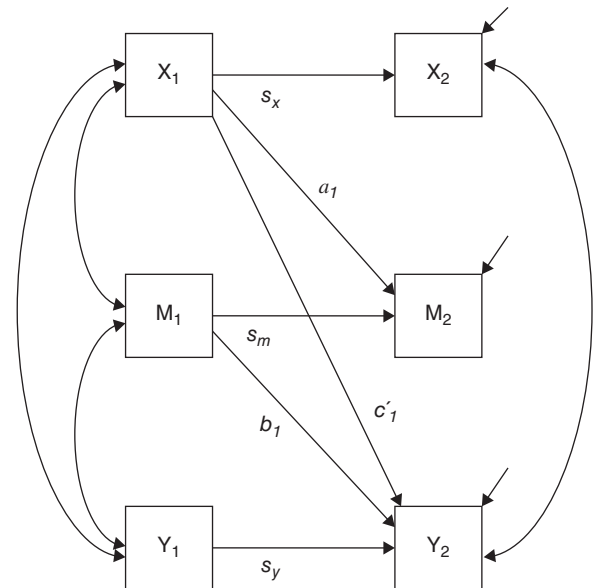


Figure 21.13 An autoregressive model with longitudinal mediation across two waves.

indicators, the repeated measures of X , M , and Y can be modeled as latent variables and the indirect effects can be estimated based on the lagged effects between the latent variables. The model in Figure 21.13 can be described by equations 21.62 to 21.64.

$$X_2 = i_{x2} + s_x X_1 + e_{x2} \quad (21.62)$$

$$M_2 = i_{m2} + a_1 X_1 + s_m M_1 + e_{m2} \quad (21.63)$$

$$Y_2 = i_{y2} + b_1 M_1 + c'_1 X_1 + s_y Y_1 + e_{y2} \quad (21.64)$$

where the variables measured at time 1 are treated as predetermined as follows:

$$X_1 = i_{x1} + e_{x1} \quad (21.65)$$

$$M_1 = i_{m1} + e_{m1} \quad (21.66)$$

$$Y_1 = i_{y1} + e_{y1} \quad (21.67)$$

where i_{x1} , i_{m1} , and i_{y1} are the means of X_1 , M_1 , and Y_1 , respectively, and e_{x1} , e_{m1} , and e_{y1} are deviations from the means. Typically, X_1 , M_1 , and Y_1 are specified to be correlated. As described previously, i_{m2} and i_{y2} in equations 21.63 and 21.64 are regression intercepts, and e_{m2} and e_{y2} are residuals. The coefficients a_1 and b_1 represent the effect of X_1 on M_2 and the effect of M_1 on Y_2 , respectively, while the coefficient c'_1 denotes the direct effect of X_1 on Y_2 . The product of the estimates of the coefficients a_1 and b_1 , i.e., $a_1 b_1$, is the estimated longitudinal indirect effect. Although the coefficient b_1 represents the relation between M and Y before X affects M , the $a_1 b_1$ estimator would be a reasonable estimate of the longitudinal indirect, because under the stationarity assumption the relation between M_1 and Y_2 (b_1) would be equal to the relation between M_2 and Y_3 that would have been measured after the second wave (Cole & Maxwell, 2003). The standard error of the estimated indirect effect can be obtained using the formulas described earlier based on the relevant path coefficients and their standard errors to conduct the significance test and compute confidence intervals.

Another example of the autoregressive mediation model with two wave data is the model in Figure 21.14 (Mackinnon, 2008). The relations relevant to the mediational pathways can be expressed by the following equations 21.68 and 21.69:

$$M_2 = i_{m2} + a_1 X_1 + a_2 X_2 + s_m M_1 + e_{m2} \quad (21.68)$$

$$Y_2 = i_{y2} + b_1 M_1 + b_2 M_2 + c'_1 X_1 + c'_2 X_2 + s_y Y_1 + e_{y2} \quad (21.69)$$

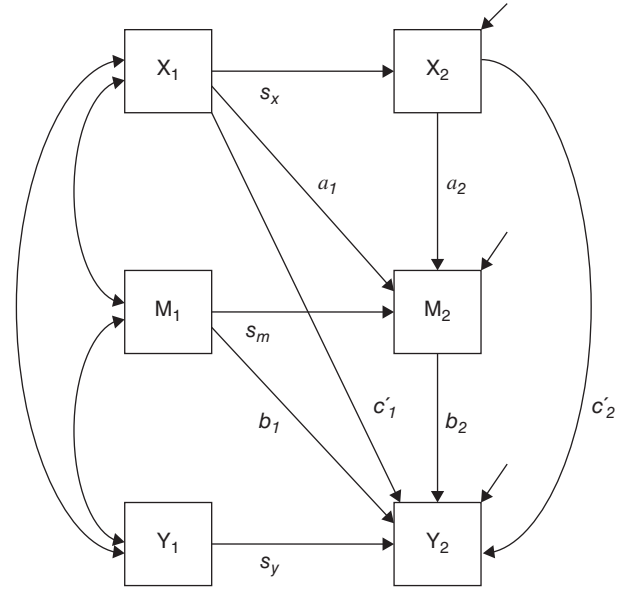


Figure 21.14 An autoregressive mediation model with longitudinal and contemporaneous mediation across two waves.

With the addition of the contemporaneous relations (a_2 , b_2 , and c'_2) to the previous model in Figure 21.13, the coefficient a_1 in the above model represents the effect of X_1 on M_2 after controlling for the effects of X_2 and M_1 , while the coefficient a_2 represents the effect of X_2 on M_2 after controlling for the effects of X_1 and M_1 . Similarly, the coefficients b_1 and b_2 indicate the effect of M_1 on Y_2 after controlling for the effects of X_1 , X_2 , M_2 , and Y_1 and the effect of M_2 on Y_2 after controlling for the effects of X_1 , X_2 , M_1 , and Y_1 , respectively. There are several ways to estimate mediated effects in the model in Figure 21.14: $a_1 b_1$, $a_2 b_2$, and $a_1 b_2$. As described earlier, the indirect effect $a_1 b_1$ is based on the across-time or lag-1 relations, while the indirect effect $a_2 b_2$ is estimated based on the contemporaneous relations among the variables measured at time 2. The indirect effect $a_1 b_2$ is based on the across-time relation between X_1 and M_2 and the contemporaneous relation between M_2 and Y_2 . Cole and Maxwell (2003) recommend the $a_1 b_1$ estimator, because the estimates $a_2 b_2$ and $a_1 b_2$ involve cross sectional relations and are likely to be biased when time is needed for the cause to exert its effect on the outcome (Gollob & Reichardt, 1987).

In the two-wave autoregressive mediation models, information on stability, stationarity, and equilibrium is quite limited, in that we cannot actually test whether these properties hold across time. However, estimating autoregressive effects is beneficial for avoiding omitted variable bias. An autoregressive effect is defined as the effect that a variable has on itself at a later time (Gollob & Reichardt, 1987) and

is denoted by the coefficients s_x , s_m , and s_y in the above models. These coefficients also estimate the stability of a variable across time (Kenny, 1979). When the dependent variable at time t is predicted by the predictors at time $t - 1$, the levels of the dependent variable at time $t - 1$ may represent unmeasured variables that are correlated with the predictors and estimating autoregressive relations can control for such unmeasured variables (Gollob & Reichardt, 1991). Without controlling for those omitted variables or confounders, the causal effects will be biased. Thus, the estimates of a_1 and a_2 in Figures 21.13 and 21.14 will be less biased when estimating s_m and so will b_1 and b_2 , when estimating s_y . These less biased coefficients in the mediational pathways will result in the indirect effect that is closer to the true effect size.

When X , M , and Y are measured repeatedly over three or more occasions, the parameterizations in the autoregressive models become more flexible. For example, the effect of M_2 on Y_3 that relied on the stationarity assumption in the two wave data (b_1 in Figures 21.13 and 21.14) can now be estimated. With three or more waves of data, estimating lag effects according to the causal order of X to M to Y is possible, although timing of measurements is a remaining issue, because it is difficult to predict how long it takes for one variable to have its optimal effect on another. Having more waves of data also provides opportunities to examine when stability, stationarity, and equilibrium are reached. It may take a long time for a system of X , M , and Y to achieve stability, stationarity, or equilibrium and sometimes it may be achieved beyond the time span the data cover. While

these characteristics can only be assumed in the cross sectional or two wave designs, researchers can check these assumptions with more waves of data by testing equivalence of parameter estimates across waves. Finally, time specific indirect effects can be estimated across multiple waves. The magnitude of effect, either the autoregressive effects or the lagged effects, may vary across time (Cole & Maxwell, 2003). The effect of X_1 on M_2 may be larger or smaller than the effect of X_2 on M_3 and the resulting indirect effects may differ depending on which time points are involved (Cole & Maxwell, 2003; Gollob & Reichardt, 1987, 1991). It might be an interesting research task to detect when the significant indirect effect arises, holds, and desists.

Figure 21.15 shows an example of the autoregressive mediation model with four waves of data, in which the relations among X , M , and Y are specified as lag-1 effects. The model in Figure 21.15 can be described in the following equations:

$$X_2 = i_{x2} + s_{x1}X_1 + e_{x2} \quad (21.70)$$

$$X_3 = i_{x3} + s_{x2}X_2 + e_{x3} \quad (21.71)$$

$$X_4 = i_{x4} + s_{x3}X_3 + e_{x4} \quad (21.72)$$

$$M_2 = i_{m2} + a_1X_1 + s_{m1}M_1 + e_{m2} \quad (21.73)$$

$$M_3 = i_{m3} + a_2X_2 + s_{m2}M_2 + e_{m3} \quad (21.74)$$

$$M_4 = i_{m4} + a_3X_3 + s_{m3}M_3 + e_{m4} \quad (21.75)$$

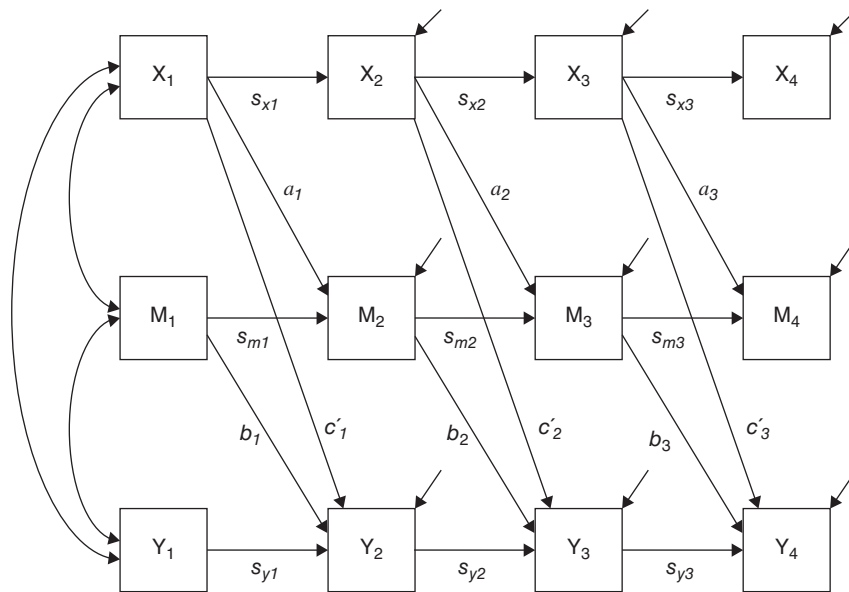


Figure 21.15 An autoregressive mediation model with longitudinal mediation across four waves.

$$Y_2 = i_{y_2} + b_1 M_1 + c'_1 X_1 + s_{y_1} Y_1 + e_{y_2} \quad (21.76)$$

$$Y_3 = i_{y_3} + b_2 M_2 + c'_2 X_2 + s_{y_2} Y_2 + e_{y_3} \quad (21.77)$$

$$Y_4 = i_{y_4} + b_3 M_3 + c'_3 X_3 + s_{y_3} Y_3 + e_{y_4} \quad (21.78)$$

Again, the variables measured at time 1 are predetermined in the same way as in equations 21.65 to 21.67. With more than two waves of data, the effect of X on M and the effect of M on Y are represented by more than one path coefficient. In Figure 21.15 and equations 21.70 to 21.78, the coefficients a_1 , a_2 , and a_3 all represent the time-specific effect of X on M . Similarly, the coefficients b_1 , b_2 , and b_3 represent the time-specific effect of M on Y controlling for X , although the coefficients b_2 and b_3 are the effects of M that were affected by X , while b_1 is the effect of M before X exerts its effect on M . The coefficients c'_1 , c'_2 , and c'_3 reflect the time-specific direct effects of X on Y .

The indirect effects for the model in Figure 21.15 can be estimated in several ways. As we did for the two wave data, one can estimate the indirect effect for each lag such as $a_1 b_1$, $a_2 b_2$, and $a_3 b_3$. Another option is to estimate the indirect effect reflecting the temporal ordering of X , M , and Y , such as $a_1 b_2$ and $a_2 b_3$. The indirect effect and its standard error can be obtained following the formulas described earlier and can be used for significance testing and construction of confidence limits. If the stationarity assumption holds across 4 waves, all the a paths will be equivalent and so will the b paths. In such case, the estimated indirect effects (i.e., $a_1 b_1$, $a_2 b_2$, $a_3 b_3$, $a_1 b_2$, $a_2 b_3$) will be the same. On the other hand, if the path coefficients vary across waves, the indirect effects will be different depending on the time points involved and specific indirect effects can be examined. For example, if four measurement occasions span across different developmental phases, the indirect effect observed at the earlier phase may differ from the indirect effect at the later phase. Differences in the indirect effects across different phases may be an interesting research question. If the autoregressive mediation model is estimated in an SEM framework, the equivalence of path coefficients across time can be tested and the parameters can be constrained accordingly before obtaining the indirect effect.

In addition to the indirect effects listed already that are based on the two longitudinal relations between $X_t \rightarrow M_{t+1}$ and $M_t \rightarrow Y_{t+1}$ (e.g., $a_1 b_1$, $a_2 b_2$, $a_3 b_3$, $a_1 b_2$, $a_2 b_3$), there are other time-specific indirect effects in the model in Figure 21.15, such as $s_{x1} a_2 b_3$ and $a_1 b_2 s_{y3}$. These individual indirect effects are time-specific indirect effects, as they represent causal mechanisms from X to M to Y

for specific time intervals. Gollob and Reichardt (1991) argued that these individual time-specific indirect effects may be of less interest to researchers because mediation may evolve gradually over the course of time, rather than occurring at a specific time point. The overall indirect effect may then provide a better estimate for the indirect effect. The *overall indirect effect* is defined as the sum of all the *individual time-specific indirect effects*, i.e., all the indirect effects from X at time 1 to Y at the last time point mediated by M at any time between the first and the last time points (Gollob & Reichardt, 1991). In the model in Figure 21.15, there are three such individual time-specific indirect effects: (1) $X_1 \rightarrow X_2 \rightarrow M_3 \rightarrow Y_4$ ($s_{x1} a_2 b_3$), (2) $X_1 \rightarrow M_2 \rightarrow M_3 \rightarrow Y_4$ ($a_1 s_{m2} b_3$), and (3) $X_1 \rightarrow M_2 \rightarrow Y_3 \rightarrow Y_4$ ($a_1 b_2 s_{y3}$). Computing the total indirect effect and its standard error, as well as the total direct effect and its standard error, is available in most SEM software programs, including EQS (Bentler & Wu, 2006), LISREL (Jöreskog & Sörbom, 2006), Proc CALIS in SAS (SAS Institute, 2014), and Mplus (Muthén & Muthén, 2014). It is worth noting that the accuracy of the total indirect effect would rely on the specified mediation model, however. For example, if the mediation model in Figure 21.15 is a true model but the measurement at time 2 is omitted, resulting in the model with data measured at time 1, 3, and 4, the estimated total indirect effect will not equal the true overall indirect effect. See Gollob and Reichardt (1991) for illustration.

We fit an autoregressive panel SEM with longitudinal mediation paths to first, third, and fifth grade ECLS-K data. Though it is possible to fit autoregressive models with unequal lags between time points (Voelkle, Oud, Davidov, & Schmidt, 2012), for the sake of simplicity, only the first, third, and fifth grade measures with constant 2-year lags are used in this example. Table 21.8 presents

TABLE 21.8 Coefficients and Standard Errors for Tests of Longitudinal Mediation in an Autoregressive Panel Model

$X1$	$M3$	$Y5$	a	b	Med
LEARN	INTERP	INTERN	0.1985*** (0.0131)	-0.0063 (0.0112)	-0.0013 (0.0022)
LEARN	READ	INTERN	0.0514*** (0.0035)	-0.1674*** (0.0204)	-0.0086*** (0.0012)
LEARN	MATH	INTERN	0.0511*** (0.0041)	-0.1270*** (0.0156)	-0.0065*** (0.0010)
EXTERN	INTERP	INTERN	-0.3229*** (0.0135)	-0.0959*** (0.0091)	0.0310*** (0.0032)
EXTERN	READ	INTERN	-0.0268*** (0.0037)	-0.2468*** (0.0203)	0.0066*** (0.0011)
EXTERN	MATH	INTERN	-0.0157*** (0.0043)	-0.1829*** (0.0155)	0.0029*** (0.0008)

coefficients and standard errors for a , b , and the longitudinal mediated effect from ADHD symptoms at first grade through interpersonal skills and academic performance at third grade on internalizing behavior at fifth grade. The results indicate significant longitudinal mediated effects of LEARN on INTERN through both READ and MATH (but not INTERP) and longitudinal effects of EXTERN on INTERN through all three mediators. However, CFI and RMSEA values indicated that these models did not fit the data well.

Another type of autoregressive mediation model is shown in Figure 21.16 with the mediational pathways described in equations 21.79 to 21.84:

$$M_2 = i_{m2} + a_1 X_1 + a_4 X_2 + s_{m1} M_1 + e_{m2} \quad (21.79)$$

$$M_3 = i_{m3} + a_2 X_2 + a_5 X_3 + s_{m2} M_2 + e_{m3} \quad (21.80)$$

$$M_4 = i_{m4} + a_3 X_3 + a_6 X_4 + s_{m3} M_3 + e_{m4} \quad (21.81)$$

$$Y_2 = i_{y2} + b_1 M_1 + b_4 M_2 + c'_1 X_1 + c'_4 X_2 + s_{y1} Y_1 + e_{y2} \quad (21.82)$$

$$Y_3 = i_{y3} + b_2 M_2 + b_5 M_3 + c'_2 X_2 + c'_5 X_3 + s_{y2} Y_2 + e_{y3} \quad (21.83)$$

$$Y_4 = i_{y4} + b_3 M_3 + b_6 M_4 + c'_3 X_3 + c'_6 X_4 + s_{y3} Y_3 + e_{y4} \quad (21.84)$$

In addition to the longitudinal relations between X , M , and Y specified in Figure 21.15, the model in Figure 21.16

includes the contemporaneous relations among X , M , and Y at each wave. These relations are estimated by the coefficients a_4 , a_5 , and a_6 for the relations between X and M at Time 2, Time 3, and Time 4, respectively, and the coefficients b_4 , b_5 , and b_6 for the relations between M and Y . The product of path coefficients for the contemporaneous relations, such as $a_4 b_4$, $a_5 b_5$, and $a_6 b_6$, provide the estimates of the contemporaneous indirect effect. When X and M have immediate effects, the longitudinal relations may miss the real effects due to the time lag between measurements. In such cases, the contemporaneous indirect effects may be closer to the true indirect effects (MacKinnon, 2008).

We fit models including both longitudinal and contemporaneous mediation pathways to the ECLS-K first through fifth grade data. The results are summarized in Table 21.9. These models fit the data better than the previous set that included only the longitudinal pathways (all CFIs > .95), but RMSEA values were in the .08–.10 range, which is still larger than desired for optimal fit. Longitudinal mediated paths ($X_1 \rightarrow M_3 \rightarrow Y_5$) involving the INTERP and READ (but not MATH) mediators were significant or marginally significant and in the expected direction. Several mediated effects involving contemporaneous a or b paths were also significant in models including the INTERP mediator and READ mediators.

Although having more than two waves for the autoregressive models may allow for the flexibility in model specification and the opportunities to obtain more detailed information about the indirect effects, it may introduce

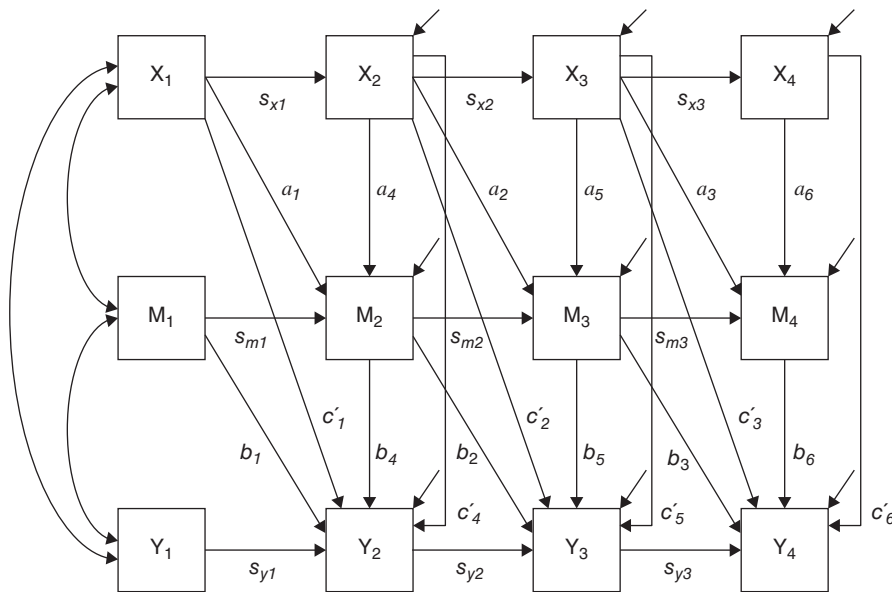


Figure 21.16 An autoregressive mediation model with longitudinal and contemporaneous mediation across four waves.

TABLE 21.9 Coefficients and Standard Errors for Tests of Longitudinal and Contemporaneous Mediation in an Autoregressive Panel Model

$X \rightarrow \text{INTERP} \rightarrow \text{INTERN}$	$X = \text{LEARN}$			$X = \text{EXTERN}$		
	a	b	Med	a	b	Med
$X1 \rightarrow M2 \rightarrow Y2$	-0.1535*** (0.0108)	-0.1326*** (0.0118)	0.0204*** (0.0023)	-0.0308* (0.0128)	-0.2076*** (0.0109)	0.0064* (0.0027)
$X1 \rightarrow M2 \rightarrow Y3$	-0.1535*** (0.0108)	0.0400*** (0.0122)	-0.0061** (0.0019)	-0.0308* (0.0128)	-0.0255* (0.0113)	0.0008† (0.0005)
$X2 \rightarrow M2 \rightarrow Y2$	0.6912*** (0.0087)	-0.1326*** (0.0118)	-0.0917*** (0.0082)	-0.6232*** (0.0110)	-0.2076*** (0.0109)	0.1294*** (0.0072)
$X2 \rightarrow M2 \rightarrow Y3$	0.6912*** (0.0087)	0.0400*** (0.0122)	0.0276** (0.0084)	-0.6232*** (0.0110)	-0.0255* (0.0113)	0.0159* (0.0070)
$X2 \rightarrow M3 \rightarrow Y3$	-0.1445*** (0.0113)	-0.1435*** (0.0124)	0.0207*** (0.0024)	-0.0057 (0.0135)	-0.2161*** (0.0110)	0.0012 (0.0029)
$X3 \rightarrow M3 \rightarrow Y3$	0.6698*** (0.0086)	-0.1435*** (0.0124)	-0.0961*** (0.0084)	-0.5954*** (0.0115)	-0.2161*** (0.0110)	0.1287*** (0.0070)
$X \rightarrow \text{READ} \rightarrow \text{INTERN}$						
$X1 \rightarrow M2 \rightarrow Y2$	0.0232*** (0.0040)	-0.0172 (0.0273)	-0.0004 (0.0006)	-0.0083 (0.0044)	-0.1257*** (0.0274)	0.0010† (0.0006)
$X1 \rightarrow M2 \rightarrow Y3$	0.0232*** (0.0040)	-0.1066*** (0.0340)	-0.0025* (0.0009)	-0.0083 (0.0044)	-0.1675*** (0.0350)	0.0014† (0.0008)
$X2 \rightarrow M2 \rightarrow Y2$	0.0618*** (0.0038)	-0.0172 (0.0273)	-0.0011 (0.0017)	-0.0365*** (0.0045)	-0.1257*** (0.0274)	-0.0046*** (0.0011)
$X2 \rightarrow M2 \rightarrow Y3$	0.0618*** (0.0038)	-0.1066*** (0.0340)	-0.0066** (0.0021)	-0.0365*** (0.0045)	-0.1675*** (0.0350)	0.0061*** (0.0015)
$X2 \rightarrow M3 \rightarrow Y3$	0.0144*** (0.0031)	0.0472 (0.0356)	0.0007 (0.0005)	-0.0022 (0.0035)	-0.0428 (0.0362)	0.0001 (0.0002)
$X3 \rightarrow M3 \rightarrow Y3$	0.0302*** (0.0030)	0.0472 (0.0356)	0.0014 (0.0011)	-0.0198*** (0.0036)	-0.0428 (0.0362)	0.0008 (0.0007)
$X \rightarrow \text{MATH} \rightarrow \text{INTERN}$						
$X1 \rightarrow M2 \rightarrow Y2$	0.0246*** (0.0047)	-0.0373 (0.0228)	-0.0009 (0.0006)	0.0048 (0.0051)	-0.1095*** (0.0229)	-0.0005 (0.0006)
$X1 \rightarrow M2 \rightarrow Y3$	0.0246*** (0.0047)	-0.0461 (0.0289)	-0.0011 (0.0007)	0.0048 (0.0051)	-0.0781** (0.0297)	-0.0004 (0.0004)
$X2 \rightarrow M2 \rightarrow Y2$	0.0569*** (0.0045)	-0.0373 (0.0228)	-0.0021 (0.0013)	-0.0400*** (0.0052)	-0.1095*** (0.0229)	0.0044*** (0.0011)
$X2 \rightarrow M2 \rightarrow Y3$	0.0569*** (0.0045)	-0.0461 (0.0289)	-0.0026 (0.0005)	-0.0400*** (0.0052)	-0.0781** (0.0297)	0.0031* (0.0013)
$X2 \rightarrow M3 \rightarrow Y3$	0.0160*** (0.0039)	-0.0190 (0.0278)	-0.0003 (0.0005)	-0.0083 (0.0044)	-0.0894** (0.0282)	0.0007 (0.0005)
$X3 \rightarrow M3 \rightarrow Y3$	0.0364*** (0.0038)	-0.0190 (0.0278)	-0.0007 (0.0010)	-0.0222*** (0.0045)	-0.0894** (0.0282)	0.0020** (0.0007)

complexity of parameterization. For example, a variant of the model in Figure 21.16 may include cross-lagged effects among X , M , and Y by adding the direct effects from M_{t-1} to X_t and Y_{t-1} to M_t based on the temporal sequence in which the variables are measured. Specifying these cross-lagged relations may reflect data more reasonably than the models in Figure 21.15 and 21.16, because X , M , and Y are related to each other. However, this model violates the causal sequence of X to M to Y specified by the mediation model because of the reversed paths from M to X and Y to M . As expected, choosing a correct model can be a tricky task with more parameters to be estimated. When the models are estimated in the SEM framework, it would be appropriate to estimate a series of nested models and conduct a series of chi-square difference tests to choose a model that better reflects the data, as well as

the theory. In addition, equality of path coefficients, such as paths for the effects of X on M (i.e., a_1 , a_2 , and a_3) can be also tested by comparing the models with and without the equality constraints.

Latent Growth Curve Mediation Models

The latent growth curve (LGC) modeling (Meredith & Tisak, 1990; Muthén & Curran, 1997; Willett & Sayer, 1994) is another approach that can be used to evaluate longitudinal mediation models when X , M , and Y are repeatedly measured across three or more time points (Cheong, 2011; Cheong, MacKinnon, & Khoo, 2003; Mackinnon, 2008). Unlike the autoregressive mediation models, where the indirect effect is based on the differences among the study participants, the indirect effect in

the LGC mediation model is conceptualized in terms of within-individual changes. In the autoregressive mediation models, the indirect effect is interpreted as the expected effect of X on Y at time t through M at time $t - 1$, if there was one unit change in X prior to time $t - 1$. Here, the changes in X , M , and Y are assumed equivalent to differences among participants in those variables. In LGC mediation models, individuals' trajectories of X , M , and Y are estimated using the repeated measures across multiple measurement time points and the indirect effect is determined based on the relations between the within-individual changes in X , M , and Y .

The capability of modeling mediation in terms of within-individual changes is important in developmental research, where the core research questions center on development, which is typically defined as within-individual change, and individual differences in development (Nesselroade, 1991). Using difference scores in a two-wave design is the simplest way of measuring individual change; however, difference scores contain minimal information about individual change (Rogosa, 1988; Rogosa, Brandt, & Zimowski, 1982; Willett & Sayer, 1994). The change patterns may follow nonlinear (e.g., quadratic or exponential) patterns or fluctuate in level between the two measurement time points. Growth curve modeling is considered a better method for investigating longitudinal within-individual changes and individual differences in those changes (Rogosa, 1988; Rogosa et al., 1982). Although growth curve models can be formulated both in the multilevel modeling framework (Raudenbush & Bryk, 2002; Goldstein, 1995) and in the SEM framework (Meredith & Tisak, 1990; Muthén & Curran, 1997; Willett & Sayer, 1994), we focus on the SEM approach in this section.

In the LGC approach to mediation, the mediation process is typically modeled using a parallel process model (Muthén & Curran, 1997; Stoolmiller, 1994). The repeated measures of X , M , and Y are modeled as distinctive growth processes and the mediation is specified in the relations among the growth factors. As it is recommended that researchers investigate the longitudinal pattern of change in each variable before examining the longitudinal relations between the variables (Raudenbush, 2001; Rogosa, 1988), the estimation of growth trajectories that best describe the changes in X , M , and Y should precede modeling mediation based on the relations among the latent growth factors. In this approach, the growth trajectories of X , M , and Y are modeled in the measurement model, where the repeated measures relate to the latent growth factors according to the specified time scale. Once the growth

trajectory shape is appropriately modeled for each X , M , and Y , the full parallel process model is specified to model the mediation process in the structural model among the latent growth factors.

Figure 21.17 shows an example of the LGC mediation model, where the growth trajectories of X , M , and Y are modeled as linear, based on repeated measures obtained with equal intervals across three waves. According to the specified time scales, the intercept factors (i.e., η_1 , η_3 , and η_5) represent the level of each variable at time 1, i.e., the initial status. The slope factors (i.e., η_2 , η_4 , and η_6) represent the change or growth rate of each variable per time unit. The mean and the variance of the intercept factor estimate the average level of initial status of each variable and the individual differences in the initial status. The mean and the variance of the latent slope factor estimate the average growth rate of each variable and the individual differences in the growth rates. Details of model specifications for LGC models can be found elsewhere (e.g., Muthén & Khoo, 1998; Willett & Sayer, 1994). The mediation process is specified in the relations among the latent factors shown in the middle of the model in Figure 21.17. More specifically, the individual differences in the initial status and the growth rates of M and Y are modeled as functions of the individual differences in the initial status and the growth rates of X and M .

The relations among the latent factors in Figure 21.17 are described in the following equations:

$$\eta_{3i} = \pi_{0m} + d_1\eta_{1i} + e_{3i} \quad (21.85)$$

$$\eta_{4i} = \pi_{1m} + a\eta_{2i} + d_2\eta_{1i} + e_{4i} \quad (21.86)$$

$$\eta_{5i} = \pi_{0y} + d_3\eta_{1i} + d_5\eta_{3i} + e_{5i} \quad (21.87)$$

$$\eta_{6i} = \pi_{1y} + b\eta_{4i} + c'\eta_{2i} + d_4\eta_{1i} + d_6\eta_{3i} + e_{6i} \quad (21.88)$$

where η_{1i} , η_{3i} , η_{5i} are the intercept factors and η_{2i} , η_{4i} , η_{6i} are slope factors for the trajectory of X , M , and Y , respectively, varying across individuals indicated by the subscript i . In the previous equations, π_{0m} , π_{0y} , π_{1m} , and π_{1y} are the regression intercepts of the growth factors and e_{3i} , e_{4i} , e_{5i} , and e_{6i} are residuals of the growth factors. Equations 21.85 and 21.87 describe the intercept factors of M and Y , respectively, as functions of the intercept factors of X and M . These directional paths in Figure 21.17 are based on the causal theory of X to M to Y . If the first measures of X , M , and Y are obtained simultaneously, they can be specified as covariances. Equation 21.86 models the slope factor of M as a function of the intercept and the slope factors of

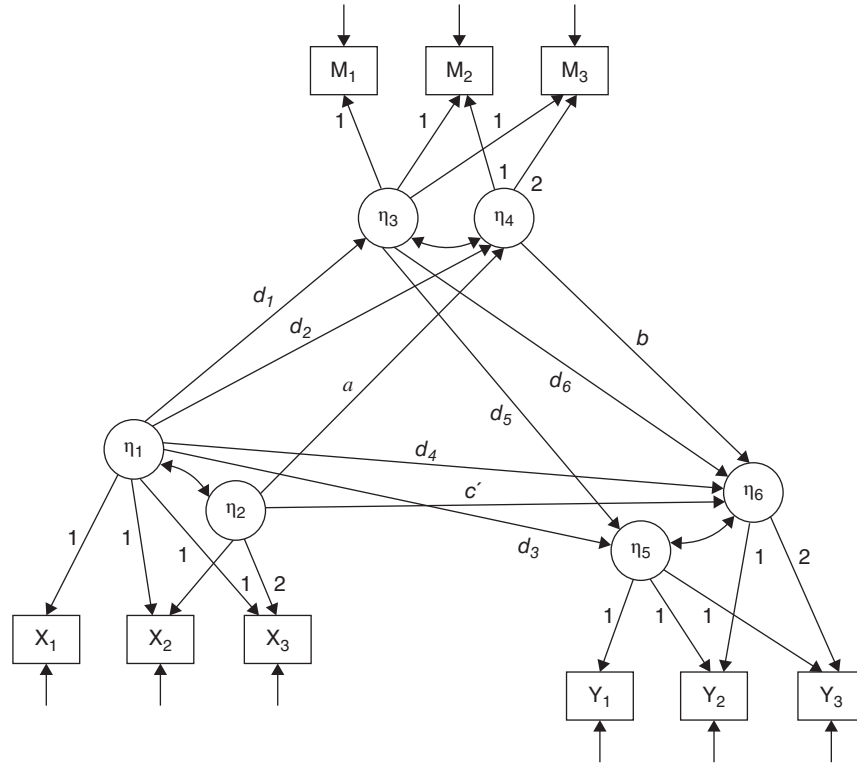


Figure 21.17 A latent growth curve mediation model across three waves.

X , while equation 21.88 specifies the slope factor of Y predicted by the intercept and the slope factors of X and M . Although there are several possible indirect effects that can be of interest in this model, the main focus is on the relations among the slope factors, as the benefit of the LGC approach is the capability of modeling mediation in terms of within-individual changes in X , M , and Y . The coefficient a represents the effect of the growth rate of X on the growth rate of M . The coefficient b represents the effect of the growth rate of M on the growth rate on Y . The product of the two coefficients (i.e., ab) provides the estimate of the indirect effect that is based on the within-individual changes, determined by the extent to which the change in X affects the change in M (a) and the extent to which the change in M affects the change in Y (b). The standard error of the indirect effect and its standard error can be obtained in the same way described earlier and used for the significance test of the indirect effect.

We fit a series of simple LGC mediation models to ECLS-K data from all 5 waves. Though preliminary models suggested that growth models did not fit the ADHD symptoms, interpersonal skill, and internalizing behavior data well, we nonetheless proceeded with the mediation models for the sake of illustration, specifying linear trajectories for the X , M , and Y variables. Unsurprisingly, we encountered

frequent convergence problems; estimation of the variance of the slope of the mediator trajectory seemed particularly problematic. This variance was estimated as negative in the $\text{LEARN} \rightarrow \text{INTERP} \rightarrow \text{INTERN}$ model and was very close to zero in the $\text{LEARN} \rightarrow \text{READ} \rightarrow \text{INTERN}$ and $\text{EXTERN} \rightarrow \text{MATH} \rightarrow \text{INTERN}$ models. It was necessary to constrain this parameter (and its covariance with the intercept of the mediator trajectory) to zero to achieve convergence in the $\text{LEARN} \rightarrow \text{MATH} \rightarrow \text{INTERN}$ and $\text{EXTERN} \rightarrow \text{INTERP} \rightarrow \text{INTERN}$ models. The $\text{EXTERN} \rightarrow \text{READ} \rightarrow \text{INTERN}$ model did not converge, even with this constraint. Both the CFI and RMSEA indicate poor fit for all models. Tests of the mediated effects involving predictor, mediator, and outcome slopes are reported in Table 21.10. For all models that converged, the estimate of the a path relating the slope of the predictor to the slope of the mediator was significant, though the relation between the slopes of LEARN and MATH was counterintuitively negative. Only two b path estimates were significant: the slope of READ was negatively related to the slope of INTERN , and the slope of INTERP was counterintuitively positively related to the slope of INTERN . Tests of the ab estimates of mediated effects show significant mediation in the $\text{LEARN} \rightarrow \text{READ} \rightarrow \text{INTERN}$ and $\text{EXTERN} \rightarrow \text{INTERP} \rightarrow \text{INTERN}$ models, but the

TABLE 21.10 Coefficients and Standard Errors in an LGM Mediation Model

<i>X</i>	<i>M</i>	<i>Y</i>	<i>a</i>	<i>b</i>	Med
LEARN	INTERP	INTERN	1.0538*** (0.0229)	−0.0501 (0.0335)	−0.0528 (0.0353)
LEARN	READ	INTERN	0.1491*** (0.0087)	−0.1429*** (0.0307)	−0.0213*** (0.0047)
LEARN	MATH	INTERN	−0.1135*** (0.0143)	−.01165 (0.1490)	0.0013 (0.0169)
EXTERN	INTERP	INTERN	−1.0272*** (0.0242)	3.7495*** (0.0507)	−3.851*** (0.1046)
EXTERN	READ	INTERN	Did not converge.		
EXTERN	MATH	INTERN	−0.1056*** (0.0113)	0.0796 (0.0409)	−0.0084 (0.0044)

mediated effect in the latter model is in a counterintuitive direction. Though LGC mediation models can be very useful in explicating mediation effects in longitudinal data, this particular technique does not seem appropriate for this particular data, as a growth framework is not well suited to capture change over time in most of the variables. As mentioned earlier, it is required to inspect the pattern of change in the study variables carefully before putting the trajectories together in a comprehensive LGC mediation model.

When the measures of *X*, *M*, and *Y* across waves are concurrent, the relations between the slope factors in the mediation model may be interpreted as correlational because the growth rates are estimated based on the measures across the same time span. However, this is less problematic than purely cross sectional data, because the growth rates are estimated based on the entire time span that the data cover and the effects of the changes in the predictors at earlier waves would be reflected in the changes in the outcome variables at later waves. An alternative method is to estimate growth trajectories in two-stage piecewise growth curve models (Raudenbush & Bryk, 2002; Khoo, 2001), where distinct growth trajectories are specified for early and later phases and the time ordering of the growth trajectories of *X*, *M*, and *Y* can be modeled accordingly. The indirect effects then can be estimated for different phases, such as the indirect effect via the earlier growth of *M* on the earlier growth of *Y*, the indirect effect via the earlier growth of *M* on the later growth of *Y*, and the indirect effect via the later growth of *M* on the later growth of *Y*.

Estimating LGC Mediation as a Multilevel Model

The LGC mediation models discussed in the previous section were estimated using structural equation modeling techniques. The SEM framework is ideally suited for such analyses, as its ability to estimate a number of simultaneous equations allows for the estimation of

predictive and outcome effects of the same variable within a single analysis. For example, in a mediational analysis, the mediator serves as an outcome variable when examining the *a* path and as a predictor variable when examining the *b* path. It is relatively simple within this framework to have one arrow pointing to a variable (i.e., that variable is an outcome) and another pointing from that variable (i.e., that variable is a predictor of another variable) in an SEM diagram and to create the appropriate equations to estimate both effects simultaneously.

The mediated effect involving latent slope parameters in Figure 21.17 could potentially also be tested within a multilevel modeling framework. The difficulty here is that latent growth in *X* is a predictor of latent growth in *M*, and that latent growth in *M* is a predictor of latent growth in *Y*. In a single multilevel model, it is simple to allow a variable to predict the growth parameter—just include that predictor in the L2 equation predicting that parameter—but there is no obvious way to expand the model to allow the growth parameter itself to predict another variable. However, after running a multilevel model, a researcher can obtain Empirical Bayesian (EB) estimates of growth parameters for each individual. These estimates combine individual-specific information from a single subject's own data and the normative trajectory for the entire sample to produce stable and reliable estimates of person-specific effects (Candel & Winkens, 2003). The EB estimates of intercepts and slopes of the *X* variable can be entered as L2 predictors of the growth parameters in a multilevel growth analysis of the mediator variable to produce estimates of the d_1 , d_2 , and *a* parameters in Figure 21.17. Similarly, EB estimates of intercepts and slopes can be obtained when running the growth analysis for the mediator variable, and these (along with the EB estimates of intercepts and slopes from the first analysis) can be entered as L2 predictors of the intercept and slope parameters in the multilevel growth analysis of the outcome variable to produce the

remaining coefficients in Figure 21.17, including the b coefficient that relates the mediator slope to the outcome slope. Conducting multiple multilevel analyses seems a bit less elegant than the single comprehensive SEM, but equivalent estimates are produced, and thus longitudinal mediation can be assessed in either framework.

Another possibility for recasting the LGC mediation model in the multilevel framework involves including all three variables (X , M and Y) in a single multivariate multilevel model (Hox, 2010; Raudenbush & Bryk, 2002). One way of conceptualizing such a model is as a three-level framework, with the variables distinguished at L1, repeated measures at L2, and individual subjects at L3. With three measurement occasions as shown in Figure 21.17, the data set would be organized with nine lines of data per subject, one for each of the X , M and Y variables at each of the three time points. The actual X , M , and Y values would be entered into a single generically named variable, V . Three dummy variables (D_1 , D_2 , and D_3) would be created to identify which observations were associated with each of the three variables, and these dummies, in conjunction with a variable indicating the time point of measurement, would uniquely identify each of the nine unique values.

The L1 model would consist of the three dummy variables, two of which will be zero for any given observation:

$$\text{L1: } V_{v ti} = \pi_{1 ti} D_{1 ti} + \pi_{2 ti} D_{2 ti} + \pi_{3 ti} D_{3 ti} \quad (21.89)$$

for variable v at time point t for subject i . Notice that there is no intercept and no error term in this equation. An intercept would imply that predictors could affect all observations equally, and we wish to keep X , M and Y distinct in this model. An error term is unnecessary because this level of the model simply sorts the variables by type, which should involve no error.

The L2 models would be:

$$\begin{aligned} \text{L2: } \pi_{1 ti} &= \beta_{10 i} + \beta_{11 i} \text{time}_{ti} + r_{1 ti} \\ \pi_{2 ti} &= \beta_{20 i} + \beta_{21 i} \text{time}_{ti} + r_{2 ti} \\ \pi_{3 ti} &= \beta_{30 i} + \beta_{31 i} \text{time}_{ti} + r_{3 ti} \end{aligned} \quad (21.90)$$

and at L3:

$$\begin{aligned} \text{L3: } \beta_{10 i} &= \gamma_{100} + u_{10 i} & \beta_{11 i} &= \gamma_{110} + u_{11 i} \\ \beta_{20 i} &= \gamma_{200} + u_{20 i} & \beta_{21 i} &= \gamma_{210} + u_{21 i} \\ \beta_{30 i} &= \gamma_{300} + u_{30 i} & \beta_{31 i} &= \gamma_{310} + u_{31 i} \end{aligned} \quad (21.91)$$

These equations at each level can be algebraically combined to give the single-equation form of the model:

$$\begin{aligned} V_{v ti} &= D_{1 ti}(\gamma_{100} + \gamma_{110} \text{time}_{ti} + r_{1 ti} + u_{10 i} + u_{11 i} \text{time}_{ti}) \\ &+ D_{2 ti}(\gamma_{200} + \gamma_{210} \text{time}_{ti} + r_{2 ti} + u_{20 i} + u_{21 i} \text{time}_{ti}) \\ &+ D_{3 ti}(\gamma_{300} + \gamma_{310} \text{time}_{ti} + r_{3 ti} + u_{30 i} + u_{31 i} \text{time}_{ti}) \end{aligned} \quad (21.92)$$

This is exactly the same as running three separate multilevel growth models on the X , M , and Y variables, but all three equations are combined into a single analysis. The advantage of the single analysis is that the L3 variance-covariance matrix of the u 's will be 6×6 and will contain off-diagonal elements that represent associations between growth in X and growth in M (a correlational version of the a path in Figure 21.17) and between growth in M and growth in Y (a correlational version of the b path in Figure 21.17).

Because there is no error term at L1, this multivariate analysis could alternately be specified as a two-level model. This is actually the more practical specification, as it is quite difficult to suppress the L1 error term or constrain it to zero in most multilevel software packages. In separate equation form, the L1 model would be

$$\begin{aligned} \text{L1: } V_{ti} &= \beta_{10 i} D_{1 ti} + \beta_{11 i} D_{1 ti} \text{time}_{ti} + \beta_{20 i} D_{2 ti} \\ &+ \beta_{21 i} D_{2 ti} \text{time}_{ti} + \beta_{30 i} D_{3 ti} \\ &+ \beta_{31 i} D_{3 ti} \text{time}_{ti} + r_{1 ti} + r_{2 ti} + r_{3 ti} \end{aligned}$$

Again, the specification contains no explicit intercept. The three dummy variables represent intercepts for their specific variables. Moreover, the time variable is entered in interaction with each of the dummy variables. These interactions represent the three slopes associated with the X , M , and Y variables. In addition, this equation includes three distinct error terms. This model could be estimated, for example, in SAS Proc Mixed (SAS Institute, 2014), which has the facility to estimate multiple L1 error terms corresponding to the distinct levels of a L1 variable with the `group =` option. A single variable distinguishing X , M , and Y with uniquely coded values would need to be created to accomplish this. The L2 specification for this model would be identical to the L3 specification of the three-level multivariate model and thus the single-equation forms of both models are identical.

In the general multivariate multilevel model, associations among slopes are correlational and thus nondirectional. Mediation analyses, however, typically involve

directional hypotheses ($X \rightarrow M$ and $M \rightarrow Y$). Examining such directional effects in the longitudinal mediation model requires the ability to include latent variables on the predictor side of the equation. The HLM program incorporates just such facility into the HLM2, HLM3, MLM, and MLM2 modules (Raudenbush, Bryk, Cheong, Congdon, & du Toit, 2011). The HLM manual provides an example in which a latent intercept is used to predict a latent slope. The method described in the manual could easily be expanded to simultaneously estimate growth parameters for more than one outcome variable and allow the slope of one to predict the slope of another. However, the latent variable modeling window in HLM currently allows each latent variable to be used as either a predictor or an outcome, but not both. So in a multivariate model including multiple measurements of X and M , this program could allow the slope of X to predict the slope of M , producing a coefficient and standard error for the a path in Figure 21.17. However, because M cannot simultaneously serve as both an outcome (on the a path from the X slope) and a predictor (on the b path to the Y slope), it would still be necessary to run a second multivariate multilevel analysis with multiple measurements of X , M , and Y to allow the slopes of X and M to predict the slope of Y to produce the b and c' coefficients in Figure 21.17. So directional coefficients can be produced, and a mediated effect can be computed and tested from these, but incorporating the entire process into a single analysis will need to wait for further development of multilevel software.

Autoregressive Latent Trajectory Mediation Models

Integrating the unique strengths of the autoregressive models and the LGC models, Bollen and his colleagues (Bollen & Curran, 2004; Bollen & Zimmer, 2010) proposed the autoregressive latent trajectory (ALT) model. The ALT model parameterizes autoregressive relations between the repeated measures as in the autoregressive models, while simultaneously estimating the growth trajectories based on the pattern of changes across time. The multivariate application of the ALT model allows for parameterization of the autoregressive relations within each variable, the lagged relations between variables at specific measurement occasions, and the relations between the within-individual changes in those variables. Applying ALT models to investigation of mediation, researchers may be able to capture time-specific indirect effects as well as indirect effects based on the growth trajectories that are estimated across the entire time span the data cover.

In the ALT model, the measures of X , M , and Y at time 1 are treated as predetermined and including this predetermined time 1 measure takes care of the biases resulting from omitted variables prior to the time span the data cover. Individual trajectories in the ALT model are estimated based on the measures from time 2 to time t . The autoregressive relations within each variable are specified between the repeated measures from time 1 to time t . Thus, the measure at time t for $t \geq 2$ is specified as the function of the variable itself measured at time $t - 1$ and the trajectory components (i.e., intercept and slope factors) estimated based on the variable measured from time 2 to time t . For example, X_2 is determined by X_1 and the intercept factor, while X_3 is determined by X_2 , the intercept factor, and the slope factor. The intercept factor in the ALT model captures the portion of the measure at time 2 that remains unexplained by the time 1 measure and the slope factor represents the growth rate estimated based on time 2 to time t measures. For an ALT model to be identified, four or more waves of data are needed (Bollen & Curran, 2006). Details about the ALT model specifications can be found elsewhere (Bollen & Curran, 2004; Bollen & Zimmer, 2010).

The model in Figure 21.18 depicts an example of the ALT mediation model, where X , M , and Y are measured across four time points. The lag-1 effects between X , M , and Y at each wave are shown in the middle of the figure relating X_{t-1} to M_t to Y_{t+1} . The mediation relations among the individual trajectories are represented by the dotted lines connecting directionally between the intercept factors and the slope factors of X , M , and Y . The time specific indirect effects are estimated based on the lagged effects between the repeated measures of X , M , and Y (e.g., $X_1 \rightarrow M_2$ and $M_2 \rightarrow Y_3$) as in the autoregressive mediation models and the indirect effect based on the within-individual changes in X , Y , and M is estimated in the relations between the slope factors (i.e., $\eta_2 \rightarrow \eta_4 \rightarrow \eta_6$) as in the LGC mediation models. These two different types of indirect effects are independent of each other, as the autoregressive and the lagged effects are net of individual trajectory components in the ALT models. As expected, model specification can be complicated and it may take several steps to determine the appropriate final model; however, the ALT models can provide unique opportunities to simultaneously estimate and separate out the time-specific and the within-individual change based indirect effects in one comprehensive model. Given the difficulties encountered in estimating the LGC mediation models in the ECLS-K data, we did not explore ALT models for our particular example.

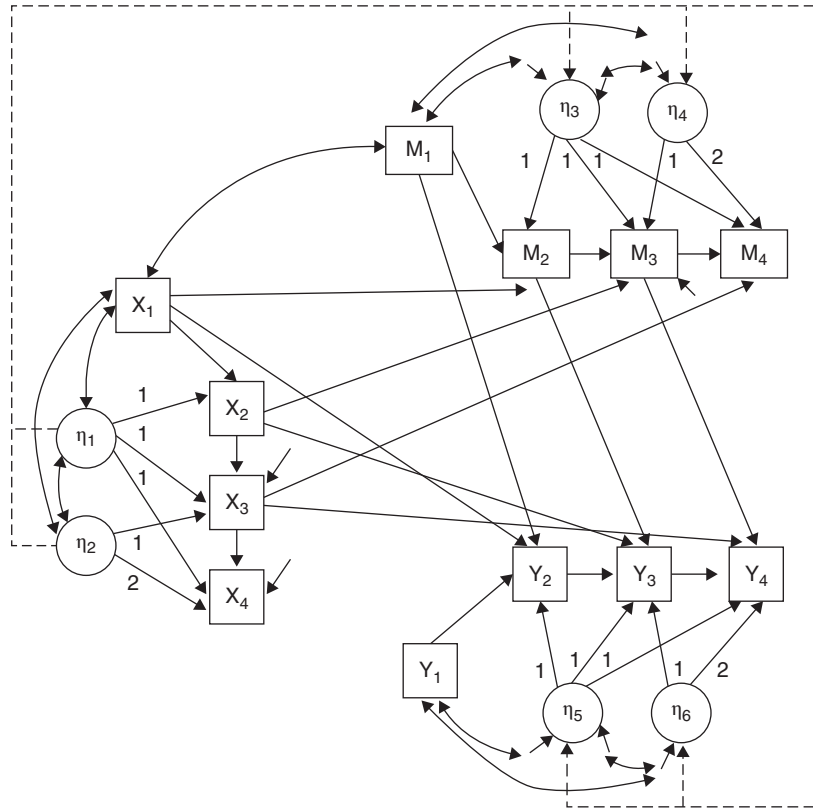


Figure 21.18 An autoregressive latent trajectory mediation model across four waves.

Latent Change Score Models for Moderation and Mediation

Before describing moderation and mediation in latent variable difference score models, a brief word is needed about manifest variable difference score models for longitudinal data. With two waves of data, MacKinnon (2008) suggests forming change scores for X , M , and Y , substituting the difference scores into the equations for the single mediator, and then testing for mediation using one of the cross sectional tests of mediation mentioned earlier. For three or more waves of data, difference scores may be computed and placed into an OLS autoregressive model to test for mediation. Although the steps for fitting the model and testing for mediation are the same as in the autoregressive mediation model, the interpretation of the results is quite different. Instead of a variable at one wave influencing another variable at a later wave, now it is the difference in scores on a variable across two waves influencing the difference in the scores of another variable across two later waves.

With a categorical moderator variable, the simplest method to test for moderation in a LCS model would be to run a multiple group analysis and compare the parameter estimates in the different groups. Depending on the model

constraints, this would test whether the proportional or constant change between groups was different or could even test whether one group changed proportionally while another changed constantly or did not change. For observed continuous time-invariant predictors, the latent change score parameters can be regressed onto both predictors and the interaction term. Interactions between latent time-invariant predictors or between a latent time-invariant predictor and a latent change score parameter are possible as well, but require more advanced methods; Marsh, Wen, and Hau (2006) provided an overview of some of these methods.

There are several ways an LCS model can be incorporated into a mediation model. Figure 21.19 shows one example of a LCS longitudinal mediation model where the mediated effect is composed of the slopes of the individual LCS models. That is, the slope of the change in X predicts the slope of the change in M , which in turn affects slope of the change in Y . This is comparable to the parallel process LGC model described by Cheong et al. (2003) because the processes are occurring at the same time. If the processes were measured after one another, the then a serial process LCS model could be specified.

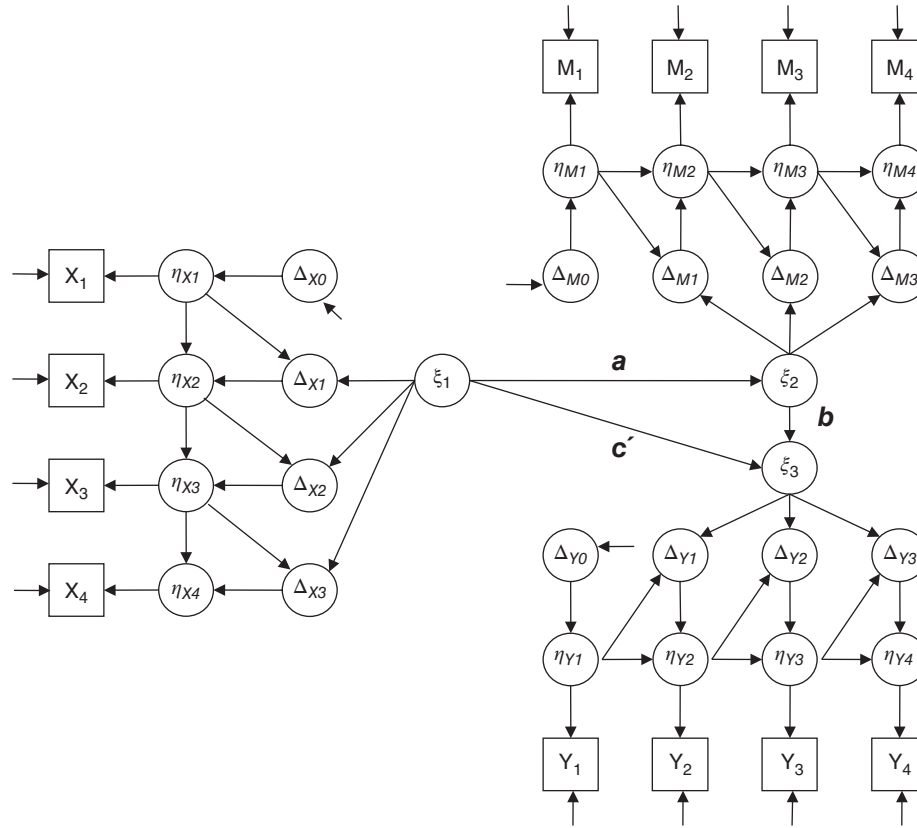


Figure 21.19 Parallel process latent change score mediation model.

MacKinnon (2008) described a different LCS longitudinal mediation model, shown in Figure 21.20. In this model, rather than have the slopes of the change in individual variables predict one another, the error-free true score η variables from one variable predict the latent change scores, Δ , in the other variable. For instance, η_{x1} predicts Δ_{m1} and η_{m2} predicts Δ_{y3} . This model is closer to an autoregressive model, because the variables only affect other variables at later waves. Note that the mean structure has been omitted from Figures 21.19 and 21.20 for ease of presentation. Additionally, an LCS model with a paths linking change in the predictor to change in the mediator (i.e., from Δ'_x s to Δ'_m s) and b paths linking change in the mediator to change in the outcome (i.e., from Δ'_m s to Δ'_y s) could plausibly be used to test longitudinal mediation.

The process of fitting the LCS mediation models in Figures 21.19 and 21.20 is similar to fitting a LGC mediation model (MacKinnon, 2008). The first step is to fit the individual LCS models for X , M , and Y . This allows the researcher to specify the best fitting LCS model for each variable, such as determining whether a dual change or

proportional change model fits better. The second step is to add the mediation paths using the structural model. As in the LGC mediation models, other paths may be appropriate, such as a path from η_{x1} to the Δ_{y2} , which is similar to the lag 1 direct effects from the autoregressive model. The Chi-square goodness of fit test, the RMSEA, and the CFI may be used to assess overall model fit.

Testing for mediation can be accomplished in several ways. First, the joint significance test could be used by examining the significance of each of the individual paths. Second, specific indirect paths can be examined by using a MODEL INDIRECT command in the Mplus program and requesting bootstrap confidence intervals. As with the autoregressive models and LGC, the overall indirect effect can be examined using EQS, SAS PROC CALIS, and LISREL, but only Mplus will test specific indirect effects for significance. For in depth examples of fitting LCS models with LISREL see McArdle (2001) and Ferrer and McArdle (2003).

We fit LCS mediation models to the first, third, and fifth grade ECLS-K data. Table 21.11 presents the results of the parallel process analysis similar to Figure 21.19. All six

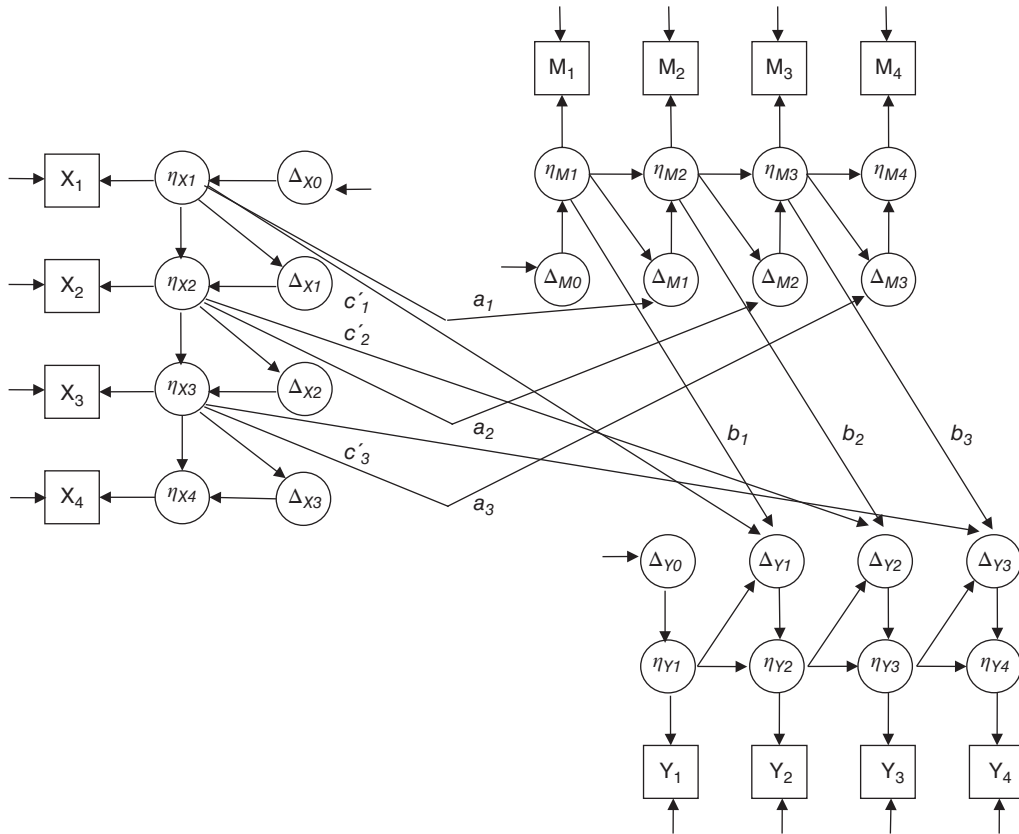


Figure 21.20 Alternate parallel process latent change score mediation model. Source: Adapted from MacKinnon (2008).

TABLE 21.11 Coefficients and Standard Errors in LCS Parallel Process Mediation Models

<i>X</i>	<i>M</i>	<i>Y</i>	<i>a</i>	<i>b</i>	Med
LEARN	INTERP	INTERN	1.1302*** (0.0231)	−0.6494*** (0.0323)	−0.7340*** (0.0394)
LEARN	READ	INTERN	0.2639*** (0.0245)	0.6345*** (0.0667)	0.1674*** (0.0235)
LEARN	MATH	INTERN	0.2226*** (0.0130)	−0.0155 (0.0153)	−0.0035 (0.0034)
EXTERN	INTERP	INTERN	1.6164*** (0.1587)	−0.0035 (0.1162)	−0.0057 (0.1878)
EXTERN	READ	INTERN	−0.2297*** (0.0201)	0.2966*** (0.0302)	−0.0681*** (0.0091)
EXTERN	MATH	INTERN	−0.1864*** (0.0226)	0.1590*** (0.0130)	−0.0296*** (0.0043)

models included significant j and k coefficients for X , M , and Y , indicating both proportional and constant change over time for all constructs. RMSEA and CFI values for all models indicated less than ideal fit, particularly for models with interpersonal skills as the mediator. The tests of mediation in Table 21.11 indicate that the slopes of change in interpersonal skills and reading performance

significantly mediated the association between the slope of change in approaches to learning and internalizing behavior. Also, the slopes of change in both academic performance variables mediated the association between the slope of change in externalizing behavior and the slope of change in internalizing behavior. The direction of the mediated effect, however, was counterintuitive for three of the four mediated pathways.

A second series of LCS mediation models was fitted to the first through fifth grade ECLS-K data in which true scores predicted latent change as in Figure 21.20. The results are reported in Table 21.12. Because three time points were included in the analysis, two distinct a and two distinct b paths were estimated in each model. a_1 and b_1 examine effects of first grade true scores on latent change from first to third grade, and a_2 and b_2 examine effects of third grade true scores on latent change from third to fifth grade. RMSEA and CFI values indicated less than ideal fit in all models, especially for those involving the interpersonal skills mediator. The tests of mediation in Table 21.12 indicate a significant $a_1 b_1$ estimate of mediation

TABLE 21.12 Coefficients and Standard Errors in LCS Mediation Models with Paths from True Scores to Latent Change

<i>X</i>	<i>M</i>	<i>Y</i>	a_1	a_2	b_1	b_2	Med $_{a1a2}$	Med $_{a1b2}$	Med $_{a2b2}$
LEARN	INTERP	INTERN	1.0144*** (0.0435)	0.9676*** (0.0469)	1.1118*** (0.1107)	0.0227 (0.0445)	1.1278*** (0.1223)	0.0230 (0.0452)	0.0220 (0.0431)
LEARN	READ	INTERN	0.0004 (0.0059)	0.0770** (0.0057)	0.0360 (0.0317)	0.0813 (0.0437)	0.0001 (0.0002)	0.0001 (0.0005)	0.0063 (0.0034)
LEARN	MATH	INTERN	-0.0170** (0.0066)	0.0143** (0.0068)	-0.0008 (0.0318)	0.0764** (0.0310)	0.0001 (0.0005)	-0.0013 (0.0007)	.0011 (.0007)
EXTERN	INTERP	INTERN	-0.9607*** (0.0179)	-0.7689*** (0.0253)	-0.0449** (0.0200)	0.3928** (0.1272)	0.0431** (0.0192)	-0.3774** (0.1224)	-0.3020** (0.0983)
EXTERN	READ	INTERN	-0.0053 (0.0061)	-0.0374*** (0.0064)	0.0286 (0.0268)	0.1117** (0.0348)	-0.0002 (.0002)	-0.0006 (.0007)	-0.0042** (.0015)
EXTERN	MATH	INTERN	0.0057 (0.0070)	-0.0137 (0.0074)	-0.0033 (0.0266)	0.0998*** (0.0266)	-0.0001 (.0002)	0.0006 (.0007)	-0.0014 (0.0008)

for LEARN $\rightarrow$ INTERP $\rightarrow$ INTERN, but the temporally ordered a_1b_2 estimate was nonsignificant. All three possible estimates of mediated effects for EXTERN $\rightarrow$ INTERP $\rightarrow$ INTERN were significant. The a_2b_2 estimate in the EXTERN $\rightarrow$ READ $\rightarrow$ INTERN model was the only significant mediated effect involving the academic performance variables.

Moderation and Mediation in Exponential Decay Models

Similar to standard multilevel models, moderation in an exponential decay multilevel growth model is accomplished by adding the moderating variable and appropriate interaction term at the correct level. If the moderator is a time-varying predictor, Z_{ti} , and interacts with another time-varying predictor, X_{ti} , that is not time, then both the predictors and the interaction term, $X_{ti}Z_{ti}$, would be added at L1 in equation 21.35.

$$\begin{aligned}
 \text{L1: } Y_{ti} &= \beta_{2i} + (\beta_{0i} - \beta_{2i}) \exp[\beta_{1i}t] \\
 &\quad + \beta_{3i}X_{ti} + \beta_{4i}Z_{ti} + \beta_{5i}Z_{ti}X_{ti} + r_{ti} \\
 \text{L2: } \beta_{0i} &= \gamma_{00} + u_{0i} \\
 \beta_{1i} &= \gamma_{10} + u_{1i} \\
 \beta_{2i} &= \gamma_{20} + u_{2i} \\
 \beta_{3i} &= \gamma_{30} + u_{3i} \\
 \beta_{4i} &= \gamma_{40} + u_{4i} \\
 \beta_{5i} &= \gamma_{50} + u_{5i}
 \end{aligned} \tag{21.93}$$

Because of the nonlinear relation between time and the outcome variable, a linear interaction between time and a time-varying covariate is difficult to conceptualize and will not be discussed for the multilevel or latent variable exponential decay models.

If the moderator is a time-invariant predictor, Z_i , of one of the growth parameters, for example β_0 , and interacts with another time-invariant predictor of the same growth parameter, X_i , then both the predictors and the interaction, X_iZ_i , between these predictors would be added to the corresponding L2 equations in equation 21.35.

$$\begin{aligned}
 \text{L1: } Y_{ti} &= \beta_{2i} + (\beta_{0i} - \beta_{2i}) \exp[\beta_{1i}t] + r_{ti} \\
 \text{L2: } \beta_{0i} &= \gamma_{00} + \gamma_{01}X_i + \gamma_{02}Z_i + \gamma_{03}X_iZ_i + u_{0i} \\
 \beta_{1i} &= \gamma_{10} + u_{1i} \\
 \beta_{2i} &= \gamma_{20} + u_{2i}
 \end{aligned} \tag{21.94}$$

L2 interactions may be of more interest for the exponential decay model as they test whether the effect of a time-invariant predictor on the initial value, rate of change, and asymptote value is dependent on the value of the moderator. For the latent variable exponential decay model with a time-invariant categorical moderator variable, the simplest method to test for moderation would be to run a multiple group analysis and compare the parameter estimates in the different groups. This would test whether the initial value, rate of change, and asymptote value were the same for the different groups. For observed continuous time-invariant predictors, the latent growth parameters can be regressed onto both the predictors and the interaction term. As with the LCS model, interactions between latent time-invariant predictors or between a latent time-invariant predictor and a latent growth parameter are possible for the latent exponential decay model, but require more advanced methods described by Marsh et al. (2006).

If one of the variables is a time-varying predictor, X_{ti} , and the other is a time-invariant predictor, Z_i , then

a cross-level interaction can be estimated by including the time-varying predictor at L1 and including the time-invariant variable as a predictor of the time-varying variable in the L2 equation in equation 21.35.

$$\begin{aligned}
 \text{L1: } Y_{ti} &= \beta_{2i} + (\beta_{0i} - \beta_{2i}) \exp[\beta_{1i}t] + \beta_{3i}X_{ti} + r_{ti} \\
 \text{L2: } \beta_{0i} &= \gamma_{00} + u_{0i} \\
 \beta_{1i} &= \gamma_{10} + u_{1i} \\
 \beta_{2i} &= \gamma_{20} + \gamma_{21}Z_i + u_{2i} \\
 \beta_{3i} &= \gamma_{30} + \gamma_{31}Z_i + u_{3i}
 \end{aligned} \tag{21.95}$$

This also allows for interactions between time-invariant predictors and time if Z_i is added as a predictor of β_1 . Note that the time-invariant predictor must also be included as an L2 predictor of β_2 in order for the first-order effect Z_i to be estimated in the model, which is necessary for the correct estimation of the interaction term $X_{ti}Z_i$.

Though not technically an interaction, differential effects based on the relations between the growth parameters can be examined using the off-diagonal elements of the covariance matrix of the random effects, τ . For example, if higher initial values were associated with higher asymptote values, then τ_{02} would be positive and significantly different from zero. The same association can be examined in the latent variable exponential decay using the covariances in the Φ matrix.

The first step in constructing an exponential decay multilevel mediation model is to rewrite the equations from the linear single mediator model in the form of equation 21.35. When M and Y decay exponentially and X is a continuous variable, X is added at L1 of equation 21.35 and the effect of X on M becomes

$$\begin{aligned}
 \text{L1: } M_{ti} &= \beta_{2i} + (\beta_{0i} - \beta_{2i}) \exp[\beta_{1i}t] + a_iX_{ti} + r_{ti} \\
 \text{L2: } \beta_{0i} &= \gamma_{00} + u_{0i} \\
 \beta_{1i} &= \gamma_{10} + u_{1i} \\
 \beta_{2i} &= \gamma_{20} + u_{2i} \\
 a_i &= \gamma_{30}
 \end{aligned} \tag{21.96}$$

where γ_{00} , γ_{10} , and γ_{20} represent the mean values across subjects for the initial value, rate of change, and asymptote for M partialling out the effect of X , respectively, and γ_{30} is the effect of X partialling out the effect of time (i.e., the natural exponential decay of M). Because M is also

continuous, X and M are added at L1 of equation 21.35 and the effect of X and M on Y becomes

$$\begin{aligned}
 \text{Level1: } Y_{ti} &= \beta_{2i} + (\beta_{0i} - \beta_{2i}) \exp[\beta_{1i}t] + b_iM_{ti} \\
 &\quad + c'_iX_{ti} + r_{ti} \\
 \text{Level2: } \beta_{0i} &= \gamma_{00} + u_{0i} \\
 \beta_{1i} &= \gamma_{10} + u_{1i} \\
 \beta_{2i} &= \gamma_{20} + u_{2i} \\
 b_i &= \gamma_{30} \\
 c'_i &= \gamma_{40}
 \end{aligned} \tag{21.97}$$

where γ_{00} , γ_{10} , and γ_{20} represent the mean values across subjects for the initial value, rate of change, and asymptote for Y partialling out the effects of X and M , respectively, γ_{30} is the effect of M partialling out the effects of time (i.e., the natural exponential decay of Y) and X , and γ_{40} is the effect of X partialling out the effects of time and M . Krull and MacKinnon (2001) call this a $1 \rightarrow 1 \rightarrow 1$ model because all three variables occur at L1.

Here the three mediation parameters, a_i , b_i , and c'_i , have all been treated as fixed effects, meaning that the mediation effects do not vary between individuals. Adding random effects for these three paths would not change the interpretations of the coefficients, but it would indicate that the effect differs among individuals, an indication of the presence of moderation. For example, if an L2 predictor was added that predicted b_i , such as a baseline value of M , this would create a cross-level interaction between that predictor and M , meaning the predictor of b_i is a moderator of the effect of M on Y . For more information on moderation of a mediated effect see Fairchild and MacKinnon (2009); for more information on mediation at different levels see Bauer et al. (2006), Kenny et al. (2003), Krull and MacKinnon (2001), and Preacher et al. (2010).

When X is a time-invariant variable, such as random assignment to a treatment, instead of entering X as an L1 covariate, X must be entered into the multilevel model at L2. The multilevel equations for the model of X on M then change from equation 21.95 to

$$\begin{aligned}
 \text{L1: } M_{ti} &= \beta_{2i} + (\beta_{0i} - \beta_{2i}) \exp[\beta_{1i}t] + r_{ti} \\
 \text{L2: } \beta_{0i} &= \gamma_{00} + a_0X_i + u_{0i} \\
 \beta_{1i} &= \gamma_{10} + a_1X_i + u_{1i} \\
 \beta_{2i} &= \gamma_{20} + a_2X_i + u_{2i}
 \end{aligned} \tag{21.98}$$

Assuming X is a dichotomous dummy coded variable, then γ_{00} , γ_{10} , and γ_{20} , are estimates of the initial value, rate of change, and asymptote of M for the group coded 0, respectively, and a_0 , a_1 , and a_2 are the differences between the two groups in X such that $\gamma_{00} + a_0$, $\gamma_{10} + a_1$, and $\gamma_{20} + a_2$ are the initial value, rate of change, and asymptote of M for the group coded 1, respectively.

The effects of M and X on Y change from equation 21.96 to

$$\begin{aligned} \text{L1: } Y_{ti} &= \beta_{2i} + (\beta_{0i} - \beta_{2i}) \exp[\beta_{1i}t_i] + b_i M_{ti} + r_{ti} \\ \text{L2: } \beta_{0i} &= \gamma_{00} + c'_0 X_i + u_{0i} \\ \beta_{1i} &= \gamma_{10} + c'_1 X_i + u_{1i} \\ \beta_{2i} &= \gamma_{20} + c'_2 X_i + u_{2i} \\ b_i &= \gamma_{30} \end{aligned} \quad (21.99)$$

Again assuming X is a dichotomous dummy coded variable, γ_{00} , γ_{10} , and γ_{20} , are estimates of the initial value, rate of change, and asymptote of Y for the group coded 0, partialling out the effect of M , respectively, c'_0 , c'_1 , and c'_2 are the differences between the two groups in X partialling out the effect of M , such that $\gamma_{00} + c'_0$, $\gamma_{10} + c'_1$, and $\gamma_{20} + c'_2$ are the asymptote value, initial value, and rate of change of Y for the group coded 1, respectively, and γ_{30} is the effect of M partialling out the effect of time (i.e., the natural exponential decay of Y) and the effect of X on the exponential decay of Y . Krull and MacKinnon (2001) call this a $2 \rightarrow 1 \rightarrow 1$ model because X is at L2 while M and Y are at L1.

Adding M as a time-varying predictor in at L1 so that the model estimates the relation between M and Y at the same point in time violates the temporal precedence assumption. An alternate method would be to introduce a lag such that M predicts Y at later time points. With a lag of one wave, M_{it} in equation 21.98 would change to $M_{(t-1)i}$, so that each measurement of M predicts the measurement of Y at the next time point. Then b_i would be the average relation across time between M at time t and Y at time $(t + 1)$, partialling out the effect of time. One issue with lags is that M must be measured prior to Y . Also, although the model makes no assumption that the measurements of M or Y are equally spaced, when the relation between M and Y is lagged, if the measurements are not equally spaced, b_i will represent the average effect across different amounts of time. For example, if M and Y were measured at 1, 3, and 12 months, introducing a lag of one wave would mean that $M_{1\text{month}}$ would predict

$Y_{3\text{months}}$, a lag of 2 months, while $M_{3\text{months}}$ would predict $Y_{12\text{months}}$, a lag of 9 months, an undesirable situation.

The value of M immediately following an intervention could also be entered as an L2 predictor of the rate of change and asymptote of Y . If X had a significant effect on the initial value of M , which in turn predicted the rate of change and asymptote of Y , then the effect of X on the rate of change and asymptote of Y would be mediated through its initial effect on M . This does not satisfy temporal precedence since the initial value of Y is used to estimate the rate of change and asymptote of Y , but it does make a stronger case for mediation than in equation 21.98.

Using the exponential decay model shown in equations 21.95 and 21.96, when X and M are added to the model at Level 1, mediation can be tested using the joint significance test by testing a and b for significance. The mediated effect $a_i b_i$ can also be tested by dividing it by the standard error

$$\begin{aligned} \text{Var}(\hat{a}\hat{b} + \hat{\sigma}_{aj,bj}) &= b^2 \text{Var}(\hat{a}) + a^2 \text{Var}(\hat{b}) + \text{Var}(\hat{a})\text{Var}(\hat{b}) \\ &+ 2ab \text{Cov}(\hat{a}\hat{b}) + \text{Cov}(\hat{a}\hat{b})^2 + \text{Var}(\hat{\sigma}_{aj,bj}) \end{aligned} \quad (21.100)$$

that includes the covariance of the a and b parameters (Bauer et al., 2006; see also Kenny et al., 2003; Krull & MacKinnon, 1999, 2001; MacKinnon, 2008). However, this assumes that the distribution of the mediated effect is normal, which is often not the case. For this reason, creating a confidence interval around the mediated effect using the percentile bootstrap may be preferred.

Testing for mediation when X is entered at L2 such as in equations 21.97 and 21.98 is slightly different than for the continuous L1 X variable model in equations 21.95 and 21.96 as there are now three individual effects of X on M , a_0 , a_1 , and a_2 . Each of these parameters represents a different effect of X on M , and because of this, three different mediated effects can be estimated, $a_0 b_i$, $a_1 b_i$, and $a_2 b_i$, all of which may be tested for significance using one of the methods described earlier. As discussed during the presentation of autoregressive models, each of these mediated effects represents a specific indirect effect, any or all of which may be of substantive interest.

Several different exponential decay mediation models are possible when using an LGC approach. The first is a direct extension of the model in equations 21.95 and 21.96. Illustrated in Figure 21.21, this model begins by fitting separate exponential decay LGC models for X , M , and Y . Then the observed M variables are regressed on the observed X variables and the observed Y variables are

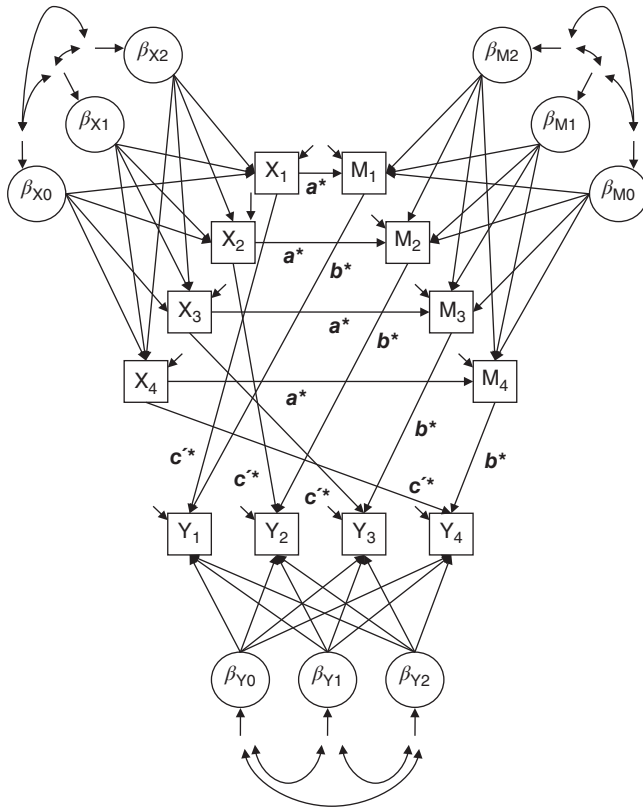


Figure 21.21 An exponential decay latent growth curve model with a continuous X variable and direct paths between observed variables.

regressed on the observed X and M variables, constraining all of the paths between sets of observed variables to be equal. For the dichotomous X variable model in equations 21.97 and 21.98, separate exponential decay LGC models are fit for M and Y , and the observed Y variables are regressed on the observed M variables, again constraining these paths to be equal. The latent growth parameters from M and Y are then regressed onto X .

Another form of exponential decay LGC mediation model is a parallel process model. As was described earlier, an exponential decay LGC model is equivalent to a linear LGC model, only with an exponential decay trajectory across time. Hence, the individual linear LGC models in the parallel process LGC mediation model may be replaced with exponential decay LGC models to form a parallel process LED mediation model, as shown in Figure 21.22.

The parallel process mediation model in Figure 21.22 only contains paths between the rate of change growth parameters. However, depending on the theory behind the model, other paths may be appropriate. For instance, if X were dichotomous and had an effect on the initial value of M , then it may be of interest to include a path between

the initial value of M and the rate of change of Y . Or, continuing with the dichotomous X example, if M reached asymptote before Y , it could be of interest to investigate whether the asymptote of M had an effect on the asymptote of Y . Testing for mediation would be conducted similarly to the parallel process LGC by testing the significance of individual paths or by testing specific indirect paths for significance. In addition, various fit indices given by most structural equation modeling software can be used to assess overall model fit.

Testing for mediation in the exponential decay LGC model is similar to testing for mediation in the multilevel exponential decay model in that the mediational parameters are tested for significance or individual indirect effects are directly tested for significance. One advantage of the latent variable framework is the addition of fit indices, which are provided by most SEM software packages and provide valuable information about the overall fit of the model and the presence of a significant mediated effect. Another advantage is the ability to model measurement error and have multiple indicators for a variable. Finally, the latent variable framework allows for the correlation of residuals and growth parameters as well as the ability to relax the equality constraints for the mediation parameters (e.g., the effect of M_1 on Y_1 does not have to equal the effect of M_2 on Y_2).

Because the ECLS-K variables in our example do not include a treatment or intervention, we did not fit illustrative exponential decay models to this data.

Moderated Mediation in SEM

Sometimes a researcher may wish to test a hypothesis about whether longitudinal mediation effects differ depending on the value of a particular variable, i.e., moderated longitudinal mediation. If the longitudinal mediation is modeled using one of the SEM approaches outlined above, an appropriate test of such moderation can be conducted using techniques for incorporating moderation into this framework, the nature of which depend on the type of moderator involved. If the moderator Z represents a categorical variable with a relatively small number of levels, such as clearly defined group membership (e.g., sex, religion, grouping on age), the multiple group approach (i.e., stacked analysis) in SEM (Jöreskog, 1971) can be used for investigating the moderated mediation effects (Little, 2013; MacKinnon, 2008). Since the independent variable and the mediator may display differential effects in the different groups, stacking groups together allows for comparisons of the parameters, with some parameters

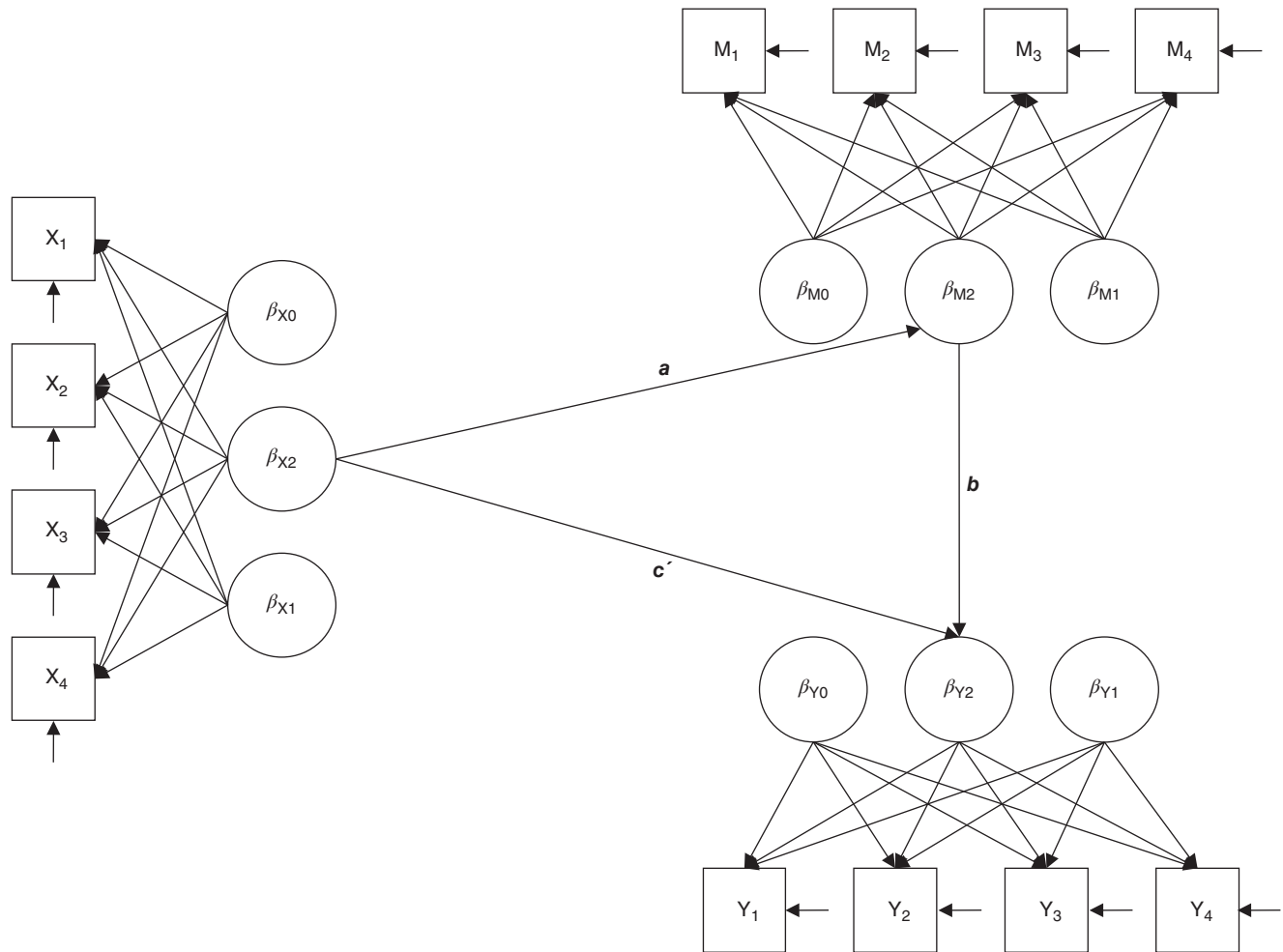


Figure 21.22 A parallel process exponential decay latent latent growth curve model with a dichotomous X variable and direct paths between latent growth parameters.

constrained to be equal, while other parameters specified to vary between the stacked groups. The parameters that differ between the groups indicate interactions between the variable providing the group membership and the variables in the model, such as X and M .

Typically, it is assumed that the same mediation model holds for the stacked groups and the equality of parameter estimates for the relations between X and M (i.e., a paths), M and Y (i.e., b paths), or X and Y (i.e., c' paths) are examined. Bollen (1989) suggested that researchers start with the model without restricting any parameters across the groups. A satisfactory fit would indicate that the same form of model holds across the stacked groups and provide a rationale for further testing the equivalence of the parameters. The moderated mediation can be investigated from the final model where the equivalent parameters are constrained and nonequivalent parameters are freely estimated. This approach can be applied to structural

equation models with observed variables and also models with latent variables for estimating the interaction effect between the latent variable and the categorical group variable (Rigdon, Schumacker, & Wothke, 1998). When X , M , or Y are latent variables, a series of hierarchical tests for invariance should be conducted for the measurement models before examining the equality of the structural relations, to assure that the measures used for the latent variables X , M , and/or Y are equivalent across the groups. If the growth trajectories of X , M , and Y are estimated for the LGC mediation, researchers should inspect if the trajectory shapes are equivalent or different across the stacked groups before testing the moderated mediation. The steps for testing measurement invariance can be found elsewhere (Horm & McArdle, 1992; Jöreskog, 1971; Meredith, 1993).

In a stacked model for moderated mediation, several hypotheses can be tested, including a test of the equality of the relation between X and M (i.e., a paths), a test of the

equality of the relation between M and Y (i.e., b paths), or a test of the equality of the relation between X and Y (i.e., c' paths) across the groups. More importantly, further analysis to test the equality of the indirect effects (i.e., ab estimators) is needed. While the tests of the equality of the relations among X , M , and Y may provide the information on the source of differential indirect effects across the groups, it is possible that the estimated indirect effects obtained in the stacked groups may not statistically differ even when the magnitude of the a paths or the b paths differ across groups. The equality of the path coefficients can be tested by comparing a series of nested models where the relevant paths are constrained to be equal across groups. The equality of the estimated indirect effects can be tested by a t test by dividing the difference in the mediated effects by the pooled standard error of the mediated effect as follows:

$$t_{a_1b_1-a_2b_2} = (a_1b_1-a_2b_2)/\sqrt{s^2_{pooled}} \quad (21.101)$$

In equation 21.100, a_1b_1 and a_2b_2 are the estimated indirect effects from each of the stacked groups and $\sqrt{s^2_{pooled}}$ is the pooled standard error of the mediated effect in each group (MacKinnon, 2008). The equality tests described for the two-group case in this section can be easily extended to cases with more than two groups. The analysis procedures described above can be carried out for each pair of groups to locate the sources of the moderation by group membership.

When the moderator Z is a continuous observed variable interacting with an observed X or M variable, the moderated mediation can be examined using the product term created to reflect the moderation of Z in the relations between X and M (i.e., a path), between M and Y (i.e., b path), or between X and Y (i.e., c' path). Again, the two variables involved in the product term (e.g., X and Z) need to be centered with respect to their means (e.g., $X - \bar{X}$ and $Z - \bar{Z}$) before creating the product term, because a product term based on the uncentered variables will be highly correlated with the first-order effect terms and the high correlations between the predictors may introduce collinearity problems (Aiken & West, 1991; Marquardt, 1980). Centering the variables will minimize or remove the correlations between the product term and the first-order effect terms. The centered variables and their product term are then entered in the model to predict M or Y to test the moderation by Z in the specific relations in the mediation model. With significant interaction effects, further probing of the interaction patterns can be conducted by obtaining simple slopes at different levels of the moderator Z as described

earlier in this chapter. Some SEM software programs such as Mplus (Muthén & Muthén, 2014) calculate the indirect effects at different levels of Z and produce the plots of indirect effects corresponding to particular values of Z .

When the moderator is a latent variable interacting with other latent variables in the model, the moderated mediation model can be investigated using the latent interaction variable. Since the use of latent variables in SEM has many advantages, including the capability of controlling for measurement errors that could obscure the interaction effects, various approaches to estimating interactions between latent variables (Arminger & Muthén, 1998; Bollen & Paxton, 1998; Jöreskog, 1998; Kenny & Judd, 1984; Klein & Moosbrugger, 2000; Marsh, Wen, & Hau, 2004; Wall & Amemiya, 2000) have been proposed in the past years. These approaches commonly specify the indicators of the latent interaction variable with the product of the indicators of the two latent variables, but they differ in the ways they specify the nonlinear constraints used for the factor loadings and variance associated with the latent interaction variable. As expected, the modeling process of these approaches may be complicated and unwieldy, and this may be the reason why such methods have not been frequently applied to the substantive research (Rigdon et al., 1998). However, the latent interaction variable approaches have the potential to be applied to estimating moderated mediation, although more research is warranted to investigate the applicability of these approaches for the models with more than one interaction term (Marsh et al., 2004).

To illustrate a basic moderated longitudinal mediation analysis, we fit a multiple groups SEM for boys and girls using the autoregressive mediation model including both lagged and contemporaneous a and b paths (as in Figure 21.16 and Table 21.9) for the LEARN $\rightarrow$ INTERP $\rightarrow$ INTERN pathway, which was the best fitting longitudinal mediation model in the first to fifth grade ECLS-K data. Comparisons of the fit of nested models revealed that several parameter estimates differed by gender. Specifically, the autoregressive path from LEARN at third grade to LEARN at fifth grade and both autoregressive paths involving the INTERP mediator were larger for boys than for girls. In addition, boys had larger error variances and disturbance variances than girls for all three variables at all time points. The only mediation effect parameters that differed by gender were the lagged a paths. Boys had stronger associations between first grade LEARN and third grade INTERP ($a_{boys} = -0.1786$, $SE = 0.0142$, $p < .0001$ vs. $a_{girls} = -0.1272$, $SE = 0.0150$, $p < .01$) and between third grade LEARN and fifth grade INTERP than girls ($a_{boys} = -0.1686$, $SE = 0.0152$, $p < .0001$ vs. $a_{girls} =$

−0.1250, SE = 0.0155, $p < .01$). The test of the difference between boys' and girls' ab estimates for the temporally ordered $X1 \rightarrow M3 \rightarrow Y5$ mediation effect was significant ($a_1b_1 - a_2b_2 = -0.0019$, SE = 0.0010, $p < .05$), indicating that the effect of approaches to learning on internalizing behavior mediated by interpersonal skill was, in fact, stronger for boys than for girls.

MODERN CAUSAL INFERENCE AND LONGITUDINAL MEDIATION MODELS

There has been a dramatic increase in research on causal inference related to mediation analysis over the last decade (Imai, Keele & Tingley, 2010; MacKinnon, 2008; Pearl, 2012, Valeri & VanderWeele, 2013). In the psychology literature, causal inference has primarily focused on the need for longitudinal rather than cross sectional relations to demonstrate mediation (Cheong et al., 2003; Cole & Maxwell, 2003, Judd & Kenny, 1981; MacKinnon, 2008; Maxwell & Cole, 2007). These articles emphasize the central role of demonstrating temporal precedence such that X occurs before and causes later change in M and M occurs before and causes later change in Y , as discussed in a recent article (Maxwell, Cole, & Mitchell, 2011) and commentary on this topic (West, 2011). Temporal precedence refers to the demonstration that for a variable to be a cause of an effect, it must occur before the effect.

The mediation model is a longitudinal model such that X causes M and M causes Y , so evidence for this longitudinal causal process is ideally obtained from longitudinal data. Mediation relations between variables measured at one time represent associations between levels of the variables such that a person with high scores on X also has higher scores on M and Y , as an example. In some situations, temporal order between the variables is reasonable such as when X represents random assignment to an intervention or measures of M are taken before measures of Y . Cole and Maxwell (2003) demonstrated algebraically the bias that can result when cross sectional relations are treated as if they are longitudinal relations. The notion that relations between levels of variables at a single time point differ from relations across time applies to any difference across any period of time. For example, the mediated effect for measures that differ one minute in time may differ from measures that differ by one year in time. It is not necessarily the cross sectional versus longitudinal nature of measures that is important. It is the match between when change occurs between variables in the true population mediation model and whether the variables are

measured when this mediation relation can be detected (MacKinnon, 2008).

Another important causal inference topic related to temporal precedence is the influence of omitted variable bias on mediation relations (MacKinnon & Pirlott, 2015). There has been an early and sustained focus of attention in the social sciences that accurate estimation of mediation depends on the consideration of other variables that may explain or alter a mediation conclusion from a research study (James, 1980, Judd & Kenny, 1981). In the medical and epidemiology literature, causal inference for mediation has primarily addressed the issue of how confounder variables may bias mediation estimation. Omitted variable bias and confounder bias are considered synonyms in this chapter representing unmeasured variables that influence mediation relations. The potential outcome model provides a way to define causal mediation analysis in the context of confounders or omitted variables. The potential outcome model for causal inference is relevant for many fields and its application has steadily grown. The beginnings of the potential outcome approach for mediation in the social sciences can be traced to an influential paper by Holland (1988). Holland applied the Rubin causal model (Rubin, 1974) to demonstrate the challenges of identifying causal mediation effects even when X represents randomization to conditions (see also Sobel, 2008). In epidemiology, Robins and Greenland (1992) described the restrictive assumptions necessary for identification of causal mediation. Pearl (2001) described and developed a solution discussed in Robins and Greenland (1992) to obtain causal mediation effects that serves as the source for a series of important recent papers on causal mediation from the potential outcome perspective (Imai, Keele & Tingley, 2010; Imai, Keele, & Yamamoto, 2010; Lynch, Cary, Gallop, & Ten Have, 2008; Pearl, 2012; Ten Have et al., 2007; Valeri & VanderWeele, 2013; VanderWeele & Vansteelandt, 2009; Vansteelandt, 2009). These new papers provide some practical guidance and software for researchers to use. All of these previous papers have focused primarily on the cross sectional mediation case of measurement of X , M , and Y , and not the longitudinal case.

The purpose of this section is to describe potential outcome model approaches for investigation of mediation effects in the context of longitudinal data. First the potential outcome model for evaluating intervention data is described, followed by applications of this model to the case of mediation. Mediation and moderation relations are described for this potential outcome model. Additional information regarding the potential outcome model for longitudinal data is outlined followed by the estimation

of causal estimators from data. Other approaches to longitudinal mediation from a modern causal inference perspective are briefly described.

Potential Outcomes Model

The potential outcomes approach to estimating causal effects (Holland, 1986, 1988; Pearl, 2009; Rubin, 1974) distinguishes between an individual's observed and counterfactual outcomes. A counterfactual outcome is an outcome for a participant in a research condition in which that participant did not serve (the condition contrary to actual condition they did serve in). For the case of two groups, let variable X be an intervention with level x ($x = 1$ for the intervention, $x = 0$ for the control) and variable Y be the outcome variable. The potential outcomes framework considers all conditions that an individual could serve in, including both intervention and control conditions, even though observed data can only be obtained for one of the two groups. In other words, these models consider all potential outcomes, so if an individual is assigned to the intervention group, the potential outcome $Y(1)$, is the observed outcome Y . The counterfactual outcome for that individual is the value for the outcome variable if that person was assigned to the control group, $Y(0)$. The individual causal effect is equal to $Y(1) - Y(0)$, even though the value in the control group cannot be observed for a participant in the intervention group. Because it is not possible for the same individual to simultaneously serve in both intervention and control groups, the individual causal effect is not estimable—a problem called the fundamental problem of causal inference. However, assuming that participants are exchangeable, then it is possible to compute a causal effect averaged across individuals in each group and this average causal effect is computed as $E[Y(1) - Y(0)]$, or the expected average difference between intervention and control groups. This average causal effect is a causal estimator under certain assumptions, primarily that individuals have been randomized to the two conditions.

The potential outcome approach applied to mediation introduces a mediating variable M with level m that mediates the relation between X and Y . The potential outcome approach to mediation demonstrates several complexities in the estimation of mediation. The additional complexity is shown in the additional effects that are now relevant when including the mediator, M , in causal sequence from X to M to Y (Hogan & Lancaster, 2004; Jo, 2008; Pearl, 2001, 2012; Robins & Greenland, 1992; VanderWeele & Vansteelandt, 2009). In addition to the average causal effect now called the total effect (TE) of X on Y , there are

five additional causal effects: the controlled direct effect (CDE) which is the direct effect of X on Y at a certain value m of M , a direct effect in the control group called the pure natural direct effect (PNDE), a direct effect in the intervention group called the total natural direct effect (TNDE), the indirect effect in the control group called the pure natural indirect effect (PNIE), and the indirect effect in the intervention group called the total natural indirect effect (TNIE). Use $Y(x, m)$ to denote the potential outcome for an individual under the intervention level x and mediator level m . X takes the value $x = 0$ for the control group, and $x = 1$ for the intervention group. If the actual value of M is m for an individual, then the counterfactual value of M for that individual is denoted as m' .

The *controlled direct effect* of intervention X on outcome Y is the direct effect of the intervention on the outcome at a fixed level m of the mediator M :

$$\text{CDE} = Y(1, m) - Y(0, m) \quad (21.102)$$

The *natural direct effect* of X on Y is different from the CDE in that M is set to the level $M(x)$, the level that would have naturally occurred under one of the conditions of X , so there are now two natural direct effects corresponding to intervention and control groups. In the case of $M(0)$, the PNDE is the effect of intervention X on outcome Y when X did not influence the mediator M (or the participants were assigned the mediator level under the control condition).

$$\text{PNDE} = Y(1, M(0)) - Y(0, M(0)) \quad (21.103)$$

In the case of $M(1)$, the TNDE is the effect of intervention X on outcome Y when X did not influence the mediator M (or the participants were assigned the mediator level under the intervention condition).

$$\text{TNDE} = Y(1, M(1)) - Y(0, M(1)) \quad (21.104)$$

Following the same approach, the *pure natural indirect effect* is the effect of the intervention on Y when changing the level m of M in the control condition. In other words, the PNIE is the effect of the intervention on outcome when the level of M is changed when X is set to 0.

$$\text{PNIE} = Y(0, M(1)) - Y(0, M(0)). \quad (21.105)$$

The *total natural indirect effect* (TNIE) is the effect of the intervention Y when changing the level of M in the

intervention condition. In other words, the TNIE is the effect of intervention on outcome when the level of M is changed when X is set to 1.

$$\text{TNIE} = Y(1, M(1)) - Y(1, M(0)) \quad (21.106)$$

The total effect is equal to the sum of the pure natural direct effect plus the total natural indirect effect, and the total effect is also equal to the sum of the pure natural indirect effect plus the total direct effect. Because there are different effects for intervention and control groups, an interaction is included in this model inasmuch as the direct and indirect effects may differ in the two groups. The direct and indirect effects are different in the two groups if the interaction between X and M in the prediction of Y is nonzero. If the interaction is zero, then the two direct effects are equal and the two indirect effects are equal and these results reduce to the traditional estimator of the direct and indirect effects. Importantly, Pearl (2001) demonstrates that the decomposition of the total effect into the natural direct and indirect effects holds even in models with interactions and nonlinear models such as logistic regression (i.e., the pure natural direct effect plus the Total Natural Indirect Effect equals the total effect).

These estimators described already are estimators of causal effects under certain assumptions. Two assumptions are required for controlled direct effect to be identified (Pearl, 2001; VanderWeele, 2010):

1. No unmeasured confounders for the relation between X and Y .
2. No unmeasured confounders for the relation between M and Y .

Two additional assumptions are required for natural direct and indirect effects to be identified:

3. No unmeasured confounders for the relation between X and M .
4. No measured or unmeasured M to Y confounders are affected by intervention.

Assumptions (1) and (3) refer to the ignorability of intervention (i.e., the intervention independent of potential outcomes for the mediator and outcome) given observed pretreatment confounders. This assumption is satisfied with randomization of X .

Assumption (2) refers to the ignorability of the mediator and the potential outcomes for Y given the observed intervention and pretreatment confounders. This ignorability

of the mediator assumption may be difficult to satisfy in actual research because randomization of M is usually not plausible for many studies. In other words, the value of the mediator is not randomly assigned to participants but is self-selected. Even when conditioning on observed confounders for the relation between M and Y , there can still be unobserved confounders that would lead to violation of this assumption. This assumption is strong and is often ignored in studies even though a causal interpretation of a mediated effect is not possible without its existence (VanderWeele & Vansteelandt, 2009; MacKinnon, 2008). Methods described in this article hold that assumption 3 is true and methods to investigate this assumption are only starting to be developed (Coffman & Zhong, 2012). Assumption 4 states that there are not confounders affected by the intervention that confound the M to Y relation. One goal of multiple mediator models is to assess these additional posttreatment confounders in a way that allows for causal estimation. However, approaches to deal with multiple mediators are an active area of current research in causal inference (VanderWeele & Vansteelandt, 2013). And in the application of the potential outcome model for longitudinal mediation, effects of earlier waves on later waves could be postintervention confounders that violate assumption 4.

The assumptions described above have also been described as sequential ignorability assumptions in the research literature on causal mediation (Imai, Keele, & Tingley, 2010; Pearl, 2001; VanderWeele & Vansteelandt, 2009). The linear regression approach as well as potential outcome methods including the natural direct and indirect effects assumes *sequential ignorability*, which consists of the ignorability of the intervention assignment and the ignorability of the mediator. In other words, random assignment of X ensures that c and a represent causal effects (with assumptions 1 and 2) but b and c' do not have a causal interpretation without further assumptions (Holland, 1988). For instance, given that the individuals are randomly assigned to conditions of an intervention, there are not confounders of the X to Y and X to M relations. However, because participants are not randomly assigned to values of M , the relation between M and Y , b , is not a causal effect because there can be confounders that account for the effect of M on Y .

Comparison of Traditional and Potential Outcome Mediation Analysis

For the case of continuous M and Y as in most social science studies, there is a direct correspondence between the

estimators of the direct and indirect effects in traditional and the potential outcome methods. If all assumptions are satisfied, traditional and potential outcome methods come to the same conclusion about direct and indirect effects if the interaction of X and M on Y is zero. So the traditional estimators of ab and c' correspond to the natural indirect and natural direct effects causal estimators. When the interaction between X and M is included in the statistical analysis, the natural indirect effects correspond to simple mediation effects in the traditional analysis where the simple mediation effects are at specific values of the mediator. However, typically the interaction of X and M is included in traditional analysis as a way to test the assumption that the causal relation of M to Y does not differ across levels of X . If the interaction is nonsignificant, it is often not included in traditional mediation analysis mainly because the relation of M to Y is often based on prior research. To clarify, if the interaction is included in the statistical model then there are simple mediation effects in traditional analysis corresponding to different b paths in the statistical analysis. These effects would reflect the mediated effect at different values of the mediator. There are corresponding effects in the potential outcome approach that refer to the effects for the control group called the pure effects and the effects for the intervention group called the total effects. For the potential outcome approach, the interaction of X and M is included in the analysis to calculate the different direct and indirect effects in each group.

Moderation in the Potential Outcome Model

As described already, the potential outcome model has different direct and indirect effects in the intervention and control groups. These additional estimators are only present if the interaction of X and M is nonzero. In this potential outcome framework, an interaction of indirect effects corresponds to testing the equality of the natural indirect effect in the two groups. Similarly, moderation effects could be obtained by testing the causal effects across subgroups where each subgroup could have different natural direct and indirect effects. A statistical test of the interaction across subgroups could be accomplished by dividing the difference between natural indirect effects across groups by a multivariate delta method standard error for the difference between the causal effects or by using a resampling method such as the bootstrap to create a confidence interval for the difference between the effects and testing whether zero is in the interval.

Potential Outcome Mediation Model with Longitudinal Data

The causal perspective on longitudinal mediation elucidates a problem in the regression analysis of data with more than one wave. The description of this problem in the causal inference literature is in the context of time-varying interventions or situations in which earlier waves of data predict later treatment and later values of the outcome (Robins & Hernan, 2009). The problems with longitudinal data from a causal inference perspective are illustrated here using a study of the effects of loneliness on depression as described in VanderWeele, Hawkley, Thisted, and Cacioppo (2011). There is evidence that loneliness and depression are separate constructs and that loneliness leads to increased depression (see VanderWeele et al., 2011) but it would be useful to estimate the size of this effect. Say that depression at the fourth time of measurement is the focus of the analysis. Depression at the fourth time point is predicted by the following variables: depression at the first, second, and third times and loneliness at the first, second, and third times. All of these variables are likely to have some effect on Time 4 depression, but there is a problem with estimating all of these effects at once. Depression at Time 2 is on the pathway from loneliness at Time 1 to depression at Time 4, so it may mediate the effect of loneliness on later depression thereby reducing the total effect of loneliness on later depression. In other words, depression at Time 2 could be a mediator of the relation of Time 1 loneliness on depression at Time 4. Because depression at Time 2 is a potential mediator, it would not be ideal to adjust effects if a researcher was interested in estimating the total effect of earlier loneliness on later depression. Depression at Time 2 is also related to both depression and loneliness at Time 3 so it is also a confounder of these variables on Time 4 depression. Because it is a possible confounder, depression at Time 2 must be adjusted to get an accurate estimate of the relation of loneliness and depression at Time 4. So depression at Time 2 should be adjusted for to remove confounding, and it should also not be adjusted for so that it does not block the total effect of loneliness at Time 1 on depression at Time 3. The correct analysis would both adjust and not adjust for Time 2 depression, which is not possible with regression analysis. Robins, Hernan, and Brumback (2000) proposed a solution to this problem by conducting an analysis that removes confounder effects so the true effects can be estimated. In this context, a model called a marginal structural model removes effects of variables that may serve as a confounder to estimate the effects of

loneliness on depression at later waves of measurement. The approach uses a weighting method to render an unconfounded data set.

In terms of an intervention, X , and a mediation model with two waves of data for M , $M1$ and $M2$, and Y , $Y1$ and $Y2$, say we are looking at the effect of the intervention, X , on the outcome $Y2$. If control is made for a variable that is a consequence of the intervention, the mediator $M2$, then the relation of X to $Y2$ will be biased. By doing this, part of the effect of the intervention is blocked. For the mediation relation, X to $M2$ to $Y2$, control for X is needed to get the correct relation of $M2$ to $Y2$, that is X is a confounder of the $M2$ to $Y2$ relation. It is not possible with a single regression model to obtain an unbiased estimate of X on $Y2$ because it must simultaneously control for $M2$ and not control for $M2$. With a structural equation model, it is possible to estimate relevant indirect effects for effects from X to $Y2$ when X is uncorrelated with $M1$ and $Y1$ because the level of X is randomized to groups. The single mediated effect in this model corresponds to X to $M2$ to $Y2$. With the marginal structural model and a nonrandomized X , the time 1 measures $M1$ and $Y1$ (before the intervention is delivered) are used to weight $M2$ to remove confounding and then a weighted analysis is used to estimate the causal effect of the unconfounded $M2$ variable on $Y2$. A similar approach is illustrated below.

Inverse Probability Weighting with Longitudinal Data

An approach to obtaining effects adjusted for confounders in longitudinal mediation models is to use observed covariates to measure confounder effects and adjust analyses to remove the confounder bias (Robins et al., 2000). The goal of these analyses is to create a pseudo-population of data in which confounder bias has been removed and then conduct analyses on this new confounder-free data set. This new confounder-free data set avoids the problem of simultaneously adjusting for and not adjusting for an intermediate variable that is also a confounder. It accomplishes the goal by separately estimating the confounder effect and other effects. A weighting strategy is used to weight observations based on how much the observations are confounded to create an unconfounded data set. A statistical model is first developed for the prediction of the mediator based on covariate information from baseline measures. These baseline measures can be the measure of M and Y at baseline as well as other available variables. The weights for each participant are obtained from the residuals of two models (1) a statistical model in which only X predicts the mediator, and (2) a statistical model in which X and all covariates are used to predict the mediator.

A weight is formed by dividing the probability of the residual in the first model by the probability of the residual in the second model. A weighted regression model is used in the analysis to predict Y with M and X (Coffman, 2011; Coffman & Zhong, 2012; Robins, Hernan, & Brumbeck, 2000; Toh & Hernan, 2008). The coefficient relating M to Y is the b coefficient with confounder bias removed and sandwich estimators of standard errors are used to create confidence intervals for the effect. This method assumes that all measures of confounder bias have been measured and included in the analysis to create the weights (thereby leading to accurate estimation of the relation of M to Y , i.e., by adjustment for potential confounders). If there are repeated measures of X , M , and Y , then weights are obtained for each predicted value of M at each time with prior values of X included as covariates to adjust for confounder bias. The weights in the final analysis for the prediction of Y are the product of the weights obtained for the prediction of each wave of Y . Weights that adjust for missing data can also be included.

Clearly, a substantial assumption of this method is that there are no unmeasured confounders, which may be difficult to satisfy in some situations. One important check for the model is to assess the extent to which there is enough overlap in the weights in each group with plots and tables to investigate the balance of the weighting procedure. To date the marginal structural model has not been widely used in psychology but it is more common in epidemiology and the health sciences. It is among the most straightforward of causal inference methods to apply. As described already, VanderWeele et al. (2011) applied the marginal structural model method to investigate the effect of loneliness on depression across five waves.

There are several methods to estimate weights for inverse probability weighting that may be especially useful when the weights vary a great deal from very small to very large (Cole & Hernan, 2008). The weights for each participant represent how much that participant's data are used in the final regression of X and M on Y . If the denominator in the weight equation becomes extremely small, the weights for some persons may be excessively large so that one person may overly influence the results of the analysis. One method to deal with this problem is the use of truncated weights to avoid some subjects having too much or too little influence on the analysis. A relative alternative approach avoids calculation of the weights by creating new data sets where confounder effects are removed using a method called *g-estimation*. *G-estimation* is not described here but is available in several sources (Loeys et al., 2014; Vansteelandt, 2009).

Causal Estimators for the ECLS-K Data Set

To illustrate the causal estimates, three variables were selected from the ECLS-K data set, an independent variable measure of externalizing behavior at first grade, a mediating measure of reading performance at time third grade, and an outcome measure of internalizing behavior at fifth grade. These measures were used as indicators of a mediating process such that externalizing behavior reduces reading performance, which then increases later depression. Covariates included the kindergarten fall and spring measures of externalizing, reading, and internalizing variables along with gender and age to adjust for potential confounding. The goal of the analysis was to estimate causal effects across waves adjusted for possible confounders where the confounders were measured at the first two waves of measurement. The program described in Valeri and VanderWeele (2013) was used to estimate the natural direct and indirect effects described earlier. To simplify the analysis, only complete cases were used in the analysis for a total of 7,355 individuals. First, the interaction of first grade externalizing behavior and first grade reading performance was not statistically significant in the prediction of Y (.011, $SE = .032$, $p < .5$) so separate estimates of direct and indirect effects at different levels of first grade externalizing behavior were not computed. In other words, the two different causal estimators of effects were not necessary because the nonsignificant interaction indicated that effects did not differ across levels of the mediator. As a result there is one natural direct effect ($NDE = .046$, $SE = .007$, $LCL = .033$, $UCL = .059$), one natural indirect effect ($NIE = .004$, $SE = .0008$, $LCL = .002$, $UCL = .005$) and the sum of these effects equals the total effect ($TE = .049$, $SE = .007$, $LCL = .036$, $UCL = .062$). All of the effects are statistically significant, which is perhaps not surprising given the large sample size. The metric of the effects is in terms of a one standard deviation change (.540 units) in externalizing at first grade. Dividing the indirect effect by the standard deviation of Y , suggests that a standard deviation change in externalizing led to an indirect effect via reading performance of about .04 standard deviations in internalizing symptoms.

FUTURE DIRECTIONS

There are many potentially fruitful directions for future research regarding longitudinal moderation and mediation. For example, Bayesian analysis of longitudinal data is one particular area where the development of new analytic methods may produce better estimates of hypothesized

effects and produce tests that more appropriately reflect substantive theory. Bayesian estimation allows researchers to incorporate prior information into the analysis, which can potentially produce more efficient estimates of moderation and mediation effects (Yuan & MacKinnon, 2009). Moreover, while many traditional estimation techniques rely on large-sample theory, Bayesian estimation is exact, making this approach particularly suitable for data with small samples (Ozechowski, 2014). This aspect of Bayesian analysis is also particularly relevant to mediation effects. Maximum likelihood estimation provides estimates of parameters and standard errors and assumes normal distributions based on large sample theory. However, the *ab* estimate of the mediated effect is known to be nonnormal (MacKinnon et al., 2007), and this problematic violation often necessitates the use of specialized techniques (e.g., bootstrapping or the application of the distribution of the product) for hypothesis testing. However, Bayesian analysis produces the whole (skewed) posterior distribution of the mediated effect and requires no such assumption. Muthén and Asparouhov (2012) presented a Bayesian approach to structural equation modeling, which could potentially include both mediation and moderation. Yuan and MacKinnon (2009) described a Bayesian approach to modeling multilevel mediation, and Ozechowski (2014) demonstrated the use of Empirical Bayes methods for testing mediation in a small sample.

Another possible focus of future research is person-centered moderation and mediation. While more typical variable-centered approaches concentrate on examining relations among variables, person-centered approaches examine patterns of responses for individuals. The person is the unit of analysis (Bauer & Shanahan, 2007), and analytic techniques such as cluster analysis, latent profile analysis, latent class analysis, and latent transition analysis can be used to empirically sort individuals into distinct groups according to their patterns of variable values and associations. For example, individuals might be classified as showing patterns either consistent or inconsistent with mediation. Collins, Graham, and Flaherty (1998) demonstrated such an analysis and show that under particular conditions, variable-centered approaches might detect mediation when there are, in fact, no differences in the proportion of individuals showing patterns consistent with mediation in control and treatment groups. Bauer & Shanahan (2007) illustrate how a person-centered approach can be used to model complex moderated effects.

The recent rapid development of new technology and methods for collecting intensive longitudinal data has resulted in a need for analytic methods suitable for

examining moderation and mediation in such large data sets. For example, current technology allows the collection of intensive ecological momentary assessment (EMA) data, where subjects repeatedly report on key variables while they go about their daily lives. An EMA data set might include hundreds of observations obtained via smartphone from a given individual over the course of a few days or perhaps just one observation per day for several weeks, months, or an entire year. Fitness tracking devices might record several physical activity and condition variables repeatedly for a period of several hours. fMRI data contain activation information about multiple voxels over the time course of an experimental session. Analysis of such rich data sets with traditional techniques can prove cumbersome (Baraldi, Wurpts, MacKinnon, & Lockhard, 2014). As the collection of such data becomes increasingly frequent, there is a clear need for the development of specific methods to explore moderation and mediation in such big data applications. Baraldi et al. (2014) described particular multilevel models that can be used to address relevant questions in EMA data and to explore the effects of personalized, technologically delivered ecological momentary interventions. Lindquist (2012) presented an extension of the mediation model for fMRI data that allows the mediating variable to be a continuous function rather than a single scalar measure and includes causal estimators of effects. The combination of intensive longitudinal data and a person-centered analytic approach may potentially allow the examination of research questions that could not be easily addressed with smaller data sets and traditional variable-centered analytic techniques. For example, groups of subjects who confirm or do not confirm the pattern of mediation at different phases of time during a study (e.g., confirming mediation at early versus later phases) could be identified and their transition patterns over the course of time can be examined to predict their status on an outcome variable.

There is also a clear need for more research on temporal design issues. Every researcher designing a longitudinal study needs to make a number of explicit decisions about the data collection schedule. This audience would benefit from clear recommendations regarding the ideal number as well as the frequency and spacing of observations for various types of research scenarios. Some research in this area has, in fact, been conducted for univariate and bivariate growth models where interest centers on testing the effects of treatment on growth (e.g., Raudenbush & Liu, 2001) or variability in growth across individuals and covariation among growth rates (e.g., Rast & Hofer, 2014). However, similar work focusing on the power to detect moderation and mediation effects has yet to be conducted.

In the interest of developing models that more accurately reflect developmental processes, future research should also focus on methods to parsimoniously represent nonlinear effects. Polynomial trends are, by far, the most common functions currently used to represent change over time, and linear models are the more frequently employed of these. However, most growth is probably not strictly linear (Fritz, 2014). Adding higher order polynomial terms may increase the realism of the model, but these additional factors are often difficult to interpret and incorporate into more complex analyses (such as those involving mediation and moderation). Cudeck and Harring (2007) described a number of nonlinear functions and discuss the considerations involved in choosing a particular nonlinear form to represent change. In addition to such nonlinear trends in an outcome variable over time, it is also possible that mediation or moderation effects might themselves be nonlinear. Our models typically assume that the moderated or mediated effect operates identically throughout the range of the predictor variable. However, such an assumption may be overly simplistic, and developing methods and recommendations for testing these effects would be a useful contribution to the literature. Semiparametric and related generalized additive modeling approaches are important directions in this work (Hastie & Tibshirani, 1990; Keele, 2008; Shen, Chou, Pentz, & Berhane, 2014).

Though the methods described in this chapter focus primarily on interindividual analysis, it is potentially useful to distinguish between intra- and interindividual processes. Most psychological studies are focused on variation between cases (Molenaar, 2004), and explaining intraindividual processes through reference to interindividual findings is a simple generalization of the ecological fallacy (Robinson, 1950). Molenaar (2004) defines the conditions under which the structure of intra- and interindividual variation will be analogous and argues that these will rarely if ever be met in real psychological processes. If a researcher wishes to explore within-person mediation, using a $1 \rightarrow 1 \rightarrow 1$ multilevel mediation model (e.g., Krull & MacKinnon, 2001), combined with judicious decisions regarding the centering of the various predictors in the model, can help to disentangle these processes. To date, however, very little work has focused on ideal methods for exploring intraindividual effects and how best to approach such questions in real world data sets.

The combination of mediation and moderation into a single analytic framework is another area that would benefit from future quantitative research and application in substantive data. Currently the most frequent method for exploring whether mediated effects might, in fact, be

moderated by a fourth variable is the rather simplistic approach of creating subgroups based on the moderator and conducting a multiple group SEM. Such an approach is well suited when the moderator is truly categorical, but artificially categorizing a continuous variable so that it can be forced into such an analysis is far from ideal. Dichotomizing a continuous variable via a median split has many undesirable consequences, including loss of power and spurious main effects and interactions (MacCallum, Zhang, Preacher, & Rucker, 2002). The development and application of more sophisticated techniques that avoid such inappropriate treatment of variables will result in much better tests of hypotheses involving mediated moderation and moderated mediation in longitudinal data. These might involve continuous variable approaches to moderation within an SEM or incorporation of person-centered methods that empirically identify subgroups of subjects based on patterns of multiple variables.

Finally, causal inference approaches in general and for mediation in particular are also being rapidly developed. Several books are now appearing that should help disseminate these methods (Hernan & Robins, in press; Morgan & Winship, 2007; Pearl, 2009), though longitudinal mediation modeling has not been widely addressed in these books. There are some recent applications with longitudinal data (Ertefaie & Stephens, 2010; Lindquist, 2012; O'Malley, 2012). Some interesting recent work attempts to combine both dynamic mediation models and modern causal inference (Aalen, Roysland, & Gran 2012; Aalen, Roysland, Gran, Kouyos, & Lange, 2014). Aalen et al. (2012) described two overlapping approaches to causal inference with longitudinal mediation corresponding to interventionist and mechanistic viewpoints. The interventionist approach focuses on estimating causal effects when there is randomization or the evaluation of the status of a system when there is and when there is not an intervention. Typically this approach focuses on the potential outcomes model as described already, although some approaches do not require counterfactuals (Dawid, 2000). The mechanistic approach focuses on how effects come about rather than how an intervention would cause effects. As described by Aalen et al. (2012, p. 832) the "mechanisms in the structure of the world are present whether humans are there to intervene or not, and hence seem to be the more fundamental part." As a compromise between interventionist and mechanistic positions, Aalen et al. (2012, p. 843) stated, "Now our modest suggestion is as follows: if a direct effect cannot reasonably be defined as a controlled or natural direct effect in the counterfactual sense because the required hypothetical manipulation of the mediator is inconceivable, then we can alternatively

view these effects as being represented by flow in a dynamic system, so that the direct effect corresponds to the flow not passing through the mediator. The indirect effect can similarly be understood as that passing through the mediator." Aalen et al. went on to describe a stochastic differential equation model for mediation that includes dynamic models for X , M , and Y processes including a dynamic model for how X , M , and Y are related. Aalen et al. apply this idea to mediation in survival analysis data. Other recent work also addresses dynamic change based on functional data analysis and mediation (Lindquist, 2012), though this approach also targets the estimation of natural and controlled effect causal estimators described previously. More application of these methods to actual data is an important future direction of this research in psychology.

Because development is longitudinal by definition, and because so many of the hypotheses developmental psychopathologists seek to test involve multiple variables interacting and affecting each other in sequence, the concepts of longitudinal moderation and mediation are particularly important in this area of research. The various analytic models presented in this chapter test how effects change over time or the mechanisms through which earlier causes produce their later effects. Application of these methods in longitudinal data will help to advance the field and achieve better understanding of the complex processes that unfold over time to produce adaptive and maladaptive behavior.

REFERENCES

- Aalen, O. O., Roysland, K., & Gran, J. M. (2012). Causality, mediation and time: A dynamic viewpoint. *Journal of the Royal Statistical Society Series A*, 175, 831–861.
- Aalen, O. O., Roysland, K., Gran, J. M., Kouyos, R., & Lange, T. (2014). Can we believe the DAGs? A comment on the relationship between causal DAGs and mechanisms. *Statistical Methods in Medical Research*. doi: 10.1177/0962280213520436
- Aiken, L. S., & West, S. G. (1991). *Multiple regression: Testing and interpreting interactions*. Thousand Oaks, CA: Sage Publications.
- Arminger, G., & Muthén, B. (1998). A Bayesian approach to non-linear latent variable models using the Gibbs sampler and the Metropolis-Hastings algorithm. *Psychometrika*, 63(3), 271–300.
- Arruda, E. H., & Krull, J. L. (2013). *The use of multilevel random coefficient models for the analysis of mediational change over time*. Paper presented at the Biannual Multilevel Modeling Conference, Utrecht, The Netherlands.
- Arruda, E. H., & Krull, J. L. (2014a). *The use of multilevel random coefficient models for the analysis of mediational change over time*. Poster presented at the annual meeting of the American Educational Research Association, Philadelphia, PA.
- Arruda, E. H., & Krull, J. L. (2014b). *The use of multilevel random coefficient models for the analysis of mediational change over time*. Poster presented at the Annual Modern Modeling Methods Conference, Storrs, CT.

- Baraldi, A. N., Wurpts, I. C., MacKinnon, D. P., & Lockhard, G. (2014). Mechanisms of behavior change from technology-based interventions. In L. Marsch, S. Lord, & J. Dallery (Eds.), *Leveraging technology to transform behavioral healthcare* (pp. 187–199). New York, NY: Oxford University Press.
- Barcikowski, R. S. (1981). Statistical power with group mean as the unit of analysis. *Journal of Educational Statistics*, 6, 267–285.
- Barnard-Brak, L., & Brak, V. (2011). Pharmacotherapy and academic achievement among children with attention-deficit/hyperactivity disorder. *Journal of Child and Adolescent Psychopharmacology*, 21, 597–603.
- Baron, R. M., & Kenny, D. A. (1986). The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51, 1173–1182.
- Bauer, D. J., Preacher, K. J., & Gil, K. M. (2006). Conceptualizing and testing random indirect effects and moderated mediation in multilevel models: New procedures and recommendations. *Psychological Methods*, 11, 142–163. doi: 10.1037/1082-989X.11.2.142
- Bauer, D. J., & Shanahan, M. J. (2007). Modeling complex interactions: Person-centered and variable-centered approaches. In T. D. Little, J. A. Bovaird, & N. A. Card (Eds.), *Modeling contextual effects in longitudinal studies* (pp. 255–284). Mahwah, NJ: Erlbaum.
- Bentler, P. M. (1990). Comparative fit indexes in structural models. *Psychological Bulletin*, 107, 238–246.
- Bentler, P. M., & Wu, E. J. C. (2006). *EQS for Windows v. 6.1* [computer software]. Encino, CA: Multivariate Software, Inc.
- Blozis, S. A. (2004). Structured latent curve models for the study of change in multivariate repeated measures. *Psychological Methods*, 9, 334–353. doi: 10.1037/1082-989X.9.3.334
- Blozis, S. A. (2007). On fitting nonlinear latent curve models to multiple variables measured longitudinally. *Structural Equation Modeling*, 14, 179–201. doi: 10.1080/10705510701293445
- Blozis, S. A., Conger, K. J., & Harring, J. R. (2007). Nonlinear latent curve models for multivariate longitudinal data. *International Journal of Behavioral Development*, 31, 340–346. doi: 10.1177/0165025407077755
- Bolger, N., & Laurenceau, J.-P. (2013). *Intensive longitudinal methods: An introduction to diary and experience sampling research*. New York, NY: Guilford.
- Bollen, K. A. (1989). *Structural equations with latent variables*. New York, NY: Wiley.
- Bollen, K. A., & Curran, P. J. (2004). Autoregressive latent trajectory (ALT) models: A synthesis of two traditions. *Sociological Methods & Research*, 32, 336–383.
- Bollen, K. A., & Curran, P. J. (2006). *Latent curve models: A structural equation perspective*. Hoboken, NJ: Wiley.
- Bollen, K. A., & Paxton, P. (1998). Two-stage least squares estimation of interaction effects. In R. E. Schumacker & G. A. Marcoulides (Eds.), *Interaction and nonlinear effects in structural equation modeling* (pp. 125–154). Mahwah, NJ: Erlbaum.
- Bollen, K. A., & Zimmer, C. (2010). An overview of the autoregressive latent trajectory (ALT) model. In K. van Montfort, J. Oud, & A. Satorra (Eds.), *Longitudinal research with latent variables* (pp. 153–176). Heidelberg, Germany: Springer.
- Browne, M. W. (1993). Structured latent curve models. In C. M. Cuadras & C. R. Rao (Eds.), *Multivariate analysis: Future directions 2* (pp. 171–198). Amsterdam, The Netherlands: North Holland.
- Burstein, L., & Miller, M. D. (1980). Regression-based analysis of multi-level educational data. *New Directions for Methodology of Social and Behavioral Sciences*, 6, 194–211.
- Candel, M. J. J. M., & Winkens B. (2003). Performance of Empirical Bayes estimators of level-2 random parameters in multilevel analysis: A Monte Carlo study for longitudinal designs. *Journal of Educational and Behavioral Statistics*, 28, 169–194.
- Cheong, J. (2011). Accuracy of estimates and statistical power for testing mediation in latent growth modeling. *Structural Equation Modeling*, 18, 195–211.
- Cheong, J., MacKinnon, D. P., & Khoo, S.-T. (2003). Investigation of mediational processes using parallel process latent growth curve modeling. *Structural Equation Modeling*, 10, 238–262. doi: 10.1207/S15328007SEM1002_5
- Cicchetti, D., & Rogosch, F. A. (1996). Equifinality and multifinality in developmental psychopathology. *Development and Psychopathology*, 8, 598–600.
- Coffman, D. L. (2011). Estimating causal effects in mediation analysis using propensity scores. *Structural equation modeling: A multidisciplinary journal*, 18(3), 357–369.
- Coffman, D. L., & Zhong, W. (2012). Assessing mediation using marginal structural models in the presence of confounding and moderation. *Psychological Methods*, 17, 642–664.
- Cohen, J., Cohen, P., West, S. G., & Aiken, L. S. (2003). *Applied multiple regression/correlation analysis for the behavioral sciences* (3rd ed.). Mahwah, NJ: Erlbaum.
- Cole, D. A., & Maxwell, S. E. (2003). Testing mediational models with longitudinal data: Questions and tips in the use of structural equation modeling. *Journal of Abnormal Psychology*, 112, 558–577.
- Cole, S. R., & Hernan, M. A. (2008). Constructing inverse probability weights for marginal structural models. *American Journal of Epidemiology*, 168, 656–664.
- Collins, L. M., Graham, J. W., & Flaherty, B. P. (1998). An alternative framework for defining mediation. *Multivariate Behavioral Research*, 33, 295–312.
- Cudeck, R. (1996). Mixed-effects models in the study of individual differences with repeated measures data. *Multivariate Behavioral Research*, 31, 371–403. doi: 10.1207/s15327906mbr3103_6
- Cudeck, R., & du Toit, S. H. C. (2002). A version of quadratic regression with interpretable parameters. *Multivariate Behavioral Research*, 37, 501–519.
- Cudeck, R., & Harring, J. R. (2007). The analysis of nonlinear patterns of change with random coefficient models. *Annual Review of Psychology*, 58, 615–637.
- Curran, P. J. (20023) Have multilevel models been structural equation models all along? *Multivariate Behavioral Research*, 38, 529–569.
- Currie, J., & Stabile, M. (2006). Child mental health in childhood and human capital accumulation: The case of ADHD. *Journal of Health Economics*, 25, 1094–1118.
- Currie, J., & Stabile, M. (2009). Mental health in childhood and human capital. In J. Gruber (Ed.), *The problems of disadvantaged youth: An economic perspective* (pp. 115–148). Chicago, IL: University of Chicago Press.
- Davidian, M., & Giltinan, D. M. (1995). *Nonlinear models for repeated measurement data*. New York, NY: Chapman and Hall/CRC Press.
- Dawid, A. P. (2000). Causal inference without counterfactuals. *Journal of the American Statistical Association*, 95(450), 407–424.
- Dearing, E., & Hamilton, L. C. (2006). Contemporary advances and classic advice for analyzing mediating and moderating variables. *Monographs of the Society for Research in Child Development*, 71, 88–104.
- Department of Health and Human Services, Administration for Children and Families. (1996–2001). *Early Head Start research and evaluation study* [computer file]. Available from <http://www.icpsr.org/>

- Drabick, D. A., & Kendall, P. C. (2010). Developmental psychopathology and the diagnosis of mental health problems among youth. *Clinical Psychology: Science and Practice*, 17, 272–208. doi: 10.1111/j.1468-2850.2010.01219.x.
- Duncan, O. D. (1969). Some linear models for two-wave, two-variable panel analysis. *Psychological Bulletin*, 72(3), 177–182.
- Duncan, T. E., Duncan, S. C., & Strycker, L. A. (2006). *An introduction to latent variable growth curve modeling: Concepts, issues, and applications* (2nd ed.). New York, NY: Lawrence Erlbaum Associates.
- Edwards, J. R., & Lambert, L. S. (2007). Methods for integrating moderation and mediation: A general analytical framework using moderated path analysis. *Psychological Methods*, 12, 1–22.
- Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. New York, NY: Chapman & Hall.
- Elder, T. E. (2010). The importance of relative standards in ADHD diagnoses: Evidence based on exact birth dates. *Journal of Health Economics*, 29, 641–656. doi: 10.1016/j.jhealeco.2010.06.003
- Enders, C. K., & Tofighi, D. (2007). Centering predictor variables in cross-sectional multilevel models: A new look at an old issue. *Psychological Methods*, 12, 121–138.
- Ertefaie, A., & Stephens, A. D. (2010). Comparing approaches to causal inference for longitudinal data: Inverse probability of treatment weighting versus propensity scores. *International Journal of Biostatistics*, 6(2), Article 14.
- Fairchild, A. J., & MacKinnon, D. P. (2009). A general model for testing mediation and moderation effects. *Prevention Science*, 10, 87–99. doi: 10.1007/s1121-008-0109-6
- Ferrer, E., & McArdle, J. J. (2003). Alternative structural models for multivariate longitudinal data analysis. *Structural Equation Modeling*, 10, 493–524.
- Fletcher, J., & Wolfe, B. (2008). Child mental health and human capital accumulation: The case of ADHD revisited. *Journal of Health Economics*, 27, 794–800.
- Frazier, P. A., Tix, A. P., & Barron, K. E. (2004). Testing moderator and mediator effects in counseling psychology research. *Journal of Counseling Psychology*, 51, 115–134.
- Frazier, T. W., Youngstrom, E. A., Glutting, J. J., & Watkins, M. W. (2007). ADHD and achievement meta-analysis of the child, adolescent, and adult literatures and a concomitant study with college students. *Journal of Learning Disabilities*, 40, 49–65.
- Fritz, M. S. (2014). An exponential decay model for mediation. *Prevention Science*, 15, 611–622. doi: 10.1007/21121-01300390-x.
- Gaub, M., & Carlson, C. L. (1997). Gender differences in ADHD: A meta-analysis and critical review. *Journal of the American Academy of Child & Adolescent Psychiatry*, 36, 1036–1045.
- Gershon, J., & Gershon, J. (2002). A meta-analytic review of gender differences in ADHD. *Journal of Attention Disorders*, 5, 143–154.
- Gogineni, A., Alsup, R., & Gillespie, D. (1995). Mediation and moderation in social work research. *Social Work Research*, 19, 57–63.
- Goldstein, H. (1995). *Multilevel statistical models* (2nd ed.). New York, NY: Wiley.
- Gollob, H. F., & Reichardt, C. S. (1987). Taking account of time lags in causal models. *Child Development*, 58(1), 80–92.
- Gollob, H. F., & Reichardt, C. S. (1991). Interpreting and estimating indirect effects assuming time lags really matter. In L. M. Collins & J. L. Horn (Eds.), *Best methods for the analysis of change: Recent advances, unanswered questions, future directions* (pp. 243–259). Washington, DC: American Psychological Association.
- Gresham, F. M., & Elliott, S. N. (1990). *Social skills rating scale: Elementary scale A ("How often")*. Circle Pines, MN: American Guidance Service.
- Hastie, T., & Tibshirani, R. (1990). *Generalized additive models*. London, UK: Chapman and Hall.
- Hayes, A. F. (2013). *Introduction to mediation, moderation and conditional process analysis: A regression-based approach*. New York, NY: Guilford Press.
- Hayes, A. F. (2014). Beyond Baron and Kenny: Statistical mediation analysis in the new millennium. *Communication Monographs*, 76, 408–420.
- Hedeker, D. (2004). An introduction to growth modeling. In D. Kaplan (Ed.), *The SAGE handbook of quantitative methodology for the social sciences* (pp. 215–234). Thousand Oaks, CA: Sage.
- Hernan, M. A., & Robins, J. M. (in press). *Causal inference*. London, UK: Chapman & Hall.
- Hoffman, L., & Stawski, R. S. (2009). Personas as contexts: Evaluating between-person and within-person effects in longitudinal analysis. *Research in Human Development*, 6, 97–120.
- Hofmann, D. A., & Gavin, M. B. (1998). Centering decisions in hierarchical linear models: Implications for research in organizations. *Journal of Management*, 24, 123–161.
- Hogan, J. W., & Lancaster, T. (2004). Instrumental variables and inverse probability weighting for causal inference from longitudinal observational studies. *Statistical Methods in Medical Research*, 13, 17–48.
- Holland, P. W. (1986). Statistics and causal inference (with comments). *Journal of the American Statistical Association*, 81, 945–970.
- Holland, P. W. (1988). Causal inference, path analysis, and recursive structural equation models. *Sociological Methodology*, 18, 449–484.
- Holmbeck, G. N. (1997). Toward terminological, conceptual, and statistical clarity in the study of mediators and moderators: Examples from the child-clinical and pediatric psychology literatures. *Journal of Consulting and Clinical Psychology*, 5, 599–610.
- Horn, J. L., & McArdle, J. J. (1992). A practical and theoretical guide to measurement invariance in aging research. *Experimental Aging Research*, 18, 117–144.
- Hox, J. J. (2010). *Multilevel analysis: Techniques and applications* (2nd ed.). New York, NY: Routledge.
- Humphreys, K. L., Katz, S. J., Lee, S. S., Hammen, C., Brennan, P. A., & Najman, J. M. (2013). The association of ADHD and depression: Mediation by peer problems and parent-child difficulties in two complementary samples. *Journal of Abnormal Psychology*, 122, 854–867.
- Humphreys, L. G. (1960). Investigations of the smipex. *Psychometrika*, 25, 313–323.
- Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods*, 15, 309–334.
- Imai, K., Keele, L., & Yamamoto, T. (2010). Identification, inference and sensitivity analysis for causal mediation effects. *Statistical Science*, 25(1), 51–71. doi:10.1214/10-STS321
- James, L. R. (1980). The unmeasured variable problem in path analysis. *Journal of Applied Psychology*, 65, 415–421.
- Jo, B. (2008). Causal inference in randomized experiments with mediational processes. *Psychological Methods*, 13(4), 314–336. doi: 10.1037/a0014207
- Jöreskog, K. G. (1971). Simultaneous factor analysis in several populations. *Psychometrika*, 36, 409–426.
- Jöreskog, K. G. (1979). Statistical estimation of structural models in longitudinal-developmental investigations. In J. R. Nesselroade & P. B. Baltes (Eds.), *Longitudinal research in the study of behavior and development* (pp. 303–352). New York, NY: Academic Press.
- Jöreskog, K. G. (1998). Interaction and nonlinear modeling: Issues and approaches. In R. E. Schumacker & G. A. Marcoulides (Eds.), *Interaction and nonlinear effects in structural equation modeling* (pp. 239–250). Mahwah, NJ: Erlbaum.

- Jöreskog, K. G., & Sörbom, D. (2006). *LISREL for Windows v. 8.7* [computer software]. Lincolnwood, IL: Scientific Software International, Inc.
- Judd, C. M., & Kenny, D. A. (1981). Process analysis estimating mediation in treatment evaluations. *Evaluation Review*, 5(5), 602–619.
- Kawabata, Y., Tseng, W. L., & Gau, S. S. F. (2012). Symptoms of attention-deficit/hyperactivity disorder and social and school adjustment: The moderating roles of age and parenting. *Journal of Abnormal Child Psychology*, 40, 177–188.
- Keele, L. (2008). *Semiparametric regression for the social sciences*. New York, NY: Wiley.
- Kenny, D. A. (1979). *Correlation and causality*. New York, NY: Wiley.
- Kenny, D. A., & Judd, C. M. (1984). Estimating the nonlinear and interactive effects of latent variables. *Psychological Bulletin*, 96, 201–210.
- Kenny, D. A., Korchmaros, J. D., & Bolger, N. (2003). Lower level mediation in multilevel models. *Psychological Methods*, 8, 115–128. doi: 10.1037/1082-989X.8.2.115
- Khoo, S. T. (2001). Assessing program effects in the presence of treatment-baseline interactions: a latent curve approach. *Psychological Methods*, 6(3), 234–257.
- Klein, A., & Moosbrugger, H. (2000). Maximum likelihood estimates of latent interaction effects with the LMS-method. *Psychometrika*, 65(4), 457–474.
- Kline, R. B. (2011). *Principles and practice of structural equation modeling* (3rd ed.). New York, NY: Guilford Press.
- Krause, M. R., Serlin, R. C., Ward, S. E., Rony, R. Y., Ezenwa, M. O., & Naab, F. (2010). Testing mediation in nursing research: Beyond Baron and Kenny. *Nursing Research*, 59, 288–294. doi: 10.1097/NNR.0b013e3181dd26b3.
- Kreft, I. G. G., de Leeuw, J., & Aiken, L. S. (1995). The effect of different forms of centering in hierarchical linear models. *Multivariate Behavioral Research*, 30, 1–21.
- Krull, J. L., & MacKinnon, D. P. (1999). Multilevel mediation modeling in group-based intervention studies. *Evaluation Review*, 23, 418–444. doi: 10.1177/0193841X9902300404
- Krull, J. L., & MacKinnon, D. P. (2001). Multilevel modeling of individual and group level mediated effects. *Multivariate Behavioral Research*, 36, 249–277. doi: 10.1207/S15327906MBR3602_06
- Lahey, B. B., Hartung, C. M., Loney, J., Pelham, W. E., Chronis, A. M., & Lee, S. S. (2007). Are there sex differences in the predictive validity of *DSM-IV* ADHD among younger children? *Journal of Clinical Child and Adolescent Psychology*, 36, 113–126.
- Laird, N. M., & Ware, H. (1982). Random-effects models for longitudinal data. *Biometrics*, 38, 963–974.
- Lee, S. S., Lahey, B. B., Owens, E. B., & Hinshaw, S. P. (2008). Few preschool boys and girls with ADHD are well-adjusted during adolescence. *Journal of Abnormal Child Psychology*, 36, 373–383.
- Lindquist, M. A. (2012). Functional causal mediation analysis with an application to brain connectivity. *Journal of the American Statistical Association*, 107, 1297–1309.
- Little, T. D. (2013). *Longitudinal structural equation modeling*. New York, NY: Guilford Press.
- Loeys, T., Moerkerke, B., De Smet, O., Byse, A., Steen, J., & Vansteelandt, S. (2014). Flexible mediation analysis in the presence of nonlinear relations: Beyond the mediation formula. *Multivariate Behavioral Research*, 48(6), 871–894.
- Longford, N. T. (1993). *Random coefficient models*. Oxford, UK: Oxford University Press.
- Lynch, K. G., Cary, M., Gallop, R., & Ten Have, T. R. (2008). Causal mediation analyses for randomized trials. *Health Services & Outcomes Research Methodology*, 8(2), 57–76.
- MacCallum, R. C., Zhang, S., Preacher, K. J., & Rucker, D. D. (2002). On the practice of dichotomization of quantitative variables. *Psychological Methods*, 7, 19–40.
- MacKinnon, D. P. (2008). *Introduction to statistical mediation analysis*. New York, NY: Lawrence Erlbaum Associates.
- MacKinnon, D. P., & Dwyer, J. H. (1993). Estimating mediated effects in prevention studies. *Evaluation Review*, 17, 144–158.
- MacKinnon, D. P., Fritz, M. S., Williams, J., & Lockwood, C. M. (2007). Distribution of the product confidence limits for the indirect effect: Program PRODCLIN. *Behavior Research Methods*, 39, 384–389.
- MacKinnon, D. P., Krull, J. L., & Lockwood, C. M. (2000). Equivalence of the mediation, confounding and suppression effect. *Prevention Science*, 1, 173–181.
- MacKinnon, D. P., Lockwood, C. M., Hoffman, J. M., West, S. G., & Sheets, V. (2002). A comparison of methods to test mediation and other intervening variable effects. *Psychological Methods*, 7, 83–104. doi:10.1037/1082-989X.7.1.83
- MacKinnon, D. P., Lockwood, C. M., & Williams, J. (2004). Confidence limits for the indirect effect: Distribution of the product and resampling methods. *Multivariate Behavioral Research*, 39, 99–128.
- MacKinnon, D. P., & Pirlott, A. G. (2015). Statistical approaches for enhancing causal interpretation of the *M* to *Y* relation in mediation analysis. *Personality and Social Psychology Review*, 1, 30–43. doi: 10.1177/1088868314542878
- MacKinnon, D. P., & Valente, M. J. (2014). Mediation from multilevel to structural equation modeling. *Annals of Nutrition and Metabolism*, 65, 196–202. doi: 10.1159/000362505
- MacKinnon, D. P., Warsi, G., & Dwyer, J. H. (1995). A simulation study of mediated effect measures. *Multivariate Behavioral Research*, 30, 41–62.
- Marquardt, D. (1980). You should standardized the predictor variables in your regression models. *Journal of the American Statistical Association*, 75, 74–103.
- Marsh, H. W., Wen, Z., & Hau, K.-T. (2004). Structural equation models of latent interactions: Evaluation of alternative estimation strategies and indicator construction. *Psychological Methods*, 9(3), 275–300.
- Marsh, H. W., Wen, Z., & Hau, K. T. (2006). Structural equation models of latent interaction and quadratic effects. In G. R. Hancock & R. O. Mueller (Eds.), *Structural equation modeling: A second course* (pp. 225–265). Greenwich, CT: Information Age Publishing.
- Martens, M. P., & Haase, R. H. (2006). Advanced applications of structural equation modeling in counseling psychology research. *Counseling Psychologist*, 34, 806–838.
- Masten, A. S., & Cicchetti, D. (2010). Editorial: developmental cascades. *Development and Psychopathology*, 22, 491–495. doi:10.1017/S0954579410000222.
- Maxwell, S. E., & Cole, D. A. (2007). Bias in cross-sectional analyses of longitudinal mediation. *Psychological Methods*, 12, 23–44.
- Maxwell, S. E., Cole, D. A., & Mitchell, M. A. (2011). Bias in cross-sectional analyses of longitudinal mediation: Partial and complete mediation under an autoregressive model. *Multivariate Behavioral Research*, 46(5), 816–841.
- Maxwell, S. E., & Delaney, H. D. (1993). Bivariate median splits and spurious statistical significance. *Psychological Bulletin*, 113, 181–190.
- McArdle, J. J. (2001). A latent difference score approach to longitudinal dynamic structural analysis. In R. Cudeck, M. du Toit, & D. Sorbom (Eds.), *Structural equation modeling: Present and future* (pp. 341–380). Lincolnwood, IL: Scientific Software International.
- McArdle, J. J., & Hamagami, F. (2001). Latent difference score structural models for linear dynamic analyses with incomplete longitudinal data. In L. M. Collins & A. G. Sayer (Eds.), *New methods for the analysis*

- of change (pp. 138–175). Washington, DC: American Psychological Association.
- McArdle, J. J., & Nesselroade, J. R. (2003). Growth curve analysis in contemporary psychological research. In J. Schinka & W. Velicer (Eds.), *Comprehensive handbook of psychology: Research methods in psychology* (Vol. 2, p. 447–480). New York, NY: Wiley.
- Meeker, Jr., W. Q., Cornwell, L. W., & Aroian, L. A. (1981). The product of two normally distributed random variables. In W. J. Kennedy & R. E. Odeh (Eds.), *Selected tables in mathematical statistics* (Vol. 7, p. 1–256). Providence, RI: American Mathematical Society.
- Meredith, W. (1993). Measurement invariance, factor analysis, and factorial invariance. *Psychometrika*, 58, 525–543.
- Meredith, W., & Tisak, J. (1990). Latent curve analysis. *Psychometrika*, 55(1), 107–122.
- Molenaar, P. C. M. (2004). A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever. *Measurement*, 2, 201–218.
- Morgan, P. L., Staff, J., Hillemeier, M. M., Farkas, G., & Maczuga, S. (2013). Racial and ethnic disparities in ADHD diagnosis from kindergarten to eighth grade. *Pediatrics*, 132, 85–93.
- Morgan, S. L., & Winship, C. (2007). *Counterfactuals and causal inference: Methods and principles for social research*. Cambridge, UK: Cambridge University Press.
- Muller, D., Judd, C. M., & Yzerbyt, V. Y. (2005). When moderation is mediated and mediation is moderated. *Journal of Personality and Social Psychology*, 89, 852–863.
- Muthén, B., & Asparouhov, T. (2012). Bayesian SEM: A more flexible representation of substantive theory. *Psychological Methods*, 17, 313–335.
- Muthén, B. O., & Curran, P. J. (1997). General longitudinal modeling of individual differences in experimental designs: A latent variable framework for analysis and power estimation. *Psychological Methods*, 2, 371–402.
- Muthén, B., & Khoo, S. T. (1998). Longitudinal studies of achievement growth using latent variable modeling. *Learning and Individual Differences*, 10(2), 73–101.
- Muthén, B. O., & Muthén, L. (2014). *Mplus v. 7.2: The comprehensive modeling program for applied researchers* [computer software]. Los Angeles, CA: Muthén & Muthén.
- Nesselroade, J. R. (1991). Interindividual differences in intraindividual changes. In L. M. Collins & J. L. Horn (Eds.), *Best methods for the analysis of change: Recent advances, unanswered questions, future directions* (pp. 92–105). Washington, DC: American Psychological Association.
- Nosek, B. A., Graham, J., Lindner, N. M., Kesebir, S., Hawkins, C. B., Hahn, C., ... Tenney, E. R. (2010). Cumulative and career-stage impact of social-personality psychology programs and their members. *Personality and Social Psychology Bulletin*, 36, 1283–1300.
- O'Malley, A. J. (2012). Instrumental variable specifications and assumptions for longitudinal analysis of mental health cost offsets. *Health Services and Outcomes Research Methodology*, 12, 254–272.
- Owens, E. B., Hinshaw, S. P., Lee, S. S., & Lahey, B. B. (2009). Few girls with childhood attention-deficit/hyperactivity disorder show positive adjustment during adolescence. *Journal of Clinical Child & Adolescent Psychology*, 38, 132–143.
- Ozechowski, T. J. (2014). Empirical Bayes MCMC estimation for modeling treatment processes, mechanism of change, and clinical outcomes in small samples. *Journal of Consulting and Clinical Psychology*, 82, 854–867.
- Pardo, A., & Román, M. (2013). Reflections on the Baron and Kenny model of statistical mediation. *Annales de Psicología*, 29, 614–623.
- Pearl, J. (2001). *Direct and indirect effects*. Proceedings of the 17th Conference on Uncertainty in Artificial Intelligence (pp. 411–420). San Francisco, CA: Morgan Kaufmann.
- Pearl, J. (2009). *Causality: Models, reasoning, and inference*. Cambridge, UK: Cambridge University Press.
- Pearl, J. (2012). The causal mediation formula: A guide to the assessment of pathways and mechanisms. *Prevention Science: The Official Journal of the Society for Prevention Research*, 13(4), 426–436.
- Pedhazur, E. J. (1997). *Multiple regression in behavioral research: Explanation and prediction* (3rd ed.). Fort Worth, TX: Harcourt Brace College Publishers.
- Pinheiro, J. C., & Bates, D. M. (2000). *Mixed-effects models in S and S-plus*. New York, NY: Springer.
- Pituch, K. A., Stapleton, L. M., & Kang, J. Y. (2006). A comparison of single sample and bootstrap methods to assess mediation in cluster randomized trials. *Multivariate Behavioral Research*, 41, 367–400.
- Preacher, K. J., & Hayes, A. F. (2004). SPSS and SAS procedures for estimating indirect effects in simple mediation models. *Behavior Research Methods, Instruments, and Computers*, 36, 717–731.
- Preacher, K. J., & Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods*, 40, 879–891.
- Preacher, K. J., Rucker, D. D., & Hayes, R. F. (2007). Addressing moderated mediation hypotheses: Theory, methods, and prescriptions. *Multivariate Behavioral Research*, 42, 185–227.
- Preacher, K. J., Zyphur, M. J., & Zhang, Z. (2010). A general multi-level SEM framework for assessing multilevel mediation. *Psychological Methods*, 15, 209–233. doi: 10.1037/a0020141
- R Development Core Team. (2013). *R: A language and environment for statistical computing (Version 2.6.1)* [Computer software]. Vienna, Austria: R Foundation for Statistical Computing.
- Rast, P., & Hofer, S. M. (2014). Longitudinal design considerations to optimize power to detect variances and covariances among rates of change: Simulation results based on actual longitudinal studies. *Psychological Methods*, 19, 133–154.
- Raudenbush, S. W. (2001). Comparing personal trajectories and drawing causal inferences from longitudinal data. *Annual Review of Psychology*, 52, 501–525.
- Raudenbush, S. W., & Bryk, A. S. (2002). *Hierarchical linear models: Applications and data analysis methods* (2nd ed.). Thousand Oaks, CA: Sage.
- Raudenbush, S. W., Bryk, A. S., Cheong, Y. F., Congdon, R. T., & du Toit, M. (2011). *HLM 7: Hierarchical linear and nonlinear modeling*. Chicago, IL: Scientific Software International.
- Raudenbush, S. W., & Liu, X. F. (2001). Effects of study duration, frequency of observation, and sample size on power in studies of group differences in polynomial change. *Psychological Methods*, 6, 387–401.
- Reisby, N., Gram, L. F., Bech, P., Nagy, A., Petersen, G. O., Ortmann, J., ... Christiansen, J. (1977). Imipramine: Clinical effects and pharmacokinetic variability. *Psychopharmacology*, 54, 263–272.
- Rigdon, E. E., Schumacker, R. E., & Wothke, W. (1998). A comparative review of interaction and nonlinear modeling. In R. E. Schumacker & G. A. Marcoulides (Eds.), *Interaction and nonlinear effects in structural equation modeling* (pp. 1–16). Mahwah, NJ: Erlbaum.
- Robins, J. M., & Greenland, S. (1992). Identifiability and exchangeability for direct and indirect effects. *Epidemiology*, 3, 143–155.
- Robins, J. M., & Hernan, M. A. (2009). Estimation of the causal effects of time-varying exposures. In G. Fitzmaurice, M. Davidian, G. Verbeke, & G. Molenberghs (Eds.), *Longitudinal data analysis* (pp. 553–599). New York, NY: Chapman & Hall.

- Robins, J. M., Hernán, M. A., & Brumback, B. (2000). Marginal structural models and causal inference in epidemiology. *Epidemiology*, 11(5), 550–560. Retrieved from <http://www.jstor.org/stable/3703997>
- Robinson, W. S. (1950). Ecological correlations and the behavior of individuals. *American Sociological Review*, 15, 351–357.
- Rogosa, D. (1988). Myths about longitudinal research. In K. W. Schaie, R. T. Campbell, W. Meredith, & S. C. Rawlings (Eds.), *Methodological issues in aging research* (pp. 171–209). New York, NY: Springer.
- Rogosa, D., Brandt, D., & Zimowski, M. (1982). A growth curve approach to the measurement of change. *Psychological Bulletin*, 92(3), 726–748.
- Rose, B. M., Holmbeck, G. N., Coakley, R. M., & Franks, E. A. (2004). Mediator and moderator effects in developmental and behavioral pediatric research. *Journal of Developmental and Behavioral Pediatrics*, 25, 58–67.
- Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66, 688–701. doi: 10.1037/h0037350
- Rutter, M. (1999). Resilience concepts and findings: Implications for family therapy. *Journal of Family Therapy*, 21, 119–144.
- SAS Inc. (2014). *SAS (Version 9.3)* [Computer software]. Cary, NC: SAS Inc.
- Schneider, H., & Eisenberg, D. (2006). Who receives a diagnosis of attention deficit/hyperactivity disorder in the United States elementary school population? *Pediatrics*, 117(4), e601–e609.
- Selig, J. P., Preacher, K. J., & Little, T. D. (2012). Modeling time-dependent association in longitudinal data: A lag as moderator approach. *Multivariate Behavioral Research*, 47, 697–716.
- Shen, E., Chou, C. P., Pentz, M. A., & Berhane, K. (2014). Quantile mediation models: A comparison of methods for assessing mediation across the outcome distribution. *Multivariate Behavioral Research*, 49, 471–485. DOI:10.1080/00273171.2014.904221
- Singer, J. D., & Willett, J. B. (2003). *Applied longitudinal data analysis: Modeling change and event occurrence*. Oxford, UK: Oxford University Press.
- Sobel, M. E. (1982). Asymptotic confidence intervals for indirect effects in structural equation models. In S. Leinhardt (Ed.), *Sociological methodology* (pp. 290–312). Washington, DC: American Sociological Association.
- Sobel, M. E. (1986). Some new results on indirect effects and their standard errors in covariance structure models. In N. Tuma (Ed.), *Sociological methodology* (pp. 159–186). Washington, DC: American Sociological Association.
- Sobel, M. E. (2008). Identification of causal parameters in randomized studies with mediating variables. *Journal of Educational and Behavioral Statistics*, 33, 230–251. doi: 10.3102/1076998607307239.
- Steiger, J. H. (1990). Structural model evaluation and modification: An interval estimation approach. *Multivariate Behavioral Research*, 25, 173–180.
- Stone, S., Brown, T., & Hinshaw, S. (2010). ADHD-related school compositional effects: An exploration. *Teachers College Record*, 112(5), 3–4.
- Stoolmiller, M. (1994). Antisocial behavior, delinquent peer association, and unsupervised wandering for boys: Growth and change from childhood to early adolescence. *Multivariate Behavioral Research*, 29, 263–288.
- Ten Have, T. R., Joffe, M. M., Lynch, K. G., Brown, G. K., Maisto, S. A., & Beck, A. T. (2007). Causal mediation analysis with rank preserving models. *Biometrics*, 63, 926–934. doi: 10.1111/j.1541-0420.2007.00766.x
- Tofighi, D., & MacKinnon, D. P. (2011). RMediation: An R package for mediation analysis confidence intervals. *Behavior Research Methods*, 43(3), 692–700. doi: 10.3758/s13428-011-0076-x
- Toh, S., & Hernan, M. A. (2008). Causal inference from longitudinal studies with baseline randomization. *International Journal of Biostatistics*, 4(1), Article 22. doi: 10.2202/1557-4679.1117.
- Tourangeau, K., Nord, C., Lê, T., Sorongon, A. G., & Najarian, M. (2009). *Early childhood longitudinal study, kindergarten class of 1998–99 (ECLS-K), combined user's manual for the ECLS-K eighth-grade and K–8 full sample data files and electronic codebooks (NCES 2009–004)*. Washington, DC: National Center for Education Statistics, Institute of Education Sciences, U.S. Department of Education.
- U.S. Department of Education, National Center for Education Statistics. (2009). Early childhood longitudinal study, kindergarten class of 1998–99 (ECLS-K) kindergarten through eighth grade full sample public-use data and documentation (DVD). (NCES 2009–005). Washington, DC: Author.
- Valeri, L., & VanderWeele, T. J. (2013). Mediation analysis allowing for exposure-mediator interactions and causal interpretation: Theoretical assumptions and implementation with SAS and SPSS macros. *Psychological Methods*, 18, 137–150.
- VanderWeele, T. J. (2010). Bias formulas for sensitivity analysis for direct and indirect effects. *Epidemiology*, 21, 540–551. doi: 10.1097/EDE.0b013e3181df191c
- VanderWeele, T. J., Hawkey, L. C., Thisted, R. A., & Cacioppo, J. T. (2011). A marginal structural model analysis for loneliness: Implications for intervention trials and clinical practice. *Journal of Consulting and Clinical Psychology* 79(2), 457–468.
- VanderWeele, T. J., & Vansteelandt, S. (2009). Conceptual issues concerning mediation, interventions and composition. *Statistics and Its Interface (Special Issue on Mental Health and Social Behavioral Science)*, 2, 457–468.
- VanderWeele, T. J., & Vansteelandt, S. (2013). Mediation analysis with multiple mediators. *Epidemiologic Methods*, 2(1), 95–115.
- Vansteelandt, S. (2009). Estimating direct effects in cohort and case-control studies. *Epidemiology*, 20, 851–860.
- Voelkle, M. C., Oud, J. H. L., Davidov, E., & Schmidt, P. (2012). An SEM approach to continuous time modeling of panel data: Relating authoritarianism and anomia. *Psychological Methods*, 17, 176–192.
- Wall, M. M., & Amemiya, Y. (2000). Estimation for polytomial structural equation models. *Journal of the American Statistical Association*, 95, 929–940.
- West, S. G. (2011). Causal inference in cross sectional and longitudinal mediational models: Introduction to the special section. *Multivariate Behavioral Research*, 46, 812–815.
- Wiley, D. E., & Wiley, J. A. (1970). The estimation of measurement error in panel data. *American Sociological Review*, 35, 112–117.
- Willett, J. B. (1997). Measuring change: What individual growth modeling buys you. In E. Amsel & K. A. Renninger (Eds.), *Change and development: Issues of theory, method, and application* (pp. 213–243). Mahwah, NJ: Lawrence Erlbaum Associates.
- Willett, J. B., & Sayer, A. G. (1994). Using covariance structure analysis to detect correlates and predictors of individual change over time. *Psychological Bulletin*, 116, 363–381.
- Winship, C., & Morgan, S. L. (1999). The estimation of causal effects from observational data. *Annual Review of Sociology*, 25, 659–707.
- Yuan, Y., & MacKinnon, D. P. (2009). Bayesian mediation analysis. *Psychological Methods*, 14, 301–322.
- Zhao, X., Lynch, J. G., & Chen, Q. (2010). Reconsidering Baron and Kenny: Myths and truths about mediation analysis. *Journal of Consumer Research*, 37, 197–206.